

Sakariya Mines & Minerals

Tanzanian Graphite Project

Graphite Flotation

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1 EXECUTIVE SUMMARY

Independent Metallurgical Operations Pty Ltd (IMO) was requested by Mr Susnata Poddar of Sakariya Mines & Minerals (Sakariya) to carry out preliminary graphite flotation testwork to determine the beneficiation amenability on both high and low sulphide bearing ores from the Tanzanian graphite deposit.

The scope of work for this testwork program is comprised of four (4) stages as follows:

1. Sample preparation and characterisation;
2. Optimisation Composite (Fresh & Oxide) testwork;
3. Variability Composite testwork; and
4. Selected Variability Composite optimisation testwork.

A total of eight (8) composites were submitted to IMO's sister company, Metallurgy Pty Ltd (Metallurgy), as ¼ Diamond Drill (DD) core.

Carbon speciation results from each composite are shown in **Table 1** with site calculated Total Graphitic Carbon (TGC) also included for comparative purposes. All composites contained primarily graphite as the carbon bearing mineral except for TDM004 which contained a 1.83% carbonate content.

Table 1: Individual Composite Carbon Speciation and Interval Calculated Assay Results

Element	Unit	TDM001	TDM002	TDM003	TDM004	TDM005	TDM006	TDM007	TDM008
TGC- Interval Estimate	%	10.0	9.0	5.0	6.0	7.0	7.0	7.6	9.7
Total Graphitic Carbon	%	10.2	8.3	5.8	6.2	7.4	8.0	8.0	7.7
Total Carbon	%	10.33	8.57	6.01	8.01	7.68	8.83	8.13	8.13
Carbonate	%	0.02	0.24	0.12	1.83	0.15	0.77	0.11	0.41
Non-Carbonate Carbon	%	10.31	8.33	5.89	6.18	7.53	8.06	8.02	7.72

Final cleaner concentrate Total Carbon (TC) grade and recovery results using the Oxide Optimisation Composite and Fresh Optimisation Composite are shown in **Table 2**. Sakariya representatives selected TDM006 as the Oxide Optimisation Composite and TDM001 as the Fresh Optimisation Composite.

Final concentrate size by assay results are shown in **Table 3** to **Table 4**.

TC grade and recovery profiles for the oxide and fresh composites are respectively presented in **Figure 1** and **Figure 2**.

All composites achieved a final concentrate TC grade of >97% which IMO considers to be a premium product as a typical saleable minimum flake TC grade is >94%. TC recoveries for all composites except for TMD004 were >89% with TMD004's low TC recovery (75.8%) due to a high carbonate mineral content.

Oxide composite flake contents were considered high with the oxide composites having a premium price flake (+150 µm) content of >57%.

The majority of the fresh composites also had high flake contents with TDM003 to TDM005 all having a premium price flake (+150 µm) content of >61%. The TDM008 (Fresh Kaolin) premium price flake (+150 µm) content was exceptional at >73%. TDM001 and TDM0002 both had lower respective premium price flake (+150 µm) contents of 34.8% and 42.5%.

Further optimisation testwork was planned on both TMD0006 and TMD001 before testing the remaining variability composites but given the high TC grade and recoveries from IMO's standard flake flotation conditions, further optimisation testwork was not conducted.

Table 2: Oxide and Fresh Composite Final Cleaner Flotation Stage TC Grade and Recovery

Sample ID	Test ID	TC	
		Grade (%)	Recovery
Oxide Composites			
TDM0006 (Optimisation Composite)	FT01	98.4	93.0%
TDM0007	FT08	97.6	94.9%
Fresh Composites			
TDM001 (Optimisation Composite)	FT02	98.6	94.4%
TDM002	FT03	98.6	89.1%
TDM003	FT05	97.2	95.2%
TDM004	FT06	97.8	75.8%
TDM005	FT07	98.2	90.9%
TDM008 (Fresh / Kaolin)	FT04	97.4	89.1%

Table 3: Oxide Composites Concentrate Size by Assay Results

Sample ID	TDM006			TDM007		
Test ID	FT01			FT08		
Size Fraction (μm)	Mass	TC		Mass	TC	
	Recovery	Grade (%)	Recovery	Recovery	Grade (%)	Recovery
500	8.7%	99.16	8.8%	1.5%	97.99	1.5%
300	28.0%	99.13	28.2%	12.7%	98.87	12.8%
180	24.2%	98.61	24.3%	30.3%	97.08	30.2%
150	6.3%	98.47	6.3%	12.7%	98.28	12.8%
106	9.9%	98.58	9.9%	19.4%	97.92	19.5%
75	6.8%	97.92	6.7%	11.7%	98.09	11.7%
38	6.7%	99.39	6.8%	8.0%	97.43	8.0%
-38	9.4%	94.89	9.1%	3.8%	91.95	3.6%
Total Con	100%	98.45	100%	100%	97.59	100%
Size Fraction (μm)	Mass	TC		Mass	TC	
	Cum. Recovery	Cum. Grade (%)	Cum. Recovery	Cum. Recovery	Cum. Grade (%)	Cum. Recovery
500	8.7%	99.16	8.8%	1.5%	97.99	1.5%
300	36.7%	99.14	37.0%	14.2%	98.77	14.4%
180	60.9%	98.93	61.2%	44.5%	97.62	44.5%
150	67.2%	98.88	67.5%	57.1%	97.77	57.2%
106	77.1%	98.85	77.4%	76.5%	97.81	76.7%
75	83.9%	98.77	84.1%	88.1%	97.84	88.4%
38	90.6%	98.82	90.9%	96.2%	97.81	96.4%
-38	100.0%	98.45	100.0%	100.0%	97.59	100.0%

Table 4: Fresh Composites Concentrate Size by Assay Results

Sample ID	TDM001			TDM002			TDM003		
Test ID	FT02			FT03			FT05		
Size Fraction (μm)	Mass	TC		Mass	TC		Mass	TC	
	Recovery	Grade (%)	Recovery	Recovery	Grade (%)	Recovery	Recovery	Grade (%)	Recovery
500	0.1%	98.67	0.1%	0.5%	98.94	0.5%	1.9%	97.86	1.9%
300	3.9%	98.67	3.9%	5.4%	98.94	5.4%	18.6%	98.76	18.9%
180	19.3%	99.03	19.4%	23.2%	99.28	23.3%	30.1%	98.98	30.7%
150	11.3%	99.41	11.4%	13.2%	99.18	13.3%	9.9%	97.83	10.0%
106	26.5%	99.07	26.6%	22.9%	99.02	23.0%	14.5%	97.33	14.5%
75	16.4%	99.01	16.5%	15.6%	98.67	15.6%	8.6%	97.02	8.6%
38	12.1%	98.47	12.1%	11.7%	99.08	11.8%	7.5%	97.17	7.5%
-38	10.4%	94.97	10.0%	7.5%	93.64	7.1%	8.8%	86.76	7.9%
Total Con	100%	98.58	100%	100%	98.64	100%	100%	97.18	100%
Size Fraction (μm)	Mass	TC		Mass	TC		Mass	TC	
	Cum. Recovery	Grade (%)	Cum. Recovery	Cum. Recovery	Grade (%)	Cum. Recovery	Cum. Recovery	Grade (%)	Cum. Recovery
500	0.1%	98.67	0.1%	0.5%	98.94	0.5%	1.9%	97.86	1.9%
300	4.0%	98.67	4.0%	5.8%	98.94	5.8%	20.5%	98.68	20.8%
180	23.2%	98.97	23.3%	29.0%	99.21	29.2%	50.6%	98.86	51.5%
150	34.6%	99.11	34.8%	42.2%	99.20	42.5%	60.5%	98.69	61.5%
106	61.1%	99.09	61.4%	65.2%	99.14	65.5%	75.0%	98.43	76.0%
75	77.5%	99.08	77.9%	80.7%	99.05	81.1%	83.6%	98.28	84.6%
38	89.6%	98.99	90.0%	92.5%	99.05	92.9%	91.2%	98.19	92.1%
-38	100.0%	98.58	100.0%	100.0%	98.64	100.0%	100.0%	97.18	100.0%

Table 4: Fresh Composites Concentrate Size by Assay Results (continued)

Sample ID	TDM004			TDM005			TDM008		
Test ID	FT06			FT07			FT04		
Size Fraction (μm)	Mass	TC		Mass	TC		Mass	TC	
	Recovery	Grade (%)	Recovery	Recovery	Grade (%)	Recovery	Recovery	Grade (%)	Recovery
500	5.4%	99.04	5.5%	4.5%	98.17	4.6%	14.1%	98.76	14.3%
300	20.2%	98.89	20.4%	25.3%	98.55	25.7%	26.3%	98.37	26.5%
180	26.4%	98.64	26.6%	27.1%	98.33	27.4%	25.1%	98.10	25.3%
150	8.4%	98.40	8.5%	8.2%	98.48	8.3%	7.2%	97.29	7.2%
106	13.5%	98.25	13.6%	11.0%	98.06	11.1%	9.8%	97.03	9.8%
75	8.5%	98.28	8.6%	6.7%	98.86	6.8%	6.0%	96.87	6.0%
38	8.3%	97.36	8.2%	7.0%	97.24	7.0%	5.2%	96.39	5.1%
-38	9.1%	91.46	8.5%	10.0%	87.39	9.0%	6.3%	89.16	5.8%
Total Con	100%	97.85	100%	100%	97.22	100%	100%	97.37	100%
Size Fraction (μm)	Mass	TC		Mass	TC		Mass	TC	
	Cum. Recovery	Grade (%)	Cum. Recovery	Cum. Recovery	Grade (%)	Cum. Recovery	Cum. Recovery	Grade (%)	Cum. Recovery
500	5.4%	99.04	5.5%	4.5%	98.17	4.6%	14.1%	98.76	14.3%
300	25.7%	98.92	26.0%	29.8%	98.49	30.2%	40.3%	98.51	40.8%
180	52.1%	98.78	52.6%	57.0%	98.42	57.7%	65.4%	98.35	66.1%
150	60.5%	98.73	61.1%	65.2%	98.42	66.0%	72.7%	98.24	73.3%
106	74.1%	98.64	74.7%	76.2%	98.37	77.1%	82.5%	98.10	83.1%
75	82.6%	98.60	83.2%	83.0%	98.41	84.0%	88.5%	98.02	89.1%
38	90.9%	98.49	91.5%	90.0%	98.32	91.0%	93.7%	97.93	94.2%
-38	100.0%	97.85	100.0%	100.0%	97.22	100.0%	100.0%	97.37	100.0%

Figure 1: Oxide Composite TC Grade Recovery Profiles

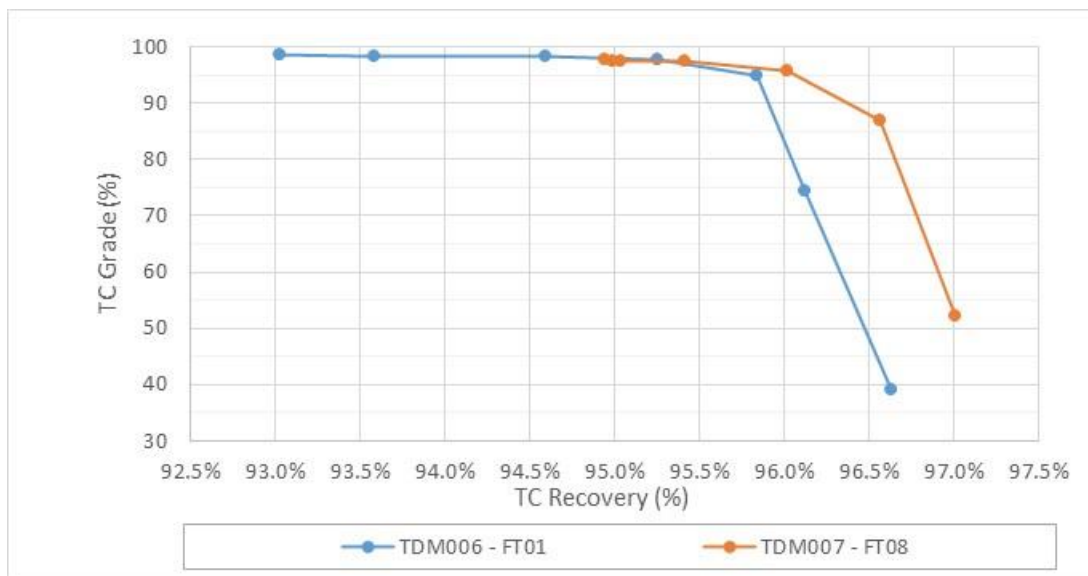
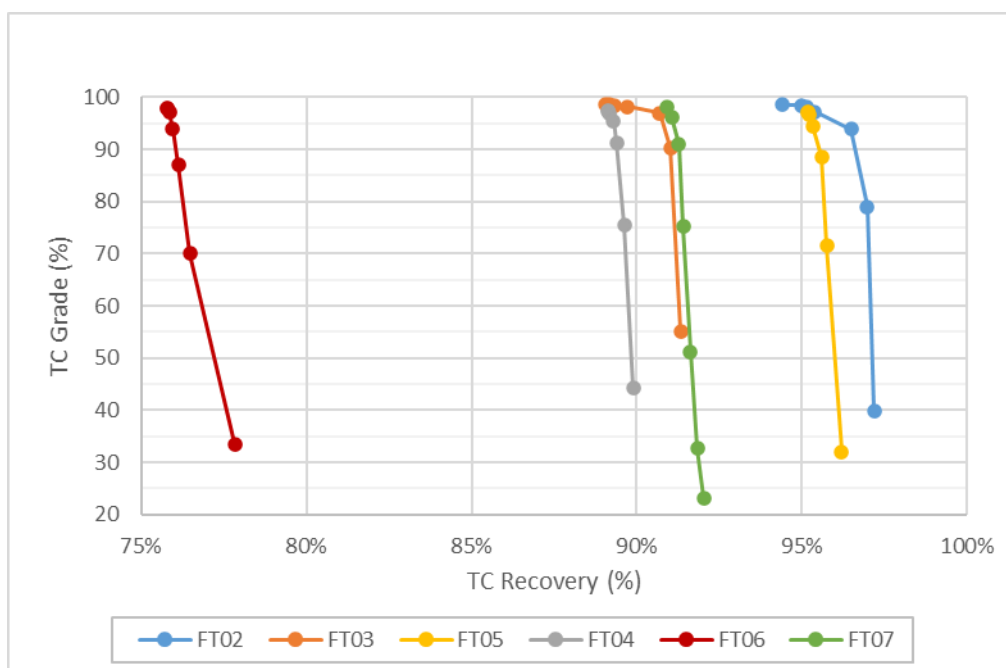


Figure 2: Fresh Composites TC Grade Recovery Profiles



1.1 Conclusions

Based on the testwork conducted to date under the Graphite Flotation Testwork Program, IMO have made the following observations:

- The Oxide composites (TDM006 and TDM007) had high TC grades and recoveries with cleaner 6 concentrate grades ranging between 97.6% to 98.4% and recoveries ranging between 93.0% and 94.9%;
- The Fresh composites (TDM001, TDM002, TDM003, TDM004, TDM005 and TDM008) had high TC grades with cleaner 6 concentrates ranging between 97.18% to 98.64%. Cleaner 6 concentrate recoveries varied between 89.1% and 95.2% with the exception of TMD004 which only achieved 75.8%;
- TMD004 TC recoveries were low due to a high carbonate content 1.8%; and
- Oxide composites with high clay content required more dilute rougher pulp densities (% solids) to provide sufficient mixing (reduced viscosity) to facilitate flotation.

1.2 Recommendations

Based on the observations on the testwork conducted to date, IMO make the following recommendations for further testwork:

- Generate Oxide and Fresh Optimisation Composites, or a single Optimisation Composites from multiple samples representative of the ore body for further process optimisation testwork. Current Oxide and Fresh Optimisation Composites utilised in this testwork program are available for further optimisation testwork if required;
- Once optimised, conduct variability testwork on composites spaced within the deposit to further understand sample process characteristics;
- Conduct bulk flotation testwork on near surface material (to reduce drilling costs) or trench material to generate a bulk concentrate for characterisation, refinery, specific application such as expandable graphite and battery testwork;
- Undertake comminution characterisation testwork on representative drill core intervals with a focus on fresh ore which generally is the harder material when processing; and
- Undertake a scoping study to determine the project economics.

2 INTRODUCTION

Sakariya has two prospects in Tanzania, located at Tanga and Lindi. These are identified with a red pin in the map below (see **Figure 3**). IMO conducted graphite beneficiation testwork on a total of eight (8) composites produced from the Tanga prospect.

Figure 3: Prospect Locations



2.1 Scope of Work

The graphite beneficiation optimisation testwork program was conducted on two (2) Optimisation Composites, a high sulphide composite (Fresh) and a low sulphide composite (Oxide). An additional six (6) composites were treated as variability composites and were subjected to the optimum flotation regime based on the sulphide head assays of each of the composites. Further optimisation work was also carried out on two (2) selected (a Fresh and an Oxide) variability composites. The scope of work consisted of the following:

- Head assay analysis of samples via XRF, carbon and sulphur analyser;
- Beneficiation optimisation testwork of two optimisation composites via rougher, intermediate regrind and cleaner stages with the aim of producing graphite concentrate >94% TC;
- Variability testwork at optimum conditions; and
- Further optimisation work on the two (2) selected (a Fresh and an Oxide) variability composites.

Testwork flowsheets can be found in **APPENDIX A**

3 SAMPLE PROVENANCE

The samples were submitted to IMO's sister company, Metallurgy Pty Ltd (Metallurgy), as ¼ Diamond Drill (DD) core. Details of the eight (8) samples received are presented in **Table 5** and were designated as individual composites as per the Met Sample ID.

The sample receipt is available in **APPENDIX B**.

Table 5: Sample Details Summary

Bore Hole ID	Met Sample ID	Depth (m)		TGC	S (Total)	Total Received Mass
		From	To	%	%	kg
TGDD1019	TDM001	73.50	75.50	10.0	5.0	3.78
		75.50	77.50			3.81
		77.50	79.50			3.74
TGDD1026	TDM002	26.50	28.70	9.0	3.0	4.20
		28.70	30.90			3.77
		30.90	33.05			3.35
TGDD1037	TDM003	51.80	53.80	5.0	3.0	3.74
		53.80	55.80			3.63
		55.80	57.80			3.58
TGDD1039	TDM004	46.30	48.30	6.0	1.0	3.55
		48.30	50.60			4.28
		50.60	52.90			4.62
TGDD1048	TDM005	57.80	61.80	7.0	4.0	4.78
		61.80	66.80			3.76
TGDD1013	TDM006	2.20	6.20	7.0	0.0	3.28
		6.20	8.20			2.84
		8.20	12.17			3.80
TGDD1016	TDM007	10.24	12.87	7.6	0.0	2.12
		12.87	15.10			1.69
		15.40	18.24			2.25
		18.24	22.24			2.49
TGDD1029	TDM008	68.30	73.30	9.7	8.0	3.48
		73.30	75.80			2.12
		77.80	81.80			3.93

3.1 Head Assay Analysis

Head assay analysis was conducted on sub samples of each composite submitted. Analysis was conducted by Intertek Group PLC (Intertek) with each sub sample undergoing the following analyses:

- X-ray Fluorescence (XRF)
- Total Carbon (TC), Total Graphitic Carbon (TGC), Carbonate (CO_3), Total Sulphur (S(t)), Sulphide Sulphur (S^{2-}) and Sulphate (SO_4^{2-}) by a Carbon and Sulphur Analyser; and
- Loss on Ignition (LOI) at both 1,000 and 425 °C.

Complete assay reports are provided in **APPENDIX C**.

Head assay results for all eight (8) composites analysed are presented in **Table 6**, the following observations have been made:

- TC assays were in line with the Sakariya provided interval TGC estimates and ranged from 6.01% to 10.33% TC;
- TGC assays ranged from 5.8% to 10.2% and were in line with the TC assays (with the exception of TDM004) indicating that TC assays are representative of the TGC content within all of the samples except for TDM004.
- Carbonate assays ranged from 0.02% to 1.83% with TDM004 being the highest at 1.83% CO_3^{2-} . Total carbon analysis on the flotation products may result in under reported recovery for TDM004 with TGC assays required to accurately report flotation results;
- Total sulphur ranges from 0.07% to 6.46%, for all composites with the exception of TDM006 and TDM007, most of the sulphur presented as sulphides. TDM006 and TDM007, the little sulphur present was present as sulphates;
- Silica (SiO_2) was the main impurity within the composites tested, ranging from 44.35% to 59.97%;
- Alumina, Iron Oxide and Potassium Oxide assays were notable and could indicate the presence of clays such as mica. Mica was also visually identified while floating.

Table 6: Interval Head Assay Analysis

Element	Unit	TDM001	TDM002	TDM003	TDM004	TDM005	TDM006	TDM007	TDM008
TGC- Interval Estimate	%	10.0	9.0	5.0	6.0	7.0	7.0	7.6	9.7
Total Graphitic Carbon	%	10.2	8.3	5.8	6.2	7.4	8.0	8.0	7.7
Total Carbon	%	10.33	8.57	6.01	8.01	7.68	8.83	8.13	8.13
Carbonate	%	0.02	0.24	0.12	1.83	0.15	0.77	0.11	0.41
Non-Carbonate Carbon	%	10.31	8.33	5.89	6.18	7.53	8.06	8.02	7.72
S	%	4.06	3.78	3.74	2.24	5.06	0.08	0.07	6.46
SO ₄ ²⁺	%	0.02	0.03	0.03	0.01	0.16	0.07	0.08	0.32
S ²⁻	%	4.04	3.75	3.71	2.23	4.90	0.01	<0.01	6.14
LOI 1,000°C	%	12.85	10.86	8.37	10.54	15.26	14.01	15.70	17.87
LOI 425°C	%	0.12	0.27	0.12	0.00	1.11	1.25	2.44	1.36
Al ₂ O ₃	%	12.76	12.70	14.60	12.38	17.09	18.64	20.52	18.80
Fe ₂ O ₃	%	6.74	6.33	7.43	5.59	6.37	5.26	7.94	7.70
K ₂ O	%	1.97	1.27	2.70	2.17	2.75	0.03	0.08	1.97
SiO ₂	%	54.42	54.57	56.83	44.35	55.94	59.97	53.90	49.22

4 GRAPHITE FLOTATION OPTIMISATION TESTWORK

Graphite flotation optimisation testwork was conducted on 2 optimisation composites, a high sulphide (Fresh) composite (TDM001) and a low sulphide (Oxide) composite (TDM006). The optimisation testwork undertaken on both composites is detailed in **Section 4.1** and **Section 4.2** of this report for the Oxide and Fresh composites respectively.

Testwork conditions trialled whilst floating the Oxide and Fresh Composites are detailed in **Table 7**. Each test was comprised of a 2 kg charge stage ground to P₁₀₀ 710 µm. Flotation testwork procedures are presented in **APPENDIX D**.

Table 7: Flotation Optimisation Conditions

Flotation Stage	Condition	Units	Oxide (FT01)	Fresh (FT02)
Roughers	Stages	#	4	4
	Cell Size	L	16*	4
	pH Target		8.5	8.5
	Total Soda Ash	g/t	2,970	85
	Total Ekofol	g/t	459	269
	Total Float Time	min	31	15
Regrind 1	Duration	min	4	4
Cleaner 1	Cell Size	L	4	2
	pH Target		8.5	8.5
	Soda Ash	g/t	155	20
	Ekofol	g/t	47	20
	Flotation Duration	min	13	8
Regrind 2	Duration	min	4	4
Cleaner 2	Cell Size	L	4	2
	pH Target		8.5	8.5
	Soda Ash	g/t	55	15
	Ekofol	g/t	22	20
	Flotation Duration	min	13	14
Regrind 3	Duration	min	4	2
Cleaner 3-6	Stages	#	4	4
	Cell Size	L	2	2
	pH Target		8.5	8.5
	Soda Ash	g/t	35	55
	Ekofol	g/t	18	18
	Flotation Duration	min	37	44
Float Totals	Regrind Duration	min	12	10
	Soda Ash	g/t	3,215	175
	Ekofol	g/t	547	327
	Flotation Duration	min	94	80

*Total volume for the whole 2 kg charge. The largest bench scale cell was 8L. The charge was therefore split into 2 x 1 kg charges and floated in 2 x 8L cells with the rougher concentrates combined to continue with the remainder of the flotation test.

4.1 Oxide Optimisation Composite

Results for the Oxide optimisation test are presented in **Table 8** and the TC grade recovery profiles are presented in **Figure 4**. The final concentrate from this test was sized, with the results listed in **Table 9** and plotted in **Figure 5** and **Figure 6**.

- The final cleaner 6 concentrate grade of 98.45% TC, which was 4.5% higher than that target grade (>94% TC), was achieved with 93.0% recovery.
- The target grade of >94% TC was attained after Cleaner 2, which had a grade of 95.00% TC at 95.8% recovery.
- The grade recovery profile indicates that continuing to clean past 3 cleaning stages with the conditions stated does little to improve the concentrate grade and reduces the overall recovery by 0.5-1.0% per cleaning stage.
- Cleaner 6 concentrate sizing analysis suggests the concentrate was sufficiently liberated with all size fractions producing grades >94% TC. Reducing the regrind times could produce a coarser product that remains above target grade.

Table 8: Oxide (TDM006) Flotation Results Summary

Product Stream	Mass	TC	
	Recovery	Grade (%)	Recovery
	FT01		
Cleaner Concentrate	7.6%	98.45	93.0%
Cleaner Tailings 6	0.1%	89.05	0.6%
Cleaner Tailings 5	0.1%	83.41	1.0%
Cleaner Tailings 4	0.1%	54.21	0.7%
Cleaner Tailings 3	0.3%	17.53	0.6%
Cleaner Tailings 2	2.3%	1.00	0.3%
Cleaner Tailings 1	9.6%	0.43	0.5%
Rougher Tailings	80.0%	0.34	3.4%
Calculated Head	100%	8.08	100%
Assayed Head		8.00	
Cumulative Results			
Cleaner Concentrate 6	7.6%	98.45	93.0%
Cleaner Concentrate 5	7.7%	98.38	93.6%
Cleaner Concentrate 4	7.8%	98.20	94.6%
Cleaner Concentrate 3	7.9%	97.64	95.3%
Cleaner Concentrate 2	8.2%	95.00	95.8%
Cleaner Concentrate 1	10.5%	74.29	96.1%
Rougher Concentrate	20.0%	38.99	96.6%
Calculated Head	100%	8.08	100%
Assayed Head		8.00	

Table 9: Oxide (TDM006) Final Cleaner Concentrate Sizing Results Summary

Size Fraction	Mass	TC	
	Recovery	Grade (%)	Recovery
µm	FT01		
500	8.7%	99.16	8.8%
300	28.0%	99.13	28.2%
180	24.2%	98.61	24.3%
150	6.3%	98.47	6.3%
106	9.9%	98.58	9.9%
75	6.8%	97.92	6.7%
38	6.7%	99.39	6.8%
-38	9.4%	94.89	9.1%
Total Concentrate	100%	98.45	100%
Size Fraction	Mass	TC	
	Cum. Recovery	Cum. Grade (%)	Cum. Recovery
500	0.1%	98.67	0.1%
300	4.0%	98.67	4.0%
180	23.2%	98.97	23.3%
150	34.6%	99.11	34.8%
106	61.1%	99.09	61.4%
75	77.5%	99.08	77.9%
38	89.6%	98.99	90.0%
-38	100.0%	98.58	100.0%

Figure 4: Oxide (TDM006) TC Grade Recovery Profile

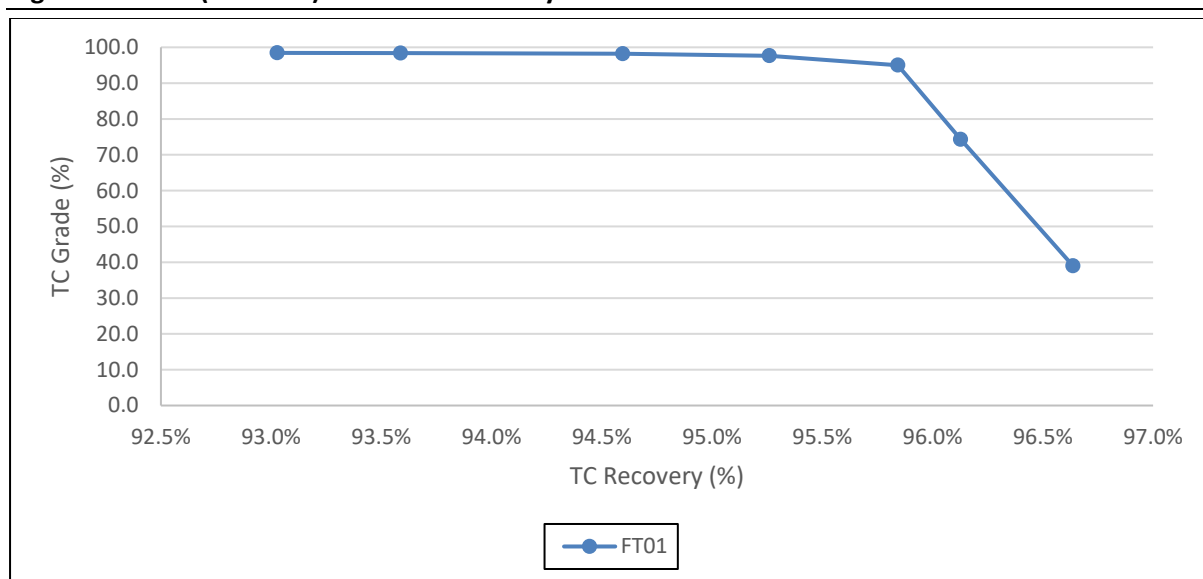


Figure 5: Oxide (TDM006) Final Cleaner Concentrate Individual Size Fraction TC Grades

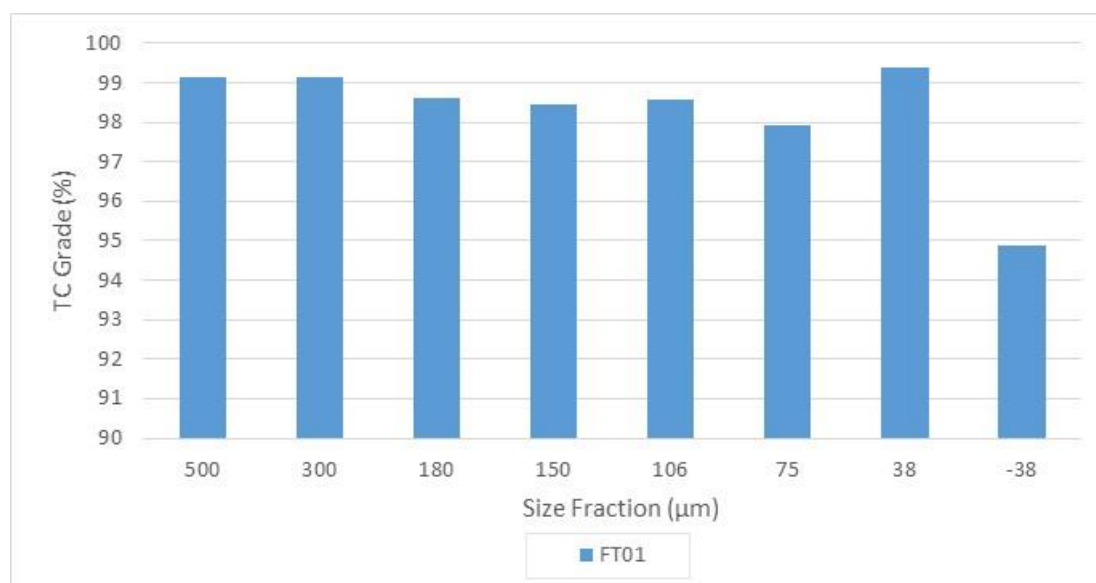
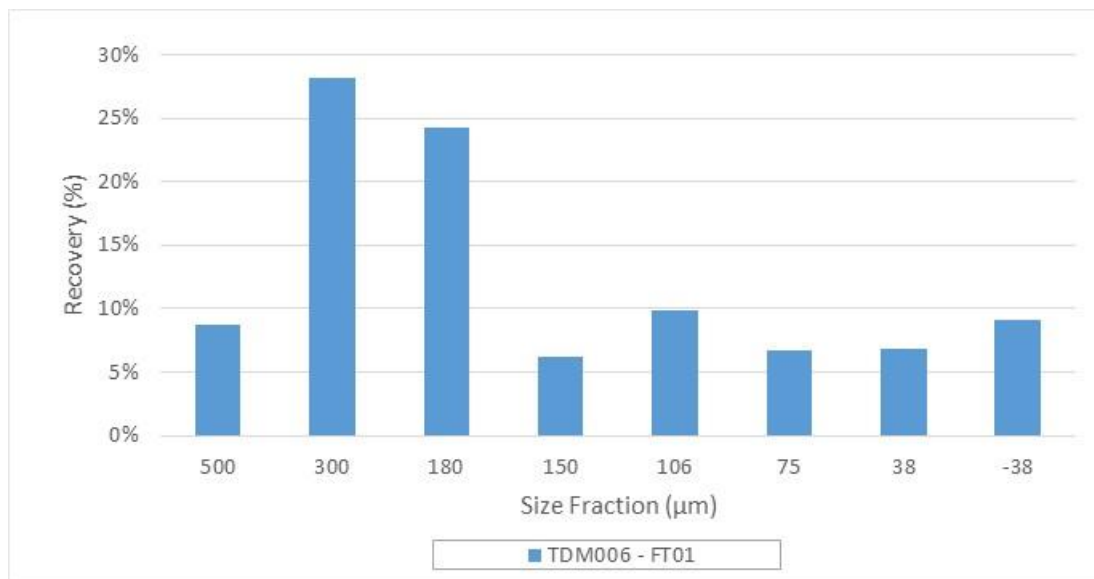


Figure 6: Oxide (TDM006) Final Cleaner Concentrate Individual Size Fraction TC Recoveries



4.2 Fresh Optimisation Composite

Results for the Fresh optimisation test are presented in **Table 10**, with the total carbon grade recovery profile presented in **Figure 7**. The concentrate from this test was sized, with the results of this sizing listed in **Table 11** and plotted in **Figure 8** and **Figure 9**.

- The final cleaner 6 concentrate grade of 98.58% TC was 4.5% higher than that target grade (>94% TC) and was achieved with 94.4% recovery.
- The target grade of >94% TC was attained after Cleaner 3, which had a grade of 97.22% TC at 95.4% recovery.
- Grade recovery profile indicates that continuing to clean past 3 cleaning stages with the conditions stated does little to improve the concentrate grade and reduces the overall recovery by ~0.5% per cleaning stage, similar to the Oxide Composite test.
- Cleaner 6 concentrate sizing analysis suggests the concentrate was sufficiently liberated with all size fractions producing grades >94% TC. Reducing the regrind times could produce a coarser product that remains above the target grade. Compared to the Oxide concentrate, the Fresh Concentrate is finer but is still considered a coarse product.

Table 10: Fresh (TDM001) Flotation Results Summary

Product Stream	Mass	TC	
	Recovery	Grade (%)	Recovery
	FT02		
Cleaner Concentrate	9.9%	98.58	94.4%
Cleaner Tailings 6	0.1%	83.57	0.6%
Cleaner Tailings 5	0.0%	39.57	0.2%
Cleaner Tailings 4	0.1%	19.88	0.3%
Cleaner Tailings 3	0.5%	23.55	1.1%
Cleaner Tailings 2	2.1%	2.40	0.5%
Cleaner Tailings 1	12.5%	0.17	0.2%
Rougher Tailings	74.8%	0.39	2.8%
Calculated Head	100%	10.35	100%
Assayed Head		10.20	
Cumulative Results			
Cleaner Concentrate 6	9.9%	98.58	94.4%
Cleaner Concentrate 5	10.0%	98.47	95.0%
Cleaner Concentrate 4	10.0%	98.23	95.1%
Cleaner Concentrate 3	10.1%	97.22	95.4%
Cleaner Concentrate 2	10.6%	93.84	96.5%
Cleaner Concentrate 1	12.7%	78.97	97.0%
Rougher Concentrate	25.2%	39.95	97.2%
Calculated Head	100%	10.35	100%
Assayed Head		10.20	

Table 11: Fresh (TDM001) Final Cleaner Concentrate Sizing Results Summary

Size Fraction	Mass	TC	
	Recovery	Grade (%)	Recovery
μm	FT02	FT02	FT02
500	0.1%	98.67	0.1%
300	3.9%	98.67	3.9%
180	19.3%	99.03	19.4%
150	11.3%	99.41	11.4%
106	26.5%	99.07	26.6%
75	16.4%	99.01	16.5%
38	12.1%	98.47	12.1%
-38	10.4%	94.97	10.0%
Total Concentrate	100%	98.58	100%
Size Fraction	Mass	TC	
	Cum. Recovery	Cum. Grade (%)	Cum. Recovery
500	0.1%	98.67	0.1%
300	4.0%	98.67	4.0%
180	23.2%	98.97	23.3%
150	34.6%	99.11	34.8%
106	61.1%	99.09	61.4%
75	77.5%	99.08	77.9%
38	89.6%	98.99	90.0%
-38	100.0%	98.58	100.0%

Figure 7: Fresh (TDM001) Grade Recovery Profile

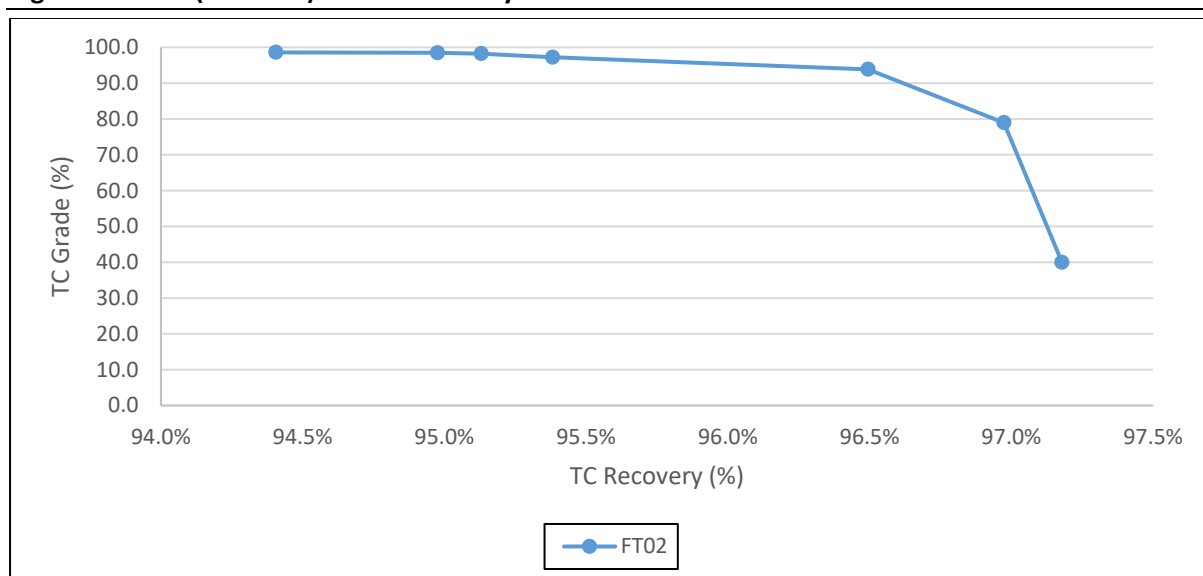


Figure 8: Fresh (TDM001) Final Cleaner Concentrate Individual Size Fraction TC Grades

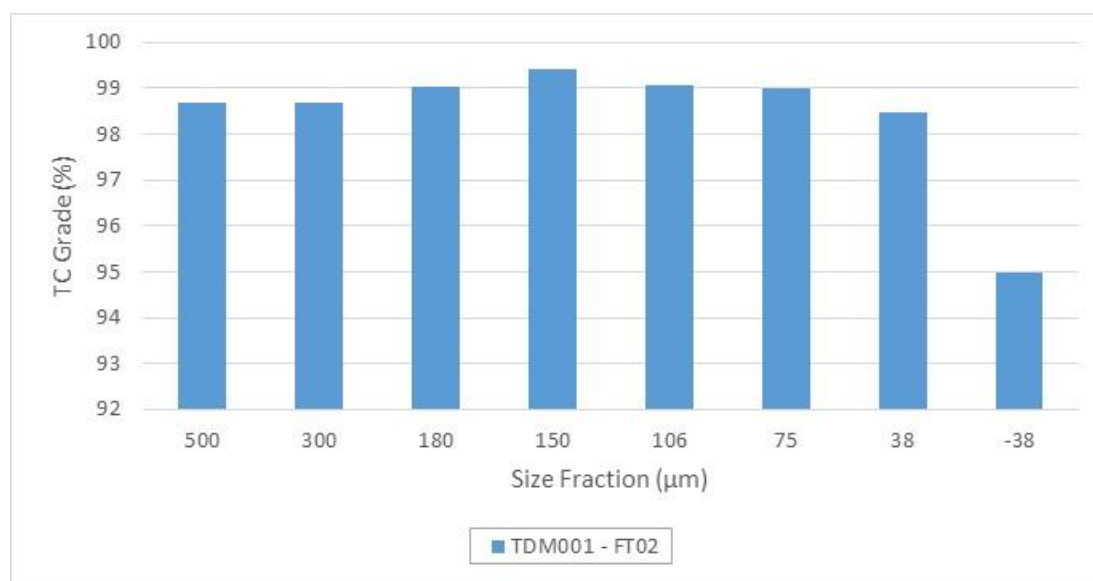
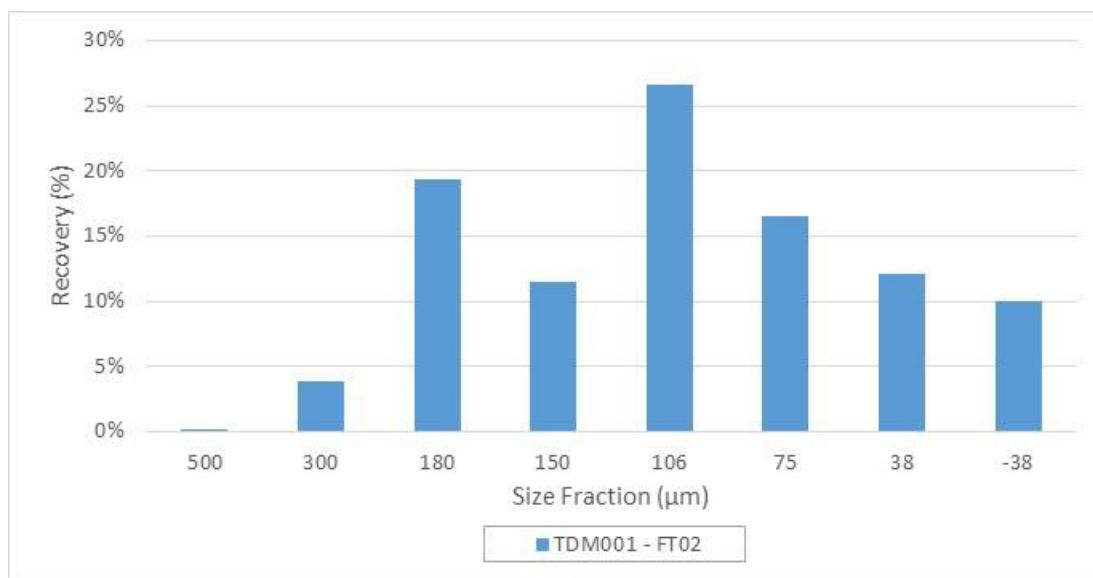


Figure 9: Fresh (TDM001) Final Cleaner Concentrate Individual Size Fraction TC Recoveries



5 VARIABILITY FLOTATION TESTWORK

The decision was made by Sakariya to test the variability composites using the conditions identified in FT01 (Oxide) and FT02 (Fresh). The variability composites were designated into Fresh and Oxide composites by the client as per **Table 12**.

Table 12: Variability Fresh/Oxide Designation

Oxide Variability	Fresh Variability
TDM007	TDM002
	TDM003
	TDM004
	TDM005
	TDM008*

**TDM008 was designated as a heavily weathered sample and IMO was informed by a client representative that the sample contained a notable volume of Kaolin clay.*

Flotation conditions utilised for the variability composites are presented in **Table 13**. The conditions for TDM008 (Fresh-Clay variability composite) were slightly altered within the roughing stages to account for the viscosity and reagent adsorption caused by the kaolin clay present in this sample. These alterations were made by an IMO Metallurgist utilising visual cues.

Table 13: Variability Flotation Conditions

Flotation Stage	Condition	Units	Oxide Variabilities	Fresh Variabilities	Fresh- Clay Variability
Composites			TDM007	TDM002 TDM003 TDM004 TDM0005	TDM008
Roughers	Stages	#	4	4	4
	Feed Mass	kg	2	2	2
	Cell Size	L	16*	4	8
	pH Target		8.5	8.5	8.5
	Total Soda Ash	g/t	3,370	85	5,645
	Total Ekofol	g/t	460	269	291
	Total Float Time	min	15.5	15	15
Regrind 1	Duration	min	4	4	4
Cleaner 1	Cell Size	L	4	2	8
	pH Target		8.5	8.5	8.5
	Soda Ash	g/t	170	20	185
	Ekofol	g/t	47	20	20
	Flotation Duration	min	13	8	8
Regrind 2	Duration	min	4	4	4
Cleaner 2	Cell Size	L	4	2	2
	pH Target		8.5	8.5	8.5
	Soda Ash	g/t	50	15	55
	Ekofol	g/t	22	20	19
	Flotation Duration	min	13	14	14
Regrind 3	Duration	min	4	2	2
Cleaner 3-6	Stages	#	4	4	4
	Cell Size	L	2	2	2
	pH Target		8.5	8.5	8.5
	Soda Ash	g/t	40	55	55
	Ekofol	g/t	18	18	56
	Flotation Duration	min	37	44	44
Float Totals	Regrind Duration	min	12	10	10
	Soda Ash	g/t	3,630	175	5,940
	Ekofol	g/t	547	327	386
	Flotation Duration	min	78.5	80	80

* - A 16 L cell was unavailable, so the rougher flotation of 2 kg was completed by separately floating two 1 kg batches in an 8 L flotation cell.

5.1 Oxide Variability Composites

The Oxide variability composite flotation results are listed alongside those for the optimisation composite in **Table 14**, and grade recovery profiles are plotted in **Figure 10**. The final concentrates from the two Oxide flotation tests were sized with the results presented in **Table 15**, **Figure 11** and **Figure 12**. Based on the results, IMO made the following observations on the Oxide variability composites:

- The Oxide variability composite (TDM007) final cleaner 6 concentrate grade of 97.59% TC was achieved with 94.9% recovery.
- The target grade of >94% TC was attained after Cleaner 2, which had a grade of 95.72% TC at 96.0% recovery.
- The Oxide variability composite (TDM007) grade recovery profile indicates that continuing to clean past 3 cleaning stages with the conditions stated does little to improve the concentrate grade.
- Cleaner 6 concentrate sizing analysis suggests the Oxide variability composite (TDM007) concentrate was sufficiently liberated with all size fractions, except -38 μm , producing grades >94% TC.
- The oxide optimisation composite (TDM006) had a higher fraction of +300 μm and +500 μm graphite flake compared to the oxide variability composite (TDM007). The optimisation composite +300 μm and +500 μm fractions recoveries were 28.2% and 8.2%, respectively, whereas those for the variability composite were 12.8% and 1.5%, respectively.
- Reducing the oxide variability composite (TDM007) grind times may increase the +300 μm and +500 μm fractions' recoveries

Table 14: Oxide Optimisation and Variability Flotation Results Summary

Lithology	Oxide					
Composite Name	Optimisation			Variability		
Sample ID	TDM006			TDM007		
Test ID	FT01			FT08		
Stage	Mass		TC	Mass		TC
	Recovery	Grade (%)	Recovery	Recovery	Grade (%)	Recovery
Cleaner Con 6	7.6%	98.45	93.0%	7.8%	97.59	94.9%
Cleaner Con 5	7.7%	98.38	93.6%	7.8%	97.57	95.0%
Cleaner Con 4	7.8%	98.20	94.6%	7.8%	97.55	95.0%
Cleaner Con 3	7.9%	97.64	95.3%	7.8%	97.43	95.4%
Cleaner Con 2	8.2%	95.00	95.8%	8.0%	95.72	96.0%
Cleaner Con 1	10.5%	74.29	96.1%	8.9%	86.90	96.6%
Rougher Con	20.0%	38.99	96.6%	14.8%	52.13	97.0%
Calculated Head	100%	8.08	100%	100%	7.97	100%
Assayed Head		8.00			8.00	

Figure 10: Oxide Optimisation (TDM006) and Variability (TDM007) Composite TC Grade Recovery Profiles

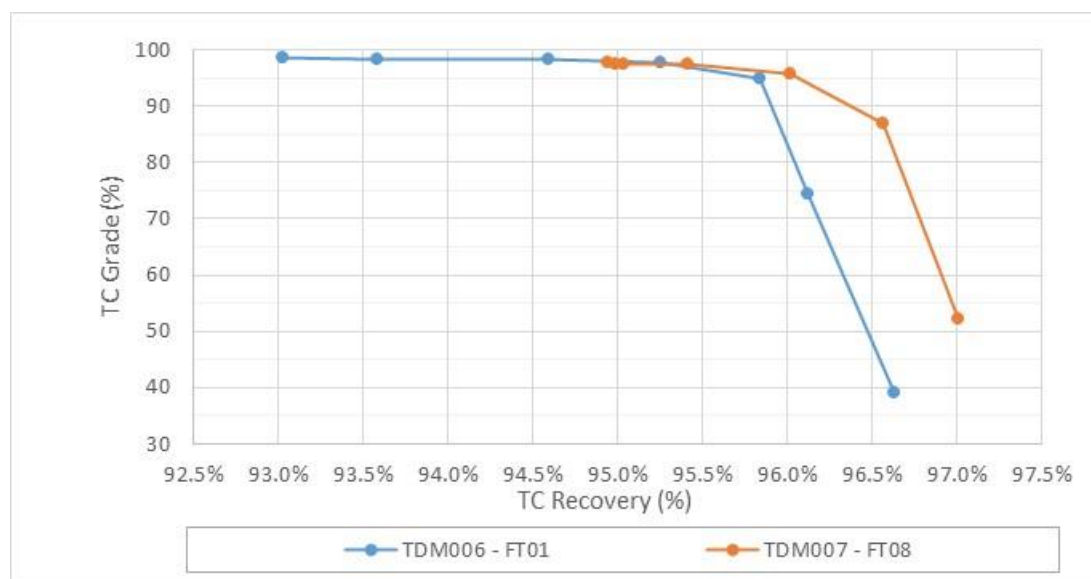


Table 15: Oxide Optimisation (TDM006) and Variability (TDM007) Composites Final Cleaner Concentrates Sizing Results Summary

Sample ID	TDM006			TDM007		
Test ID	FT01			FT08		
Size Fraction (μm)	Mass	TC		Mass	TC	
	Recovery	Grade (%)	Recovery	Recovery	Grade (%)	Recovery
500	8.7%	99.16	8.8%	1.5%	97.99	1.5%
300	28.0%	99.13	28.2%	12.7%	98.87	12.8%
180	24.2%	98.61	24.3%	30.3%	97.08	30.2%
150	6.3%	98.47	6.3%	12.7%	98.28	12.8%
106	9.9%	98.58	9.9%	19.4%	97.92	19.5%
75	6.8%	97.92	6.7%	11.7%	98.09	11.7%
38	6.7%	99.39	6.8%	8.0%	97.43	8.0%
-38	9.4%	94.89	9.1%	3.8%	91.95	3.6%
Total Con	100%	98.45	100%	100%	97.59	100%
Size Fraction (μm)	Mass	TC		Mass	TC	
	Cum. Recovery	Cum. Grade (%)	Cum. Recovery	Cum. Recovery	Cum. Grade (%)	Cum. Recovery
500	8.7%	99.16	8.8%	1.5%	97.99	1.5%
300	36.7%	99.14	37.0%	14.2%	98.77	14.4%
180	60.9%	98.93	61.2%	44.5%	97.62	44.5%
150	67.2%	98.88	67.5%	57.1%	97.77	57.2%
106	77.1%	98.85	77.4%	76.5%	97.81	76.7%
75	83.9%	98.77	84.1%	88.1%	97.84	88.4%
38	90.6%	98.82	90.9%	96.2%	97.81	96.4%
-38	100.0%	98.45	100.0%	100.0%	97.59	100.0%

Figure 11: Oxide Optimisation (TDM006) and Variability (TDM007) Composites Final Cleaner Concentrates Individual Size TC Grades

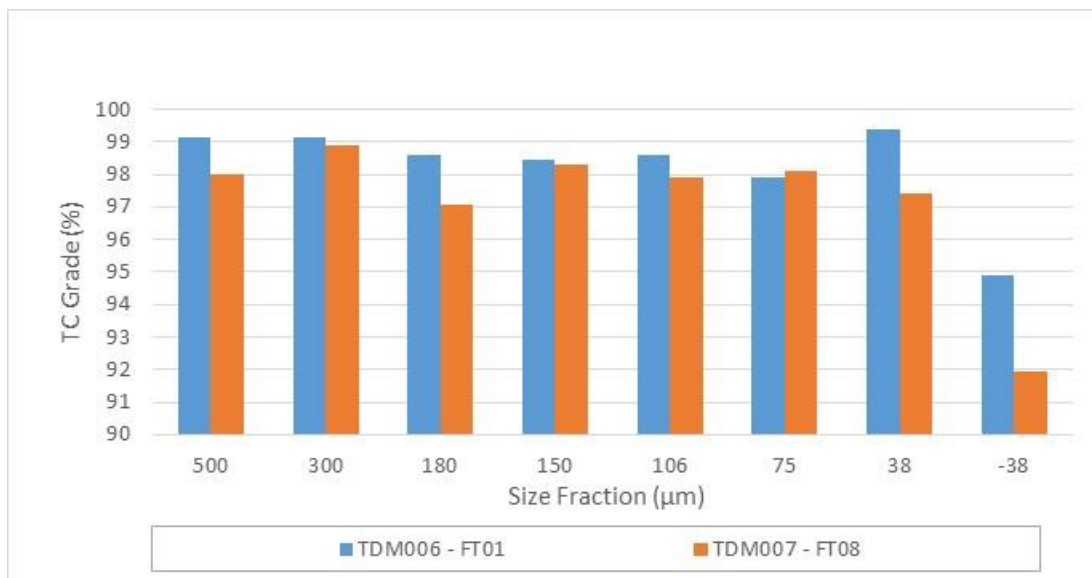
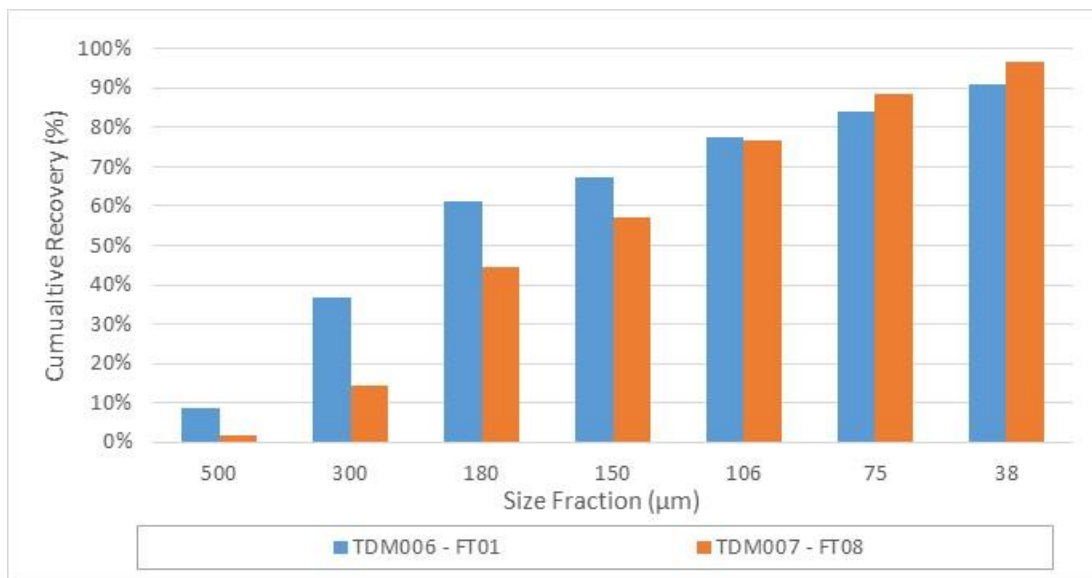


Figure 12: Oxide Optimisation (TDM006) and Variability (TDM007) Composite Final Cleaner Concentrate Individual Size Fraction TC Recoveries



5.2 Oxide Variability Composite TDM007 – Effect of Reducing Regrind Times

A further optimisation test (FT10), using reduced regrind times, was undertaken on the variability composite (TDM007) to increase the +300 µm and +500 µm fractions' concentrate mass recoveries. The conditions for this test (FT10) and FT08 which FT10 was based on are provided in **Table 16**.

Table 16: Oxide Variability (TDM007) Flotation Conditions

Stage	Parameter	Lithology	Oxide	
		Sample ID	TDM007	
		Units / Test ID	FT08	FT10
Feed	Grind Size P100	µm	710	
Roughers	No of Stages	#	4	
	pH Target		8.5	
	Total Soda Ash	g/t	3,370	3,255
	Total Ekofol	g/t	460	460
	Total Float Time	min	15.5	16
Regrind 1	Duration	min	4	3
Cleaner 1	pH Target		8.5	
	Soda Ash	g/t	170	260
	Ekofol	g/t	47	47
	Flotation Duration	min	13	
Regrind 2	Duration	min	4	3
Cleaner 2	pH Target		8.5	
	Soda Ash	g/t	50	90
	Ekofol	g/t	22	22
	Flotation Duration	min	13	
Regrind 3	Duration	min	4	3
Cleaner 3-6	Stages	#	4	
	pH Target		8.5	
	Soda Ash	g/t	40	60
	Ekofol	g/t	18	18
	Flotation Duration	min	37	
Float Totals	Regrind Duration	min	12	9
	Soda Ash	g/t	3630	3665
	Ekofol	g/t	547	547
	Flotation Duration	min	78.5	79

Results of Oxide variability composite flotation tests with varying grind times are listed in **Table 17** and grade recovery profiles are plotted in **Figure 13**. The final concentrates from the two Oxide flotation tests were sized with the results presented in **Table 18** and **Figure 14** to **Figure 16**. Based on the results, IMO made the following observations on the Oxide Variability Composites:

- In the rougher stage, the FT08 rougher concentrate had a 52.13% TC grade at 97.0% TC recovery whereas the FT10 rougher concentrate only had 34.06% TC grade at 98.6% TC recovery. FT10 had a higher mass recovery (23.1%) than FT08 (14.8%) which has resulted in the rougher concentrate TC grade being diluted by gangue minerals;
- The target grade of >94% TC was attained after Cleaner 2 when using the longer regrind time (in FT08) but attained after Cleaner 4 when using the shorter regrind time (in FT10). However, this is more likely attributable to the FT08 rougher concentrate having a higher grade (52.13% TC) compared to FT10 (34.06% TC);
- Both 12 minute and 9 minute regrind times had TC grades > 94% for all fractions except the >38 μm fraction (see **Table 18** and **Figure 14**); and
- Decreasing the regrind time from 12 minutes to 9 minutes had minimal impact on the final concentrate +150 μm cumulative graphite mass distribution, increasing it from 57.3% to 58.3%, respectively (see **Table 18** and **Figure 16**).

Table 17: Oxide Variability Composite (TDM007) - Effect of Regrind Time

Lithology	Oxide					
Sample ID	TDM007					
Test ID	FT08			FT10		
Total Regrind Time (min)	12			9		
Stage	Mass		TC	Mass		TC
	Recovery	Grade (%)	Recovery	Recovery	Grade (%)	Recovery
Cleaner Con 6	7.8%	97.59	94.9%	8.0%	96.59	96.7%
Cleaner Con 5	7.8%	97.57	95.0%	8.0%	96.31	96.8%
Cleaner Con 4	7.8%	97.55	95.0%	8.1%	95.44	97.1%
Cleaner Con 3	7.8%	97.43	95.4%	8.4%	92.51	97.3%
Cleaner Con 2	8.0%	95.72	96.0%	9.5%	81.87	97.8%
Cleaner Con 1	8.9%	86.90	96.6%	12.9%	60.80	98.2%
Rougher Con	14.8%	52.13	97.0%	23.1%	34.06	98.6%
Calculated Head	100%	7.97	100%	100%	7.97	100%
Assayed Head		8.00			8.00	

Figure 13: Oxide Variability Composite (TDM007) - Effect of Regrind Time on TC Grade Recovery Profile

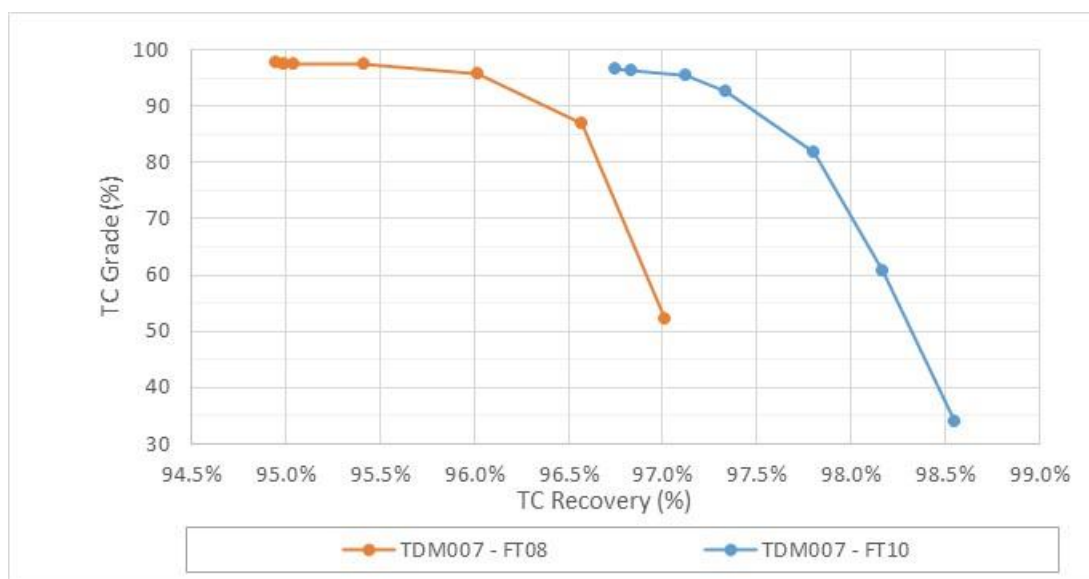


Table 18: Variability Composite (TDM007) - Effect of Regrind Time on Concentrate Sizing

Sample ID	TDM007					
Test ID	FT08			FT10		
Regrind Time (min)	12			9		
Size Fraction (µm)	Mass	TC		Mass	TC	
	Recovery	Grade (%)	Recovery	Recovery	Grade (%)	Recovery
500	1.5%	97.99	1.5%	1.9%	97.85	1.9%
300	12.7%	98.87	12.8%	13.4%	97.45	13.5%
180	30.3%	97.08	30.2%	30.2%	97.15	30.4%
150	12.7%	98.28	12.8%	12.5%	96.73	12.5%
106	19.4%	97.92	19.5%	17.7%	96.89	17.8%
75	11.7%	98.09	11.7%	11.0%	96.60	11.0%
38	8.0%	97.43	8.0%	8.4%	95.76	8.3%
-38	3.8%	91.95	3.6%	5.0%	90.41	4.7%
Total Con	100%	97.59	100%	100%	96.59	100%
Size Fraction (µm)	Mass	TC		Mass	TC	
	Cum. Recovery	Cum. Grade (%)	Cum. Recovery	Cum. Recovery	Cum. Grade (%)	Cum. Recovery
500	1.5%	97.99	1.5%	1.9%	97.85	1.9%
300	14.2%	98.77	14.4%	15.3%	97.50	15.4%
180	44.5%	97.62	44.5%	45.5%	97.27	45.8%
150	57.1%	97.77	57.2%	57.9%	97.15	58.3%
106	76.5%	97.81	76.7%	75.6%	97.09	76.0%
75	88.1%	97.84	88.4%	86.6%	97.03	87.0%
38	96.2%	97.81	96.4%	95.0%	96.92	95.3%
-38	100.0%	97.59	100.0%	100.0%	96.59	100.0%

Figure 14: Oxide Variability Composite (TDM007) Final Cleaner Concentrates Individual Size Fraction TC Grades – Effect of Grind Time

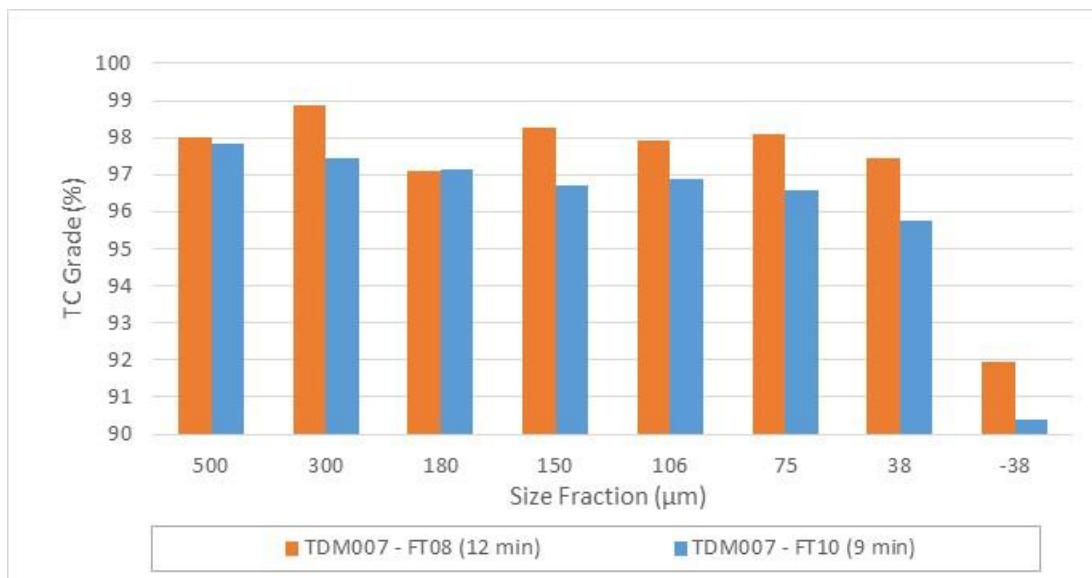


Figure 15: Oxide Variability Composite (TDM007) Final Cleaner Concentrate Individual Size Fraction TC Recoveries– Effect of Grind Time

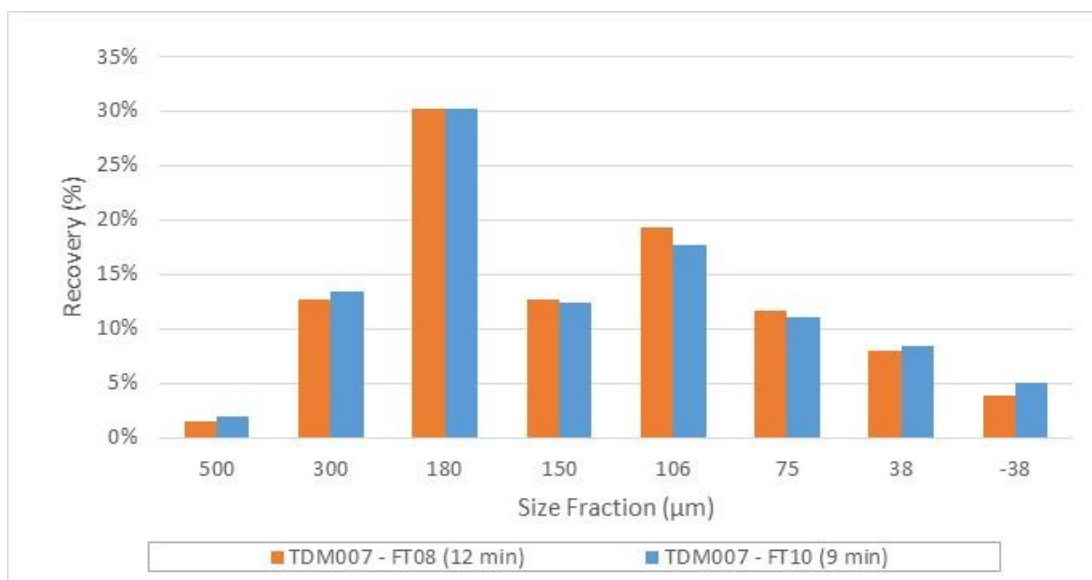
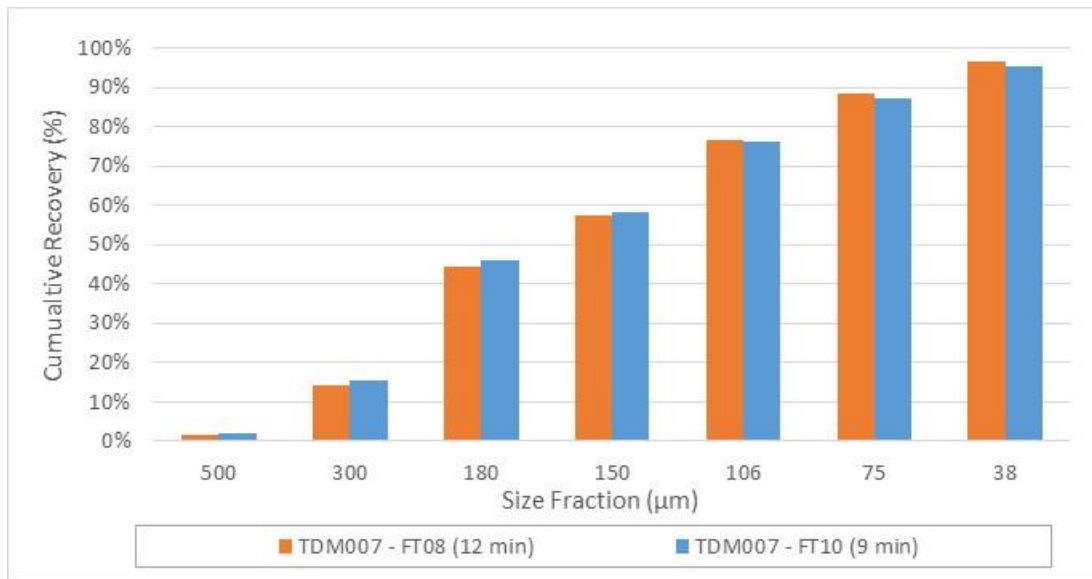


Figure 16: Oxide Variability Composite (TDM007) Final Cleaner Concentrate Cumulative Mass Recovery Size Distribution – Effect of Grind Time



5.3 Fresh Variability Composites

The Fresh variability composites flotation results are listed alongside those for the Fresh optimisation composite in **Table 19**. Grade recovery profiles for all variability composites are plotted in **Figure 17**, except for TDM004 which had a lower recovery and is plotted in **Figure 18**. The final concentrates from the six (6) Fresh flotation tests were sized with the results presented in **Table 20**, **Figure 19** and **Figure 20**. Based on the results, IMO made the following observations on the Fresh variability composites:

- The Fresh composites final cleaner 6 concentrate grades were all higher than the target grade of 94% TC and ranged from 97.18% TC to 98.64% TC. However final cleaner 6 concentrate recoveries varied more widely, ranging from 75.8% for TDM004 to 95.2% for TDM003.
- The number of cleaner stages required to attain the target grade of >94% TC ranged from two (for TDM002) to five (for TDM0005), with the other Fresh composites requiring either 3 or 4 stages.
- Cleaner 6 concentrate sizing analysis suggests the Fresh variability composites concentrates were all sufficiently liberated, with all size fractions, except -38 µm, producing grades >94% TC.
- Fresh cleaner 6 concentrates cumulative recoveries for size fractions greater than 300 µm ranged from 4.0% to 40.8% and cumulative recoveries for size fractions greater than 180 µm ranged from 23.3% to 66.1%.

Table 19: Fresh Variability Flotation Results Summary

Lithology	Fresh								
Sample ID	TDM001			TDM002			TDM003		
Test ID	FT02			FT03			FT05		
Stage	Mass	TC		Mass	TC		Mass	TC	
	Recovery	Grade (%)	Recovery	Recovery	Grade (%)	Recovery	Recovery	Grade (%)	Recovery
Cleaner Con 6	9.9%	98.58	94.4%	7.7%	98.64	89.1%	5.9%	97.18	95.2%
Cleaner Con 5	10.0%	98.47	95.0%	7.7%	98.57	89.2%	5.9%	97.10	95.2%
Cleaner Con 4	10.0%	98.23	95.1%	7.7%	98.50	89.3%	5.9%	96.65	95.2%
Cleaner Con 3	10.1%	97.22	95.4%	7.8%	98.26	89.7%	6.1%	94.51	95.3%
Cleaner Con 2	10.6%	93.84	96.5%	8.0%	96.94	90.7%	6.5%	88.58	95.6%
Cleaner Con 1	12.7%	78.97	97.0%	8.6%	90.38	91.0%	8.1%	71.57	95.7%
Rougher Con	25.2%	39.95	97.2%	14.1%	55.14	91.3%	18.1%	32.05	96.2%
Calculated Head	100%	10.35	100%	100.0%	8.51	100.0%	100.0%	6.03	100.0%
Assayed Head		10.20			8.30			5.80	

Note: Highlighted cells indicate the first cleaner concentrate for each sample where TC > 94%.

Table 19: Fresh Variability Flotation Results Summary (continued)

Lithology	Fresh						Fresh / Kaolin		
Sample ID	TDM004			TDM005			TDM008		
Test ID	FT06			FT07			FT04		
Stage	Mass	TC		Mass	TC		Mass	TC	
	Recovery	Grade (%)	Recovery	Recovery	Grade (%)	Recovery	Recovery	Grade (%)	Recovery
Cleaner Con 6	6.0%	97.85	75.8%	7.5%	98.17	90.9%	7.2%	97.37	89.1%
Cleaner Con 5	6.0%	97.80	75.8%	7.7%	96.27	91.1%	7.2%	97.31	89.1%
Cleaner Con 4	6.1%	97.14	75.9%	8.2%	90.98	91.3%	7.3%	97.03	89.2%
Cleaner Con 3	6.3%	93.85	76.0%	9.9%	75.14	91.4%	7.4%	95.54	89.3%
Cleaner Con 2	6.8%	87.04	76.1%	14.6%	51.18	91.6%	7.7%	91.35	89.4%
Cleaner Con 1	8.5%	69.98	76.5%	22.9%	32.64	91.8%	9.4%	75.39	89.6%
Rougher Con	18.1%	33.47	77.8%	32.4%	23.13	92.0%	16.1%	44.19	89.9%
Calculated Head	100.0%	7.77	100.0%	100.0%	8.15	100.0%	100.0%	7.90	100.0%
Assayed Head		6.20			7.40			7.70	

Note: Highlighted cells indicate the first cleaner concentrate for each sample where TC > 94%.

Figure 17: Fresh Optimisation and Variability Composites TC Grade Recovery Profiles

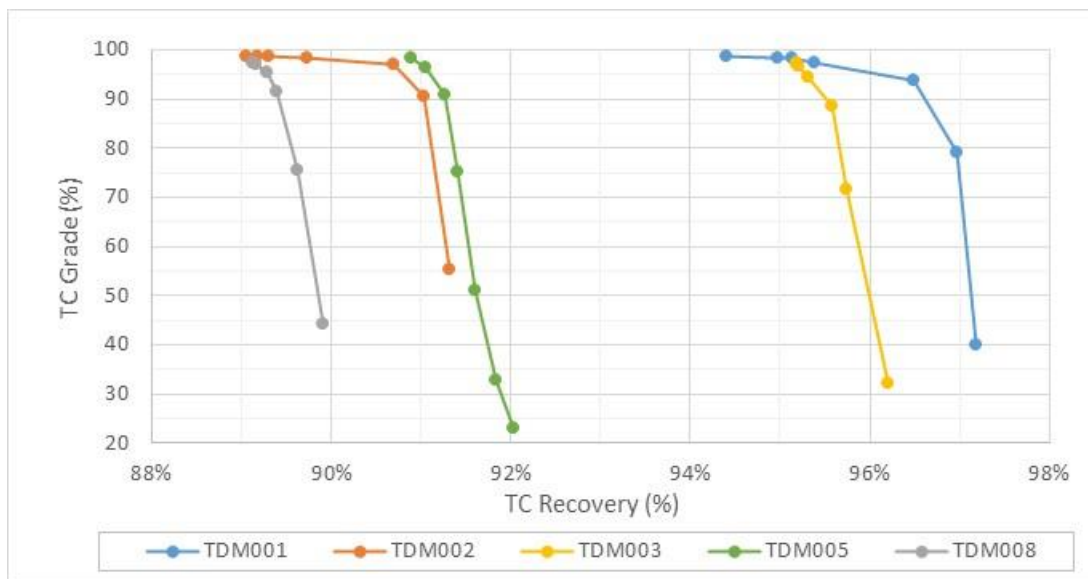


Figure 18: Fresh Variability Composite TDM004 TC Grade Recovery Profile

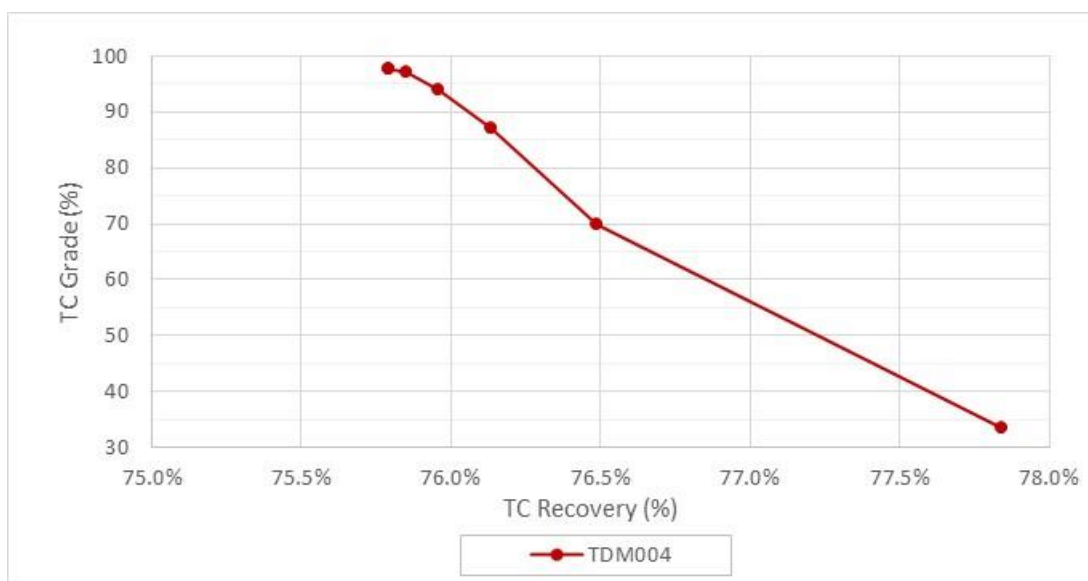


Table 20: Fresh Optimisation (TDM001) and Variability Composites Concentrate Sizing Results Summary

Sample ID	TDM001			TDM002			TDM003		
Test ID	FT02			FT03			FT05		
Size Fraction (μm)	Mass	TC		Mass	TC		Mass	TC	
	Recovery	Grade (%)	Recovery	Recovery	Grade (%)	Recovery	Recovery	Grade (%)	Recovery
500	0.1%	98.67	0.1%	0.5%	98.94	0.5%	1.9%	97.86	1.9%
300	3.9%	98.67	3.9%	5.4%	98.94	5.4%	18.6%	98.76	18.9%
180	19.3%	99.03	19.4%	23.2%	99.28	23.3%	30.1%	98.98	30.7%
150	11.3%	99.41	11.4%	13.2%	99.18	13.3%	9.9%	97.83	10.0%
106	26.5%	99.07	26.6%	22.9%	99.02	23.0%	14.5%	97.33	14.5%
75	16.4%	99.01	16.5%	15.6%	98.67	15.6%	8.6%	97.02	8.6%
38	12.1%	98.47	12.1%	11.7%	99.08	11.8%	7.5%	97.17	7.5%
-38	10.4%	94.97	10.0%	7.5%	93.64	7.1%	8.8%	86.76	7.9%
Total Con	100%	98.58	100%	100%	98.64	100%	100%	97.18	100%
Size Fraction (μm)	Mass	TC		Mass	TC		Mass	TC	
	Cum. Recovery	Grade (%)	Cum. Recovery	Cum. Recovery	Grade (%)	Cum. Recovery	Cum. Recovery	Grade (%)	Cum. Recovery
500	0.1%	98.67	0.1%	0.5%	98.94	0.5%	1.9%	97.86	1.9%
300	4.0%	98.67	4.0%	5.8%	98.94	5.8%	20.5%	98.68	20.8%
180	23.2%	98.97	23.3%	29.0%	99.21	29.2%	50.6%	98.86	51.5%
150	34.6%	99.11	34.8%	42.2%	99.20	42.5%	60.5%	98.69	61.5%
106	61.1%	99.09	61.4%	65.2%	99.14	65.5%	75.0%	98.43	76.0%
75	77.5%	99.08	77.9%	80.7%	99.05	81.1%	83.6%	98.28	84.6%
38	89.6%	98.99	90.0%	92.5%	99.05	92.9%	91.2%	98.19	92.1%
-38	100.0%	98.58	100.0%	100.0%	98.64	100.0%	100.0%	97.18	100.0%

Table 20: Fresh Optimisation (TDM001) and Variability Composites Concentrate Sizing Results Summary (continued)

Sample ID	TDM004			TDM005			TDM008		
Test ID	FT06			FT07			FT04		
Size Fraction (μm)	Mass	TC		Mass	TC		Mass	TC	
	Recovery	Grade (%)	Recovery	Recovery	Grade (%)	Recovery	Recovery	Grade (%)	Recovery
500	5.4%	99.04	5.5%	4.5%	98.17	4.6%	14.1%	98.76	14.3%
300	20.2%	98.89	20.4%	25.3%	98.55	25.7%	26.3%	98.37	26.5%
180	26.4%	98.64	26.6%	27.1%	98.33	27.4%	25.1%	98.10	25.3%
150	8.4%	98.40	8.5%	8.2%	98.48	8.3%	7.2%	97.29	7.2%
106	13.5%	98.25	13.6%	11.0%	98.06	11.1%	9.8%	97.03	9.8%
75	8.5%	98.28	8.6%	6.7%	98.86	6.8%	6.0%	96.87	6.0%
38	8.3%	97.36	8.2%	7.0%	97.24	7.0%	5.2%	96.39	5.1%
-38	9.1%	91.46	8.5%	10.0%	87.39	9.0%	6.3%	89.16	5.8%
Total Con	100%	97.85	100%	100%	97.22	100%	100%	97.37	100%
Size Fraction (μm)	Mass	TC		Mass	TC		Mass	TC	
	Cum. Recovery	Grade (%)	Cum. Recovery	Cum. Recovery	Grade (%)	Cum. Recovery	Cum. Recovery	Grade (%)	Cum. Recovery
500	5.4%	99.04	5.5%	4.5%	98.17	4.6%	14.1%	98.76	14.3%
300	25.7%	98.92	26.0%	29.8%	98.49	30.2%	40.3%	98.51	40.8%
180	52.1%	98.78	52.6%	57.0%	98.42	57.7%	65.4%	98.35	66.1%
150	60.5%	98.73	61.1%	65.2%	98.42	66.0%	72.7%	98.24	73.3%
106	74.1%	98.64	74.7%	76.2%	98.37	77.1%	82.5%	98.10	83.1%
75	82.6%	98.60	83.2%	83.0%	98.41	84.0%	88.5%	98.02	89.1%
38	90.9%	98.49	91.5%	90.0%	98.32	91.0%	93.7%	97.93	94.2%
-38	100.0%	97.85	100.0%	100.0%	97.22	100.0%	100.0%	97.37	100.0%

Figure 19: Fresh Composites Final Cleaner Concentrate Individual Size TC Grades

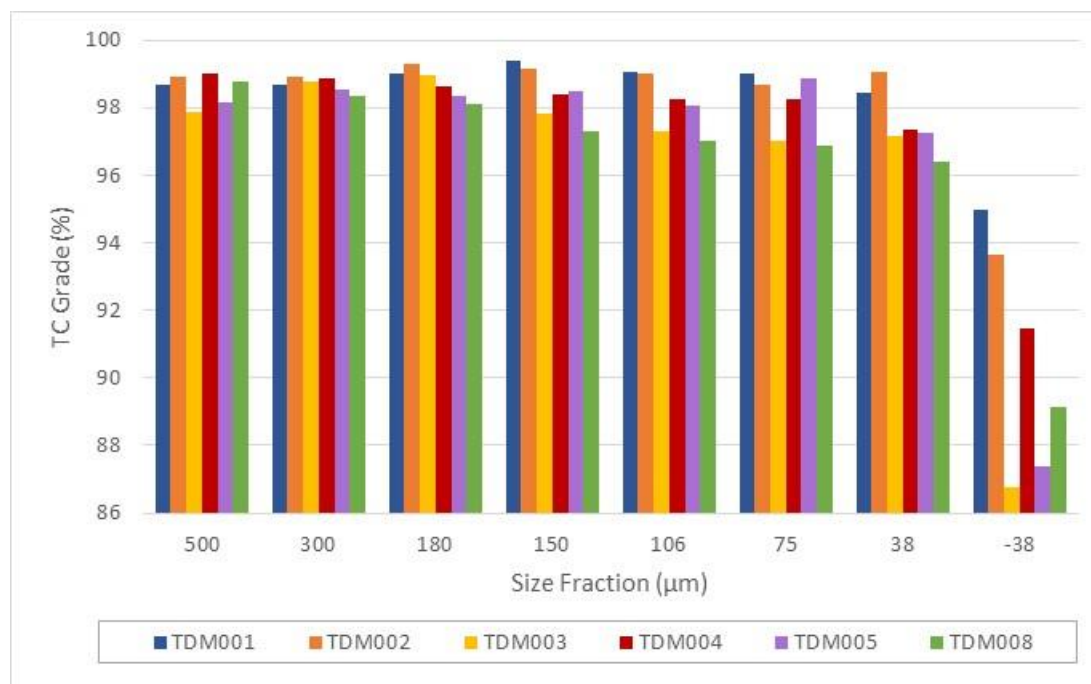
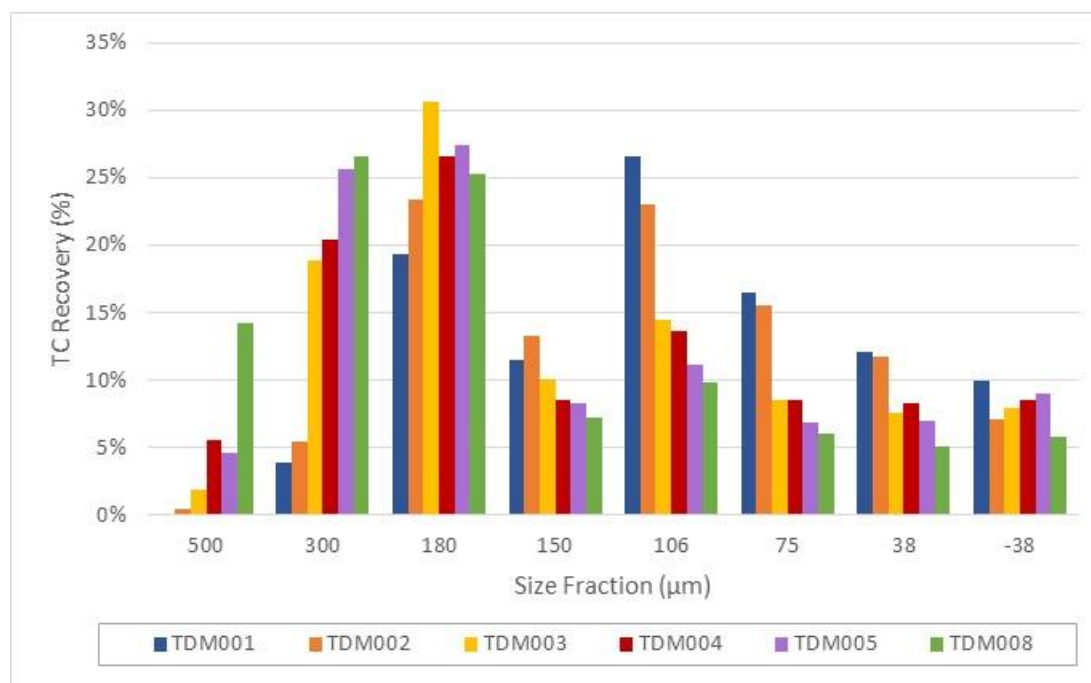


Figure 20: Fresh Composites Final Cleaner Concentrate Individual Size Fraction TC Recoveries



5.4 Fresh Variability Composite TDM005 – Effect of Reducing Feed Grind Size

Amongst the Fresh composites, the TDM005 composite had a relatively low TC rougher recovery (92.0%). The TDM005 composite also required the highest number of cleaner stages before attaining > 94% TC. TDM005 required five (5) cleaner stages whereas the other Fresh composites required between 2 and 4 cleaner stages. In addition, TDM005 had the lowest rougher concentrate TC grade of 23.1% compared to the other Fresh composites whose rougher grades ranged from 32.0% TC to 55.1%.

To increase the TC grade and recovery of the TDM005 rougher and cleaner concentrates the feed grind size was reduced from 710 μm to 500 μm with the hope that reducing the grind size may liberate more graphite. The flotation conditions for each test at P₁₀₀ 710 μm (FT07) and P₁₀₀ 500 μm (FT09) are listed in **Table 21**.

Table 21: TDM005 Flotation Conditions

Stage	Parameter	Lithology	Fresh	
		Sample ID	TDM005	
		Units / Test ID	FT07	FT09
Feed	Grind Size P100	µm	710	500
Roughers	No of Stages	#	4	4
	pH Target		8.5	8.5
	Total Soda Ash	g/t	1305	1320
	Total Ekofol	g/t	269	269
	Total Float Time	min	15	15
Regrind 1	Duration	min	4	4
Cleaner 1	pH Target		8.5	8.5
	Soda Ash	g/t	190	215
	Ekofol	g/t	20	20
	Flotation Duration	min	8	8
Regrind 2	Duration	min	4	4
Cleaner 2	pH Target		8.5	8.5
	Soda Ash	g/t	115	115
	Ekofol	g/t	20	20
	Flotation Duration	min	14	14
Regrind 3	Duration	min	2	2
Cleaner 3-6	Stages	#	4	4
	pH Target		8.5	8.5
	Soda Ash	g/t	100	115
	Ekofol	g/t	56	56
	Flotation Duration	min	44	44
Float Totals	Regrind Duration	min	10	10
	Soda Ash	g/t	1710	1765
	Ekofol	g/t	365	365
	Flotation Duration	min	80	80

TDM005 flotation results with 710 µm and 500 µm feed sizes are presented in **Table 22** and grade recovery profiles are plotted in **Figure 21**. The 710 µm and 500 µm final cleaner 6 concentrates were sized, with the results presented in **Table 23** and **Figure 22**. The results indicate:

- Decreasing the feed grind size had minimal effect on increasing the rougher concentrate grade, increasing it from 23.13% to 24.23% but significantly increased the recovery from 92.0% to 99.7%;
- Decreasing the feed grind size decreased the final cleaner 6 concentrate grade from 98.17% to 96.68% but increased the recovery from 91.1% to 97.7%;
- The number of cleaner stages required to attain TC > 94% was the same (5 stages) for both feed grind sizes. The cleaner 5 concentrate grades and recoveries at 710 μm and 500 μm were, respectively, 96.27% TC at 91.1% and 94.33% TC at 97.7%; and
- Reducing the initial grind size had minimal impact on the concentrate size distribution with the cumulative portion of +150 μm material (68.7%) at the finer grind size marginally higher than at the coarser grind size (65.2%).

Table 22: TDM005 Flotation Results

Lithology	Fresh					
Sample ID	TDM005					
Test ID	FT07			FT09		
Feed Size P ₁₀₀ (μm)	710			500		
Stage	Mass	TC		Mass	TC	
	Recovery	Grade (%)	Recovery	Recovery	Grade (%)	Recovery
Cleaner Con 6	7.5%	98.17	90.9%	8.1%	96.68	97.5%
Cleaner Con 5	7.7%	96.27	91.1%	8.3%	94.33	97.7%
Cleaner Con 4	8.2%	90.98	91.3%	9.0%	87.48	98.1%
Cleaner Con 3	9.9%	75.14	91.4%	11.2%	70.57	98.8%
Cleaner Con 2	14.6%	51.18	91.6%	16.0%	49.46	99.2%
Cleaner Con 1	22.9%	32.64	91.8%	22.9%	34.70	99.5%
Rougher Con	32.4%	23.13	92.0%	32.9%	24.23	99.7%
Calculated Head	100.0%	8.15	100.0%	100.0%	7.99	100.0%
Assayed Head		7.40			7.40	

Figure 21: TDM005 Grade Recovery Profiles at 710 μm and 500 μm Grind Sizes

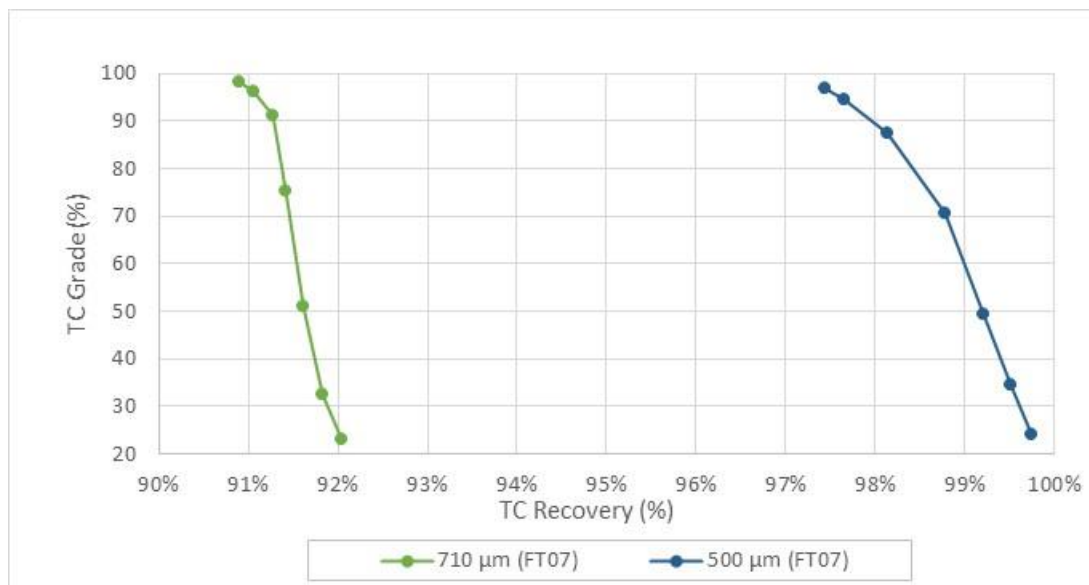
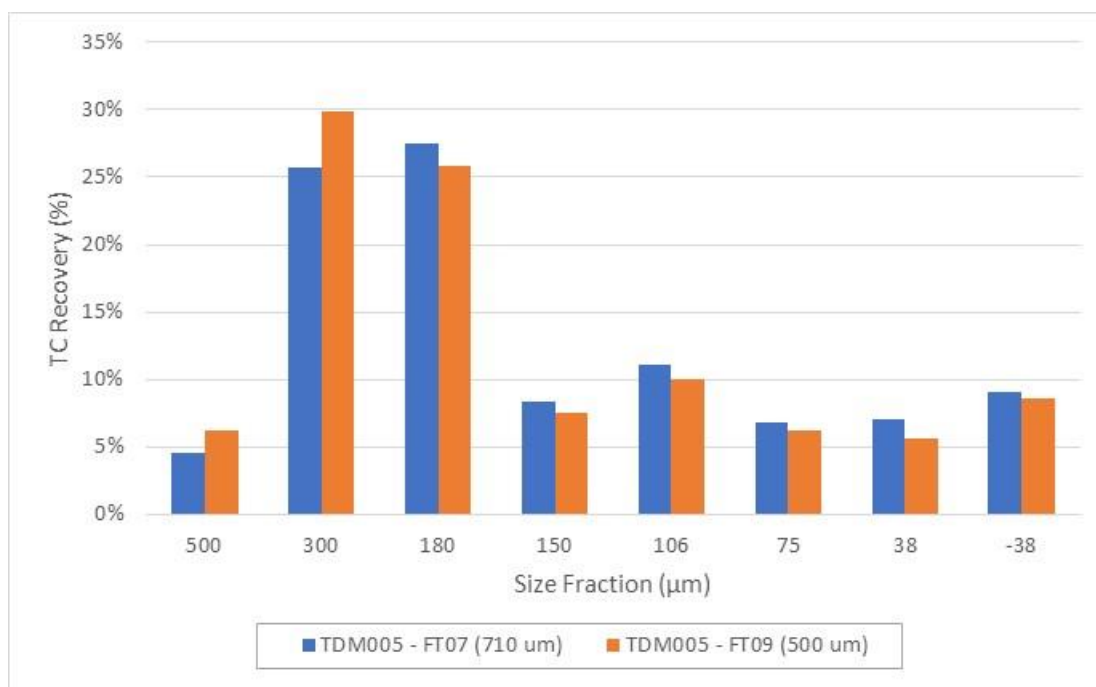


Table 23: Fresh CompositeTDM005 Final Cleaner Concentrate Sizing Results – Effect of Feed Grind Size

Sample ID	TDM005					
Test ID	FT07			FT09		
Grind Size P ₁₀₀ (μm)	710			500		
Size Fraction (μm)	Mass	TC		Mass	TC	
	Recovery	Grade (%)	Recovery	Recovery	Grade (%)	Recovery
500	4.5%	98.17	4.6%	6.2%	98.19	6.3%
300	25.3%	98.55	25.7%	29.5%	98.04	29.9%
180	27.1%	98.33	27.4%	25.6%	97.77	25.8%
150	8.2%	98.48	8.3%	7.5%	97.46	7.6%
106	11.0%	98.06	11.1%	9.9%	97.39	10.0%
75	6.7%	98.86	6.8%	6.2%	97.13	6.2%
38	7.0%	97.24	7.0%	5.7%	96.46	5.7%
-38	10.0%	87.39	9.0%	9.5%	87.04	8.6%
Total Con	100%	97.22	100%	100%	96.68	100.0%
Size Fraction (μm)	Mass	TC		Mass	TC	
	Cum. Recovery	Cum. Grade (%)	Cum. Recovery	Cum. Recovery	Cum. Grade (%)	Cum. Recovery
500	4.5%	98.17	4.6%	6.2%	98.19	6.3%
300	29.8%	98.49	30.2%	35.6%	98.07	36.2%
180	57.0%	98.42	57.7%	61.2%	97.94	62.0%
150	65.2%	98.42	66.0%	68.7%	97.89	69.6%
106	76.2%	98.37	77.1%	78.6%	97.83	79.5%
75	83.0%	98.41	84.0%	84.8%	97.78	85.8%
38	90.0%	98.32	91.0%	90.5%	97.69	91.4%
-38	100.0%	97.22	100.0%	100.0%	96.68	100.0%

Figure 22: Oxide Composite TDM005 Final Cleaner Concentrate Individual Size Fraction TC Recoveries



6 CONCLUSIONS AND RECOMMENDATIONS

6.1 Conclusions

Based on the testwork conducted to date under the Graphite Flotation Testwork Program, IMO have made the following observations:

- The Oxide composites (TDM006 and TDM007) had high TC grades and recoveries with cleaner 6 concentrate grades ranging between 97.6% to 98.4% and recoveries ranging between 93.0% and 94.9%;
- The Fresh composites (TDM001, TDM002, TDM003, TDM004, TDM005 and TDM008) had high TC grades with cleaner 6 concentrates ranging between 97.18% to 98.64%. Cleaner 6 concentrate recoveries varied between 89.1% and 95.2% with the exception of TMD004 which only achieved 75.8%;
- TMD004 TC recoveries were low due to a high carbonate content 1.8%; and
- Oxide composites with high clay content required more dilute rougher pulp densities (% solids) to provide sufficient mixing (reduced viscosity) to facilitate flotation.

6.2 Recommendations

Based on the observations on the testwork conducted to date, IMO make the following recommendations for further testwork:

- Generate Oxide and Fresh Optimisation Composites, or a single Optimisation Composite from multiple samples representative of the ore body for further process optimisation testwork. Current Oxide and Fresh Optimisation Composites utilised in this testwork program are available for further optimisation testwork if required;
- Once optimised, conduct variability testwork on composites spaced within the deposit to further understand sample process characteristics;
- Conduct bulk flotation testwork on near surface material (to reduce drilling costs) or trench material to generate a bulk concentrate for characterisation, refinery, specific application such as expandable graphite and battery testwork;
- Undertake comminution characterisation testwork on representative drill core intervals with a focus on fresh ore which generally is the harder material when processing; and
- Undertake a scoping study work determine the project economics.

IMO Pty Ltd ABN 33 084 725 557

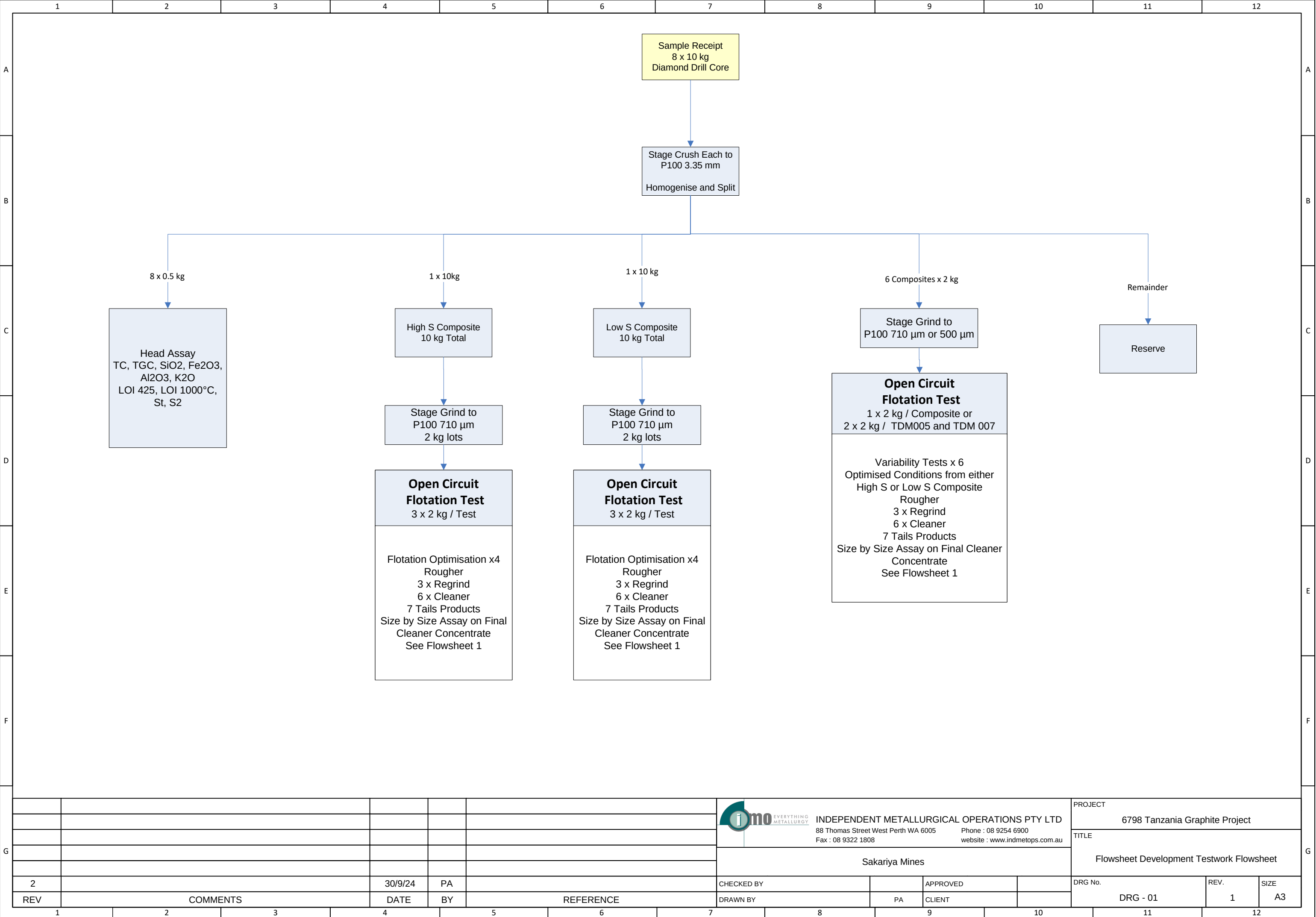
88 Thomas Street. West Perth, WA 6005

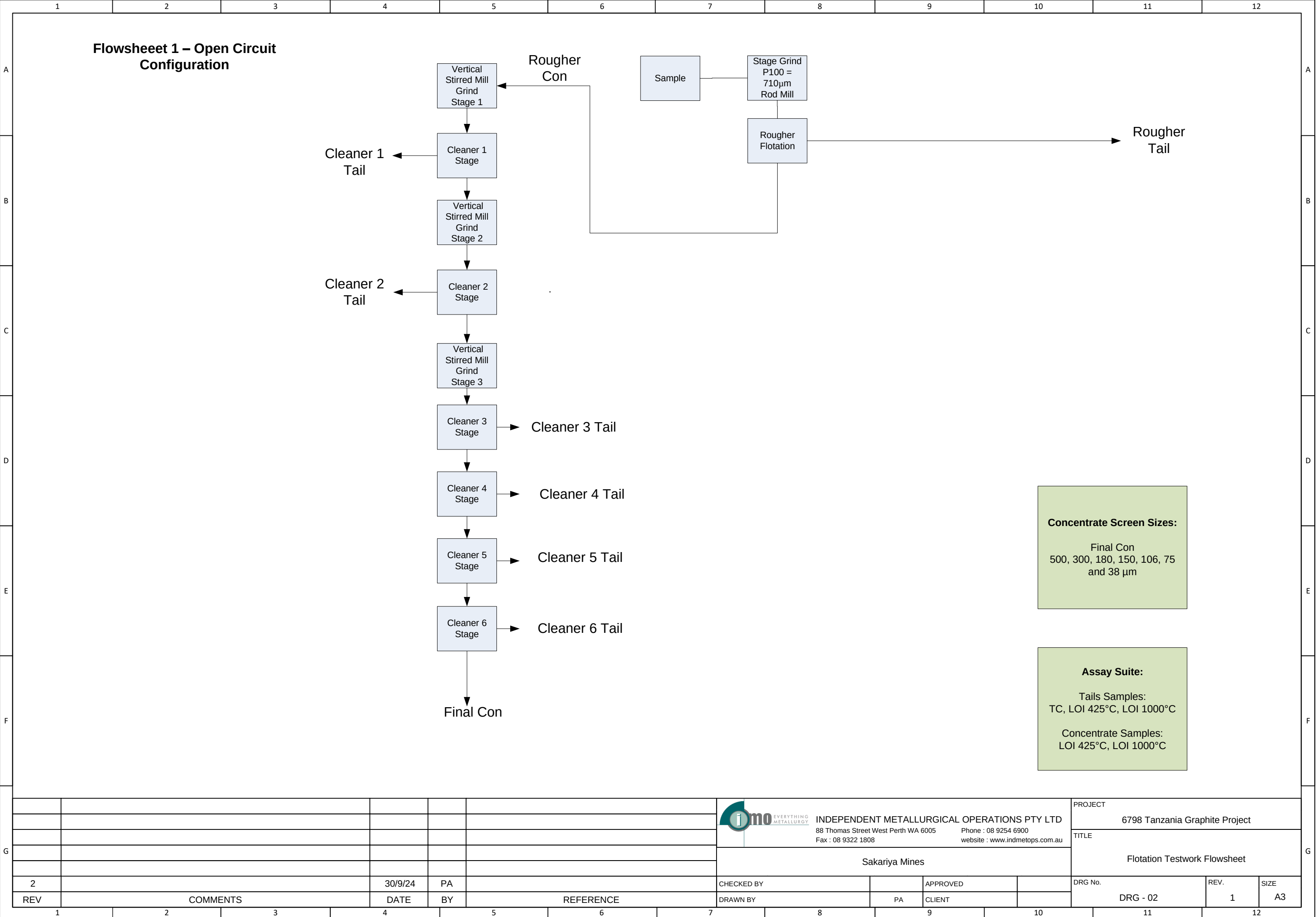
T: 61 8 9254 6900 E: admin@indmetops.com.au

Document Status

Date	Revision	Status / Comments	Prepared By	Reviewed By
12/11/2025	0	Draft, reviewed internally	J. McGinnity	P. Adamini

APPENDIX A TESTWORK FLOWSHEETS





APPENDIX B SAMPLE RECEIPT



METALLURGY PTY LTD

ABN: 24 009 466 791

64-70 Kurnall Road, Welshpool, Western Australia, 6106

Telephone +61 8 9458 3286 Fax + 61 8 9322 1808

admin@metlab.com.au

SAMPLE RECEIPT SHEET

Submitted By:		Date:	08-Jan-25
Testwork Title:		Client code:	M815
Page:		AQIS Paper Reqd:	
Sample ID		Carrier/Con. Note:	
Received From:	Sakariya Mines		
Client Contact:	Hollie		
Received By:	Bec		

Box Number	Hole Number	Meterage		Weight (kg)
		From	To	
Sample ID: TDM001	TGDD1019			
Bag 1		73.5	75.5	3.733
Bag 2		75.5	77.5	3.813
Bag 3		77.5	79.5	3.742
Sample ID: TDM002	TGDD1026			
Bag 1		26.5	28.7	4.195
Bag 2		28.7	30.9	3.770
Bag 3		30.9	33.05	3.346
Sample ID: TDM003	TGDD1037			
Bag 1		51.8	53.8	3.741
Bag 2		53.8	55.8	3.634
Bag 3		55.8	57.8	3.576
Sample ID: TDM004	TGDD1039			
Bag 1		46.3	48.3	3.554
Bag 2		48.3	50.6	4.275
Bag 3		50.6	52.9	4.621
Sample ID: TDM005	TGDD1048			
Bag 1		57.8	61.8	4.779
Bag 2		61.8	66.8	3.763
Sample ID: TDM006	TGDD1013			
Bag 1		2.2	6.2	3.280
Bag 2		6.2	8.2	2.840
Bag 3		8.2	12.17	3.797
Sample ID: TDM007	TGDD1016			
Bag 1		10.24	12.87	2.123
Bag 2		12.87	15.1	1.691
Bag 3		15.4	18.24	2.246
Bag 4		18.24	22.24	2.491
Sample ID: TDM008	TGDD1029			
Bag 1		68.3	73.3	3.476
Bag 2		73.3	75.8	2.115
Bag 3		77.8	81.8	3.927
Drum 1 sub-total				46.000
Drum 2 sub-total				36.528
Total				82.53

APPENDIX C HEAD ASSAYS

Client:	Sakariya Mines
Client ID:	6798
Date:	4/03/2025
Sample:	Head Assay Analysis
Prepared by:	Hollie Harrison



Element	Unit	LDL	TDM001	TDM002	TDM003	TDM004	TDM005	TDM006	TDM007	TDM008
TGC- Interval Estimate	%		10.0	9.0	5.0	6.0	7.0	7.0	7.6	9.7
Total Graphitic Carbon	%	0.1	9.5	8.2	6.3	5.8	7.4	8.0	8.0	7.7
Total Graphitic Carbon Rpt	%	0.01	10.2	8.3	5.8	6.2	NC	NC	NC	NC
Total Carbon	%	0.01	10.38	8.94	5.93	8.11	7.68	8.83	8.13	8.13
Non-Carbonate Carbon	%	0.01	10.31	7.86	5.81	6.94	7.53	8.06	8.02	7.72
Carbonate	%	0.01	0.07	1.08	0.12	1.17	0.15	0.77	0.11	0.41
Total Carbon Rpt	%	0.01	10.33	8.57	6.01	8.01	NC	NC	NC	NC
Non-Carbonate Carbon Rpt	%	0.01	10.31	8.33	5.89	6.18	NC	NC	NC	NC
Carbonate Rpt	%	0.01	0.02	0.24	0.12	1.83	NC	NC	NC	NC
S	%	0.01	4.06	3.78	3.74	2.24	5.06	0.08	0.07	6.46
SO ₄ ²⁺	%	0.01	0.02	0.03	0.03	0.01	0.16	0.07	0.08	0.32
S ²⁻	%	0.01	4.04	3.75	3.71	2.23	4.90	0.01	<0.01	6.14
LOI 1,000°C	%	0.01	12.85	10.86	8.37	10.54	15.26	14.01	15.7	17.87
LOI 425°C	%	0.01	0.12	0.27	0.12	0.00	1.11	1.25	2.44	1.36
Al ₂ O ₃	%	0.01	12.76	12.7	14.6	12.38	17.09	18.64	20.52	18.8
Fe ₂ O ₃	%	0.01	6.74	6.33	7.43	5.59	6.37	5.26	7.94	7.7
K ₂ O	%	0.01	1.97	1.27	2.7	2.17	2.75	0.03	0.08	1.97
SiO ₂	%	0.01	54.42	54.57	56.83	44.35	55.94	59.97	53.9	49.22
BaO	%	0.01	0.15	0.14	0.1	0.05	0.15	0.09	0.05	0.42
CaO	%	0.01	6.03	8.22	4.55	15.18	0.06	0.02	0.01	0.22
Cr ₂ O ₃	%	0.005	0.027	0.025	0.025	0.024	0.027	0.032	0.026	0.025
MgO	%	0.01	3.26	3.77	2.82	4.93	1.39	0.03	0.04	0.27
MnO	%	0.01	0.08	0.1	0.07	0.08	0.02	<0.01	<0.01	<0.01
Na ₂ O	%	0.01	0.77	0.61	1.27	0.92	0.23	0.09	0.05	0.16
P ₂ O ₅	%	0.002	0.306	0.295	0.252	0.251	0.073	0.154	0.21	0.29
SO ₃	%	0.002	3.618	1.434	6.34	2.781	7.055	0.119	0.015	8.241
TiO ₂	%	0.01	0.67	0.65	0.69	0.58	0.63	0.97	1.02	1.05
Total	%	0.01	100.27	99.79	99.94	97.24	100.37	99.5	99.69	98.35
V ₂ O ₅	%	0.005	0.14	0.155	0.146	0.098	0.27	0.135	0.109	0.158
ZrO ₂	%	0.01	0.02	0.02	0.02	0.02	0.04	0.03	0.03	0.03

APPENDIX D FLOTATION PROCEDURES

Rougher-Regrind-Cleaner Flotation Procedure

A. Sample Preparation

1. P₁₀₀ 3.35 mm Stage Crush

- 2.1 Stage crush the sample (10 kg) down to 100% passing 3.35 mm.
- 2.2 Using either a rotary or riffle splitter, mix and split the crushed samples into 2 kilogram lots.

2. P₁₀₀ 710 µm Stage Grind

- 2.1 Stage grind the P₁₀₀ 3.35 mm sample (2 kg) in the rod mill at 50 % w/w solids (2 L water) to P₁₀₀ 710 µm, further details are provided below:
 - 2.1.1 Wet screen the material through a 710 µm screen. Keep the < 710 µm as a slurry.
 - 2.1.2 Take the oversize (>710 µm) and grind it in the rod mill at 50 % w/w solids.
 - 2.1.3 Wet screen ground material over the 710 µm screen and combine the < 710 µm with the <710 µm from the initial screen stage.
 - 2.1.4 Repeat Steps 2.1.2 and 2.1.3 until between 95 and 100% of the material is <710 µm with the majority of the >710 µm material present as flake graphite. Combine this material with the <710 µm slurry.
- 2.2 Filter the combined slurry using a pressure filter, and retain the wet cake on filter paper.
- 2.3 Weigh the wet cake.

B. Rougher-Regrind-Cleaner Flotation Stages

Table 1 presents an overview of the typical rougher, regrind and cleaner stages. Ekofol additions and flotation times for each flotation test are presented Appendix A. Procedures for each of the rougher, regrind and cleaner operations are presented overleaf.

Rougher-Regrind-Cleaner Flotation Procedure

Table 1: Rougher, Regrind and Cleaner Conditions

Operation	Grind Time (min)	Float Time (min)	Ekofol (μL)
Rougher 1		1.0	223
Rougher 2		1.5	150
Rougher 3		5.0	150
Rougher 4		7.0	75
Stirred Mill Regrind Stage 1	4.0		
Cleaner 1		7.5	45
Stirred Mill Regrind Stage 2	4.0		
Cleaner 2		13.5	45
Stirred Mill Regrind Stage 3	2.0		
Cleaner 3		18.0	45
Cleaner 4		10.0	40
Cleaner 5		9.0	35
Cleaner 6		7.0	10

1. Rougher Flotation Stage 1

- 1.1. Transfer the wet cake to a 4 L flotation cell. Using a wash bottle containing tap water, rinse all the wet cake from the filter paper into the flotation cell.
- 1.2. Start the agitator and mix the pulp.
- 1.3. Adjust the water level in the flotation cell, using tap water, so that the level is approximately 5 cm below the flotation lip.
- 1.4. Check the pH and Eh of the slurry and record on the logsheet.
- 1.5. Adjust to pH 8.5 by adding soda ash and allow to condition for 1 minute
- 1.6. Add Ekofol (223 μL for Stage 1) and allow to condition for 1 minute.
- 1.7. Record the pH and Eh.
- 1.8. Slowly add air by means of the air valve to get the required froth level. The froth should be as close to the lip height as possible but not too high that it flows over the flotation cell without manual scraping.

Rougher-Regrind-Cleaner Flotation Procedure

- 1.9 Collect the froth into a clean tray. Froth removal is done by steadily and evenly removing the froth by means of a scraper, into a clean tray. Ensure that the whole area of the cell is scraped once and that the scrape rate is being maintained at 10 scrapes per minute or 1 scrape every 6 seconds.
- 1.10 During the test continually wash down the sides of the cell with water as well as adding water to maintain the correct volume of pulp in the cell.
- 1.11 Continue flotation and froth removal for the required flotation time (Stage 1 = 1 minute).
- 1.12 After the required flotation time turn off the air.
- 1.13 Record the pH and Eh of the remaining rougher tails slurry.

2. Rougher Flotation Stages 2-4

- 2.1 Adjust the rougher tails slurry from the previous rougher flotation stage to pH 8.5 by the addition of soda ash.
- 2.2 Repeat Rougher Stage 1 Steps 1.6 to 1.15 using the following Ekofol additions and flotation times.

Table 2: Rougher Stages 2-4 Float Times and Ekofol Additions

Rougher Stage	Flotation Time (min)	Ekofol (μL)
2	1.5	150
3	5	150
4	7	75

- 2.3 Combine each of the rougher flotation stage 1-4 concentrates into a single tray.

Rougher-Regrind-Cleaner Flotation Procedure

After completing Rougher Stage 4:

- 2.4 Filter the rougher tails using a Buchner funnel and flask with vacuum filtration.
- 2.5 Dry the rougher tails wet cake at 100°C and weigh after cooling to ambient temperature.
- 2.6 Submit dried rougher tails for total graphitic carbon assay.

3. Rougher Concentrate Regrind

- 3.1 To the Ultra Fine Grind Pin Mill transfer the combined rougher flotation concentrate and grinding media in alternating layers.
- 3.2 Grind the rougher concentrate for 4 minutes.
- 3.3 Remove ground rougher concentrate and media from the mill.
- 3.4 Separate ground concentrate from the media by screening through a 2 mm screen.
- 3.5 Filter the ground concentrate using a pressure filter.
- 3.6 Collect and weigh the ground concentrate wet cake.

4. Cleaner 1 Flotation

- 4.1 Transfer the rougher concentrate wet cake into a 2 L flotation cell.
- 4.2 Adjust slurry to pH 8.5 by addition of soda ash.
- 4.3 Repeat Rougher Stage 1 Steps 1.6 -1.15 using Ekofol addition and flotation time listed in **Table 1**.
- 4.4 Filter the Cleaner 1 tails using a Buchner funnel and flask with vacuum filtration.
- 4.5 Dry the Cleaner 1 tails wet cake at 100°C and weigh after cooling to ambient temperature.
- 4.6 Submit the dried Cleaner 1 tails for total graphitic carbon assay.

Rougher-Regrind-Cleaner Flotation Procedure

5. Cleaner 1 Concentrate Regrind

- 5.1 Repeat as per Steps 3.1 to 3.6 using Cleaner 1 concentrate instead of combined rougher concentrate.

6. Cleaner 2 Flotation

- 6.1 Transfer the Cleaner 1 concentrate wet cake into a 2 L flotation cell.
- 6.2 Adjust slurry to pH 8.5 by addition of soda ash.
- 6.3 Repeat Rougher Stage 1 Steps 1.6 -1.15 using Ekofol addition and flotation time listed in **Table 1**.
- 6.4 Filter the Cleaner 2 tails using a Buchner funnel and flask with vacuum filtration.
- 6.5 Dry the Cleaner 2 tails wet cake at 100°C and weigh after cooling to ambient temperature.
- 6.6 Submit dried Cleaner 2 tails for total graphitic carbon assay.

7. Cleaner 2 Concentrate Regrind

- 7.1 Repeat as per steps 3.1 to 3.6 using Cleaner 2 concentrate instead of combined rougher concentrate.

8. Cleaner 3 - 6 Flotation

- 8.1 Transfer the previous cleaner concentrate wet cake into a 2 L flotation cell.
- 8.2 Adjust slurry to pH 8.5 by addition of soda ash.
- 8.3 Repeat Rougher Stage 1 Steps 1.6 -1.15 using the appropriate Ekofol addition and flotation time listed in **Table 1**.
- 8.4 Filter the cleaner tails after each cleaner 3 - 6 flotation stage using a Buchner funnel and flask with vacuum filtration.
- 8.5 Dry each flotation stage's cleaner tails wet cake at 100°C and weigh after cooling to ambient temperature.
- 8.6 Submit dried cleaner tails for total graphitic carbon assay.

Rougher-Regrind-Cleaner Flotation Procedure

9. Cleaner Concentrate Sizing and Drying

- 9.1. Wet screen the concentrate at 38 μm using a vibrating screen.
- 9.2. Pressure filter the -38 μm material and dry at 100°C.
- 9.3. Dry the +38 μm material at 100°C.
- 9.4. Dry screen the +38 μm concentrate over a screen stack with sizes ranging from 500 μm to 38 μm .
- 9.5. Weigh and determine the portion of graphite concentrate on each screen.
Assay each size fraction for total graphitic carbon.

APPENDIX E FLOTATION RESULTS

	PROJECT	M815
	CLIENT	IMO - Sakariya Mines
	JOB REQUEST	JR002
	SAMPLE	M815 JR001 Comp 6 2kg
	TEST No	FT01
	MASS	1000g (DWE)
	GRIND SIZE	P100 710µm
	TEST DESCRIPTION	TGC 6-stage Cleaner Flotation
	DATE	7-Mar-25
	TECHNICIAN	0

REAGENT SCHEME

FT01	Time			Reagents					Monitors				
Operation	Grind Time (mins)	Cond Time (mins)	Float Time (mins)	Soda Ash	EkofoI			pH Target	pH - Start	pH - Finish	Eh (mV Ag /AgCl)	Cell Size (L)	Impellor Speed (rpm)
Concentration				g/t	g/t				8.50				
Rougher 1	0	1, 1	3	2375	145			8.50	8.52	8.27	202	16	900
Rougher 2	0	1, 1	4	250	90			8.50	8.50	8.40	137	16	900
Rougher 3	0	1, 1	14	220	135			8.50	8.48	8.36	79	16	900
Rougher 4	0	1, 1	10	125	90			8.50	8.51	8.42	215	16	900
Stirred Mill Regrind Stage 1	4.0	Obtain wet mass											
Cleaner 1	0	1, 1	13	155	47			8.50	8.40	8.18	147	4	900
Stirred Mill Regrind Stage 2	4.0	Obtain wet mass											
Cleaner 2	0	1, 1	13	55	22			8.50	8.47	8.28	187	4	900
Stirred Mill Regrind Stage 3	4.0	Obtain wet mass											
Cleaner 3	0	1, 1	12	20	11			8.50	8.44	8.29	238	2	900
Cleaner 4	0	1, 1	10	5	4			8.50	8.56	8.26	171	2	900
Cleaner 5	0	1, 1	8	5	2			8.50	8.55	8.27	153	2	900
Cleaner 6	0	1, 1	7	5	0			8.50	8.54	8.25	140	2	900

Product	Weight	
	(g)	(%)
Cleaner Con	151.80	7.64%
Cleaner Tail 6	1.01	0.05%
Cleaner Tail 5	1.94	0.10%
Cleaner Tail 4	1.97	0.10%
Cleaner Tail 3	5.34	0.27%
Cleaner Tail 2	45.80	2.30%
Cleaner Tail 1	190.30	9.58%
Rougher Tail	1589.00	79.96%
Total	1987.2	100%

DISTRIBUTION

Product	Weight		C		LOI 1000°C		LOI 425°C					
	(g)	(%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)
Cleaner Con	151.8	7.6%	98.4	93.0%	98.74	49.7%	0.3	1.5%		#DIV/0!		#DIV/0!
Cleaner Tail 6	1.0	0.1%	89.1	0.6%		0.0%		0.0%		#DIV/0!		0.0%
Cleaner Tail 5	1.9	0.1%	83.4	1.0%	83.94	0.5%	1.6	0.1%		#DIV/0!		0.0%
Cleaner Tail 4	2.0	0.1%	54.2	0.7%	58.11	0.4%	1.9	0.1%		#DIV/0!		0.0%
Cleaner Tail 3	5.3	0.3%	17.5	0.6%	25.04	0.4%	2.7	0.5%		#DIV/0!		#DIV/0!
Cleaner Tail 2	45.8	2.3%	1.0	0.3%	11.74	1.8%	2.3	3.4%		#DIV/0!		#DIV/0!
Cleaner Tail 1	190.3	9.6%	0.4	0.5%	11.18	7.1%	2.0	12.2%		#DIV/0!		#DIV/0!
Rougher Tail	1589.0	80.0%	0.3	3.4%	7.61	40.1%	1.6	82.2%		#DIV/0!		#DIV/0!
Total	1987.2	20.0%		100%		100%		100%		#DIV/0!		#DIV/0!
Calc. Head Grade			8.08		15.18		1.53		0.00		0.00	
Head Grade			8.00		14.01		1.25					
CUMULATIVE DATA												
Cleaner Con	151.8	7.6%	98.4	93.0%	98.7	49.7%	0.3	1.5%		#DIV/0!		#DIV/0!
Cleaner Tail 6	152.8	7.7%	98.4	93.6%	98.1	49.7%	0.3	1.5%		#DIV/0!		#DIV/0!
Cleaner Tail 5	154.8	7.8%	98.2	94.6%	97.9	50.2%	0.3	1.6%		#DIV/0!		#DIV/0!
Cleaner Tail 4	156.7	7.9%	97.6	95.3%	97.4	50.6%	0.3	1.7%		#DIV/0!		#DIV/0!
Cleaner Tail 3	162.1	8.2%	95.0	95.8%	95.0	51.1%	0.4	2.2%		#DIV/0!		#DIV/0!
Cleaner Tail 2	207.9	10.5%	74.3	96.1%	76.7	52.8%	0.8	5.6%		#DIV/0!		#DIV/0!
Cleaner Tail 1	398.2	20.0%	39.0	96.6%	45.4	59.9%	1.4	17.8%		#DIV/0!		#DIV/0!
Total				96.6%		59.9%		17.8%		#DIV/0!		#DIV/0!



PROJECT	M815
CLIENT	IMO - Sakariya Mines
JOB REQUEST	JR002
SAMPLE	M815 JR001 Comp 6 2kg
TEST No	FT01
MASS	1000g (DWE)
GRIND SIZE	P100 710µm
TEST DESCRIPTION	TGC 6-stage Cleaner Flotation
Sizing Test	SxA

Grade					
Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	13.20	8.7%	99.16	0.18	99.34
300	42.50	28.0%	99.13	0.15	99.28
180	36.80	24.2%	98.61	0.21	98.82
150	9.50	6.3%	98.47	0.29	98.76
106	15.00	9.9%	98.58	0.27	98.85
75	10.30	6.8%	97.92	0.36	98.28
38	10.20	6.7%	99.39	0.42	99.81
-38	14.30	9.4%	94.89	0.91	95.80
Calc Head	151.80	100.0%	98.45	0.29	98.74

Elemental Distribution

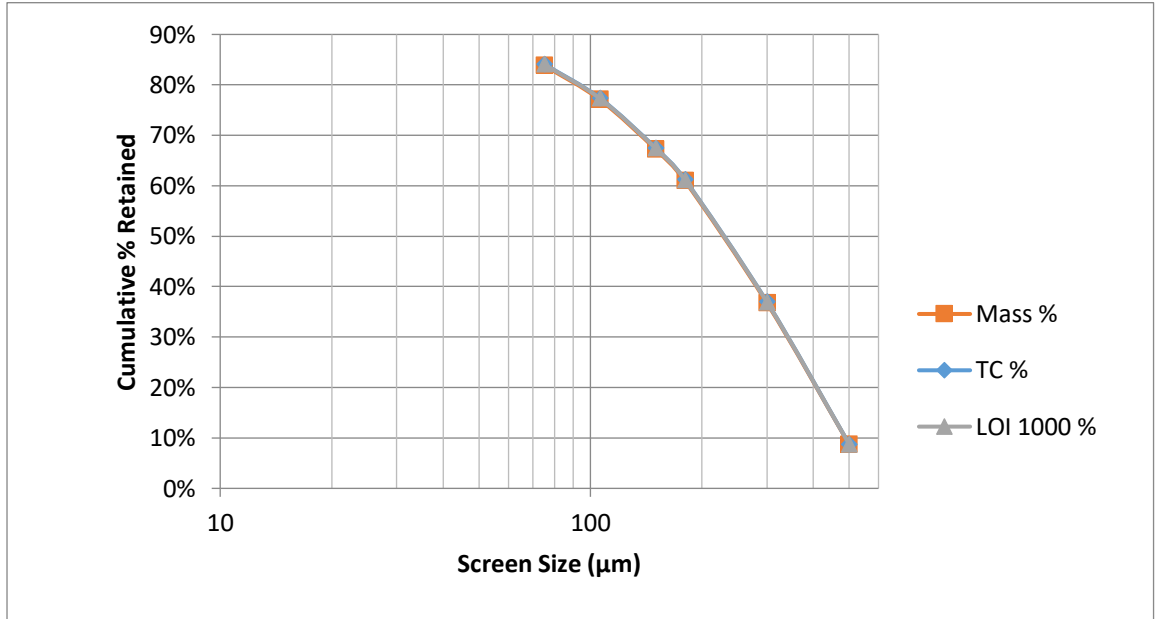
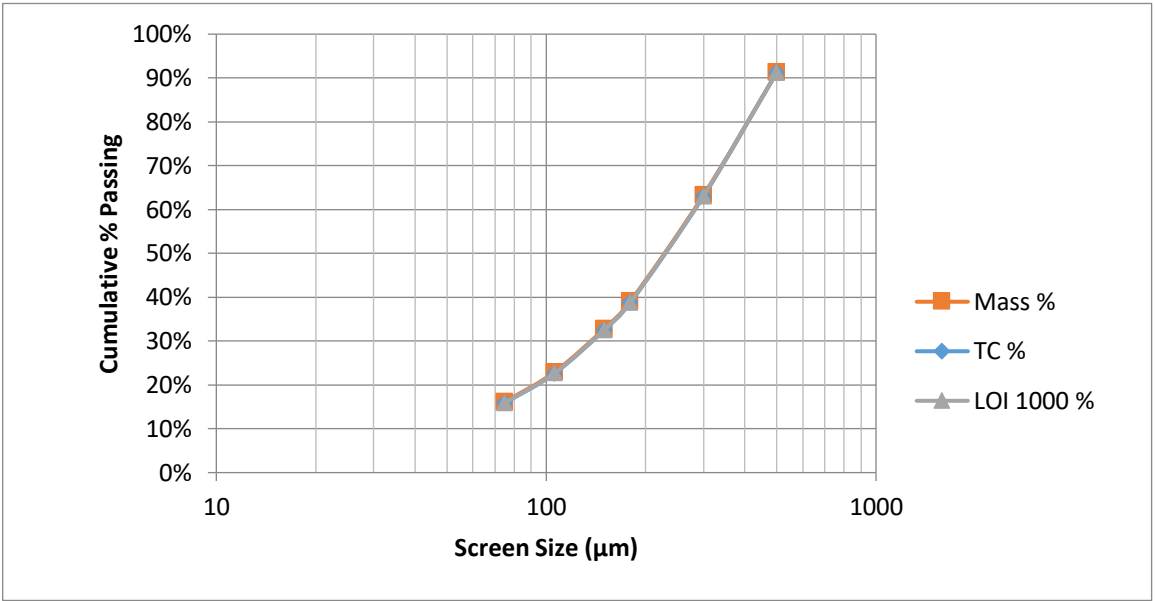
Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	13.2	8.7%	8.8%	5.4%	8.7%
300	42.5	28.0%	28.2%	14.4%	28.2%
180	36.8	24.2%	24.3%	17.4%	24.3%
150	9.5	6.3%	6.3%	6.2%	6.3%
106	15.0	9.9%	9.9%	9.1%	9.9%
75	10.3	6.8%	6.7%	8.4%	6.8%
38	10.2	6.7%	6.8%	9.7%	6.8%
-38	14.3	9.4%	9.1%	29.4%	9.1%
Total	151.8	100.0%	100.0%	100.0%	100.0%

Cumulative % Passing

Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	138.6	91.3%	91.2%	94.6%	91.3%
300	96.1	63.3%	63.0%	80.2%	63.1%
180	59.3	39.1%	38.8%	62.8%	38.8%
150	49.8	32.8%	32.5%	56.6%	32.6%
106	34.8	22.9%	22.6%	47.4%	22.7%
75	24.5	16.1%	15.9%	39.1%	15.9%
38	14.3	9.4%	9.1%	29.4%	9.1%
-38	0.0	0.0%	0%	0%	0%

Cumulative % Retained

Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	13.2	8.7%	8.8%	5.4%	8.7%
300	55.7	36.7%	37.0%	19.8%	36.9%
180	92.5	60.9%	61.2%	37.2%	61.2%
150	102.0	67.2%	67.5%	43.4%	67.4%
106	117.0	77.1%	77.4%	52.6%	77.3%
75	127.3	83.9%	84.1%	60.9%	84.1%
38	137.5	90.6%	90.9%	70.6%	90.9%
-38	151.8	100.0%	100.0%	100.0%	100.0%



 METALLURGY Quality Metallurgical Testwork	PROJECT	M815
	CLIENT	IMO - Sakariya Mines
	JOB REQUEST	JR003
	SAMPLE	M815 JR001 Comp 1 2kg
	TEST No	FT02
	MASS	2000 g (DWE)
	GRIND SIZE	P100 710µm
	TEST DESCRIPTION	TGC 6-stage Cleaner Flotation
	DATE	0-Jan-00
	TECHNICIAN	0

REAGENT SCHEME

FT02	Time			Reagents					Monitors				
Operation	Grind Time (mins)	Cond Time (mins)	Float Time (mins)	Soda Ash	Ekofol			pH Target	pH - Start	pH - Finish	Eh (mV Ag /AgCl)	Cell Size (L)	Impellor Speed (rpm)
Concentration				g/t	g/t				8.50				
Rougher 1	0	1, 1	1	45	100			8.50	8.51	8.37	35	4	900
Rougher 2	0	1, 1	2	10	67			8.50	8.50	8.34	33	4	900
Rougher 3	0	1, 1	5	20	67			8.50	8.56	8.28	49	4	900
Rougher 4	0	1, 1	7	10	34			8.50	8.51	8.24	64	4	900
Stirred Mill Regrind Stage 1	4.0	Obtain wet mass											
Cleaner 1	0	1, 1	8	20	20			8.50	8.50	8.02	63	2	800
Stirred Mill Regrind Stage 2	4.0	Obtain wet mass											
Cleaner 2	0	1, 1	14	15	20			8.50	8.52	8.06	79	2	800
Stirred Mill Regrind Stage 3	2.0	Obtain wet mass											
Cleaner 3	0	1, 1	18	25	20			8.50	8.84	7.98	44	2	800
Cleaner 4	0	1, 1	10	10	16			8.50	8.51	7.96	75	2	800
Cleaner 5	0	1, 1	9	10	16			8.50	8.59	7.93	75	2	800
Cleaner 6	0	1, 1	7	10	4			8.50	8.50	7.97	100	2	800

Product	Weight	
	(g)	(%)
Cleaner Con	196.60	9.91%
Cleaner Tail 6	1.40	0.07%
Cleaner Tail 5	0.80	0.04%
Cleaner Tail 4	2.60	0.13%
Cleaner Tail 3	9.70	0.49%
Cleaner Tail 2	41.00	2.07%
Cleaner Tail 1	247.20	12.46%
Rougher Tail	1485.10	74.84%
Total	1984.4	100%

DISTRIBUTION

Product	Weight		C		LOI 1000°C		LOI 425°C					
	(g)	(%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)
Cleaner Con	196.6	9.9%	98.6	94.4%	98.97	73.5%	0.4	14.6%		0.0%		#DIV/0!
Cleaner Tail 6	1.4	0.1%	83.6	0.6%	86.86	0.5%	1.5	0.4%		0.0%		0.0%
Cleaner Tail 5	0.8	0.0%	39.6	0.2%		0.0%		0.0%		0.0%		0.0%
Cleaner Tail 4	2.6	0.1%	19.9	0.3%	22.53	0.2%	0.9	0.4%	21.6	0.9%		0.0%
Cleaner Tail 3	9.7	0.5%	23.6	1.1%	25.42	0.9%	0.7	1.3%	24.7	3.7%		#DIV/0!
Cleaner Tail 2	41.0	2.1%	2.4	0.5%	5.22	0.8%	0.4	3.1%	4.8	3.1%		#DIV/0!
Cleaner Tail 1	247.2	12.5%	0.2	0.2%	3.68	3.4%	0.2	7.9%	3.5	13.4%		#DIV/0!
Rougher Tail	1485.1	74.8%	0.4	2.8%	3.69	20.7%	0.3	72.2%	3.4	78.9%		#DIV/0!
Total	1984.4	25.2%		100%		100%		100%		100%		#DIV/0!
Calc. Head Grade			10.35		13.35		0.27		3.25		0.00	
Head Grade			10.20		12.85		0.12					
CUMULATIVE DATA												
Cleaner Con	196.6	9.9%	98.6	94.4%	99.0	73.5%	0.4	14.6%		0.0%		#DIV/0!
Cleaner Con 5	198.0	10.0%	98.5	95.0%	98.9	73.9%	0.4	15.0%		0.0%		#DIV/0!
Cleaner Con 4	198.8	10.0%	98.2	95.1%	98.5	73.9%	0.4	15.0%		0.0%		#DIV/0!
Cleaner Con 3	201.4	10.1%	97.2	95.4%	97.5	74.1%	0.4	15.5%		0.9%		#DIV/0!
Cleaner Con 2	211.1	10.6%	93.8	96.5%	94.2	75.1%	0.4	16.8%		4.6%		#DIV/0!
Cleaner Con 1	252.1	12.7%	79.0	97.0%	79.7	75.9%	0.4	19.9%		7.6%		#DIV/0!
Rougher Con	499.3	25.2%	40.0	97.2%	42.1	79.3%	0.3	27.8%		21.1%		#DIV/0!
Total				97.2%		79.3%		27.8%		21.1%		#DIV/0!



PROJECT	M815
CLIENT	IMO - Sakariya Mines
JOB REQUEST	JR003
SAMPLE	M815 JR001 Comp 1 2kg
TEST No	FT02
MASS	2000 g (DWE)
GRIND SIZE	P100 710µm
TEST DESCRIPTION	TGC 6-stage Cleaner Flotation
Sizing Test	SxA

		Grade			
Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	0.20	0.1%	98.67	0.35	99.02
300	7.60	3.9%	98.67	0.35	99.02
180	37.90	19.3%	99.03	0.19	99.22
150	22.30	11.3%	99.41	0.28	99.69
106	52.10	26.5%	99.07	0.24	99.31
75	32.30	16.4%	99.01	0.33	99.34
38	23.80	12.1%	98.47	0.50	98.97
-38	20.40	10.4%	94.97	1.32	96.29
Calc Head	196.60	100.0%	98.58	0.40	98.97

Elemental Distribution

Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	0.2	0.1%	0.1%	0.1%	0.1%
300	7.6	3.9%	3.9%	3.4%	3.9%
180	37.9	19.3%	19.4%	9.2%	19.3%
150	22.3	11.3%	11.4%	8.0%	11.4%
106	52.1	26.5%	26.6%	16.0%	26.6%
75	32.3	16.4%	16.5%	13.6%	16.5%
38	23.8	12.1%	12.1%	15.2%	12.1%
-38	20.4	10.4%	10.0%	34.4%	10.1%
Total	196.6	100.0%	100.0%	100.0%	100.0%

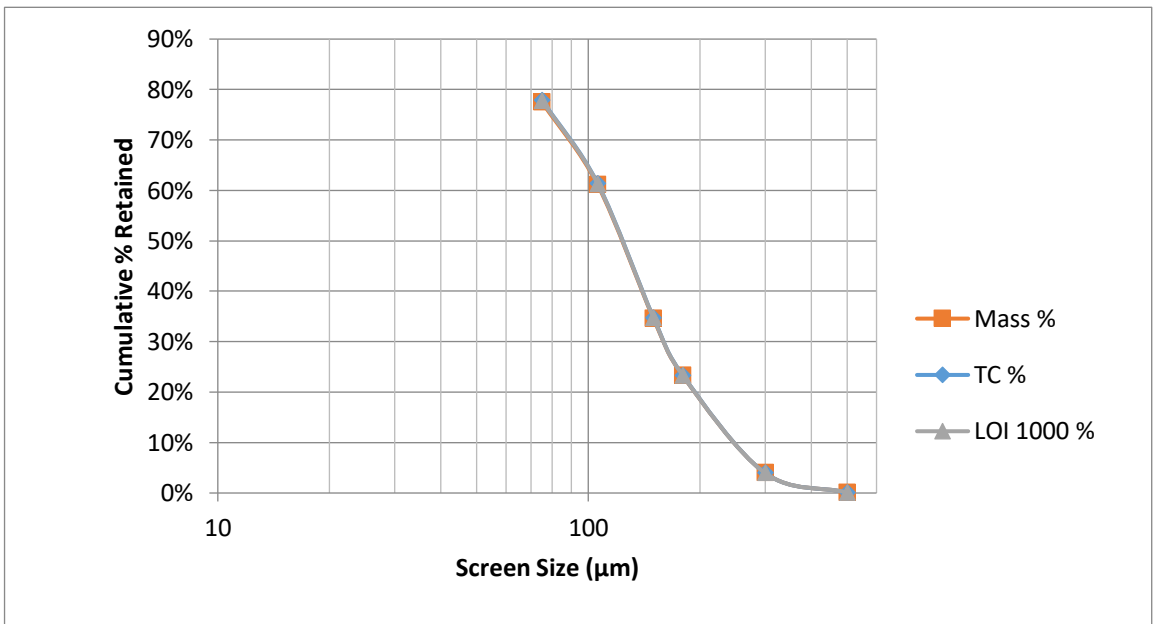
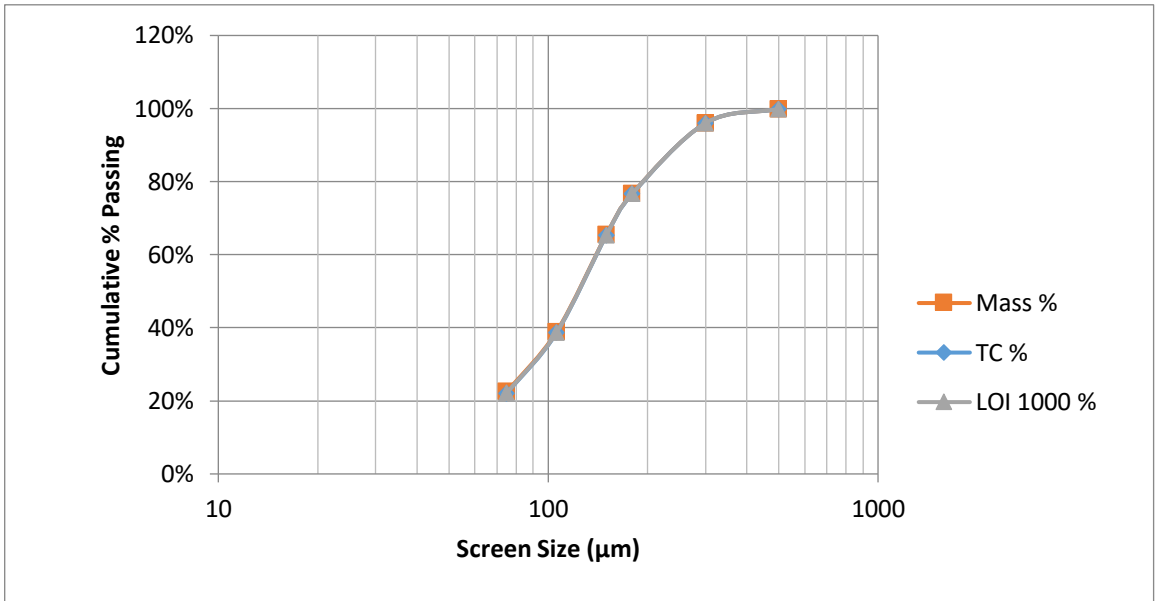
Combined for assay

Cumulative % Passing

Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	196.4	99.9%	99.9%	99.9%	99.9%
300	188.8	96.0%	96.0%	96.5%	96.0%
180	150.9	76.8%	76.7%	87.3%	76.7%
150	128.6	65.4%	65.2%	79.3%	65.3%
106	76.5	38.9%	38.6%	63.3%	38.7%
75	44.2	22.5%	22.1%	49.7%	22.2%
38	20.4	10.4%	10.0%	34.4%	10.1%
-38	0.0	0.0%	0%	0%	0%

Cumulative % Retained

Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	0.2	0.1%	0.1%	0.1%	0.1%
300	7.8	4.0%	4.0%	3.5%	4.0%
180	45.7	23.2%	23.3%	12.7%	23.3%
150	68.0	34.6%	34.8%	20.7%	34.7%
106	120.1	61.1%	61.4%	36.7%	61.3%
75	152.4	77.5%	77.9%	50.3%	77.8%
38	176.2	89.6%	90.0%	65.6%	89.9%
-38	196.6	100.0%	100.0%	100.0%	100.0%



	PROJECT	M815
	CLIENT	IMO - Sakariya Mines
	JOB REQUEST	JR004
	SAMPLE	M815 JR001 Comp 2 2kg (stage ground to 710µm)
	TEST No	FT03
	MASS	2000 g (DWE)
	GRIND SIZE	P100 710µm
	TEST DESCRIPTION	TGC 6-stage Cleaner Flotation
	DATE	27-Mar-25
	TECHNICIAN	Jermaine

REAGENT SCHEME

FT03	Time			Reagents					Monitors				
Operation	Grind Time (mins)	Cond Time (mins)	Float Time (mins)	Soda Ash	Ekofol			pH Target	pH - Start	pH - Finish	Eh (mV Ag /AgCl)	Cell Size (L)	Impellor Speed (rpm)
Concentration				g/t	g/t				8.50				
Rougher 1	0	1, 1	1	340	100			8.50	8.48	8.23	96	4	900
Rougher 2	0	1, 1	2	60	67			8.50	8.44	8.29	133	4	900
Rougher 3	0	1, 1	5	45	67			8.50	8.47	8.27	117	4	900
Rougher 4	0	1, 1	7	45	34			8.50	8.48	8.33	127	4	900
Stirred Mill Regrind Stage 1	4	Obtain wet mass											
Cleaner 1	0	1, 1	8	20	20			8.50	8.55	8.20	187	2	900
Stirred Mill Regrind Stage 2	4	Obtain wet mass											
Cleaner 2	0	1, 1	14	10	20			8.50	8.48	8.07	209	2	900
Stirred Mill Regrind Stage 3	2	185.0											
Cleaner 3	0	1, 1	18	15	20			8.50	8.63	8.01	185	2	900
Cleaner 4	0	1, 1	10	10	18			8.50	8.52	8.01	169	2	900
Cleaner 5	0	1, 1	9	10	16			8.50	8.55	8.08	163	2	900
Cleaner 6	0	1, 1	7	5	4			8.50	8.49	8.09	157	2	900

Product	Weight	
	(g)	(%)
Cleaner Con	152.70	7.69%
Cleaner Tail 6	0.31	0.02%
Cleaner Tail 5	0.33	0.02%
Cleaner Tail 4	1.10	0.06%
Cleaner Tail 3	3.80	0.19%
Cleaner Tail 2	12.10	0.61%
Cleaner Tail 1	109.80	5.53%
Rougher Tail	1706.20	85.90%
Total	1986.3	100%

DISTRIBUTION

Product	Weight		C		LOI 1000°C		LOI 425°C					
	(g)	(%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)
Cleaner Con	152.7	7.7%	98.64	89.1%	98.87	70.1%	0.22	3.5%				
Cleaner Tail 6	0.3	0.0%	64.51	0.1%	3.57	0.0%	0.36	0.0%				
Cleaner Tail 5	0.3	0.0%	64.51	0.1%	3.57	0.0%	0.36	0.0%				
Cleaner Tail 4	1.1	0.1%	64.51	0.4%	3.57	0.0%	0.36	0.0%				
Cleaner Tail 3	3.8	0.2%	43.42	1.0%	23.66	0.4%	1.21	0.5%				
Cleaner Tail 2	12.1	0.6%	4.57	0.3%	7.91	0.4%	0.56	0.7%				
Cleaner Tail 1	109.8	5.5%	0.46	0.3%	2.90	1.5%	0.31	3.5%				
Rougher Tail	1706.2	85.9%	0.86	8.7%	3.47	27.5%	0.53	91.9%				
Total	1986.3	100.0%		100%		100%		100%				
Calc. Head Grade			8.51		10.84		0.50					
Head Grade			8.30		10.86		0.27					
CUMULATIVE DATA												
Cleaner Con	152.7	7.7%	98.6	89.1%	98.9	70.1%	0.2	3.5%				
Cleaner Tail 6	153.0	7.7%	98.6	89.2%	98.7	70.1%	0.2	3.5%				
Cleaner Tail 5	153.3	7.7%	98.5	89.3%	98.5	70.1%	0.2	3.5%				
Cleaner Tail 4	154.4	7.8%	98.3	89.7%	97.8	70.2%	0.2	3.5%				
Cleaner Tail 3	158.2	8.0%	96.9	90.7%	96.0	70.6%	0.2	4.0%				
Cleaner Tail 2	170.3	8.6%	90.4	91.0%	89.8	71.0%	0.3	4.7%				
Cleaner Tail 1	280.1	14.1%	55.1	91.3%	55.7	72.5%	0.3	8.1%				
Total				91.3%		72.5%		8.1%				

Combined for Assay



PROJECT	M815
CLIENT	IMO - Sakariya Mines
JOB REQUEST	JR004
SAMPLE	M815 JR001 Comp 2 2kg (stage ground to 710µm)
TEST No	FT03
MASS	2000 g (DWE)
GRIND SIZE	P100 710µm
TEST DESCRIPTION	TGC 6-stage Cleaner Flotation
Sizing Test	SxA

			Grade		
Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	0.70	0.5%	98.94	0.27	99.21
300	8.20	5.4%	98.94	0.27	99.21
180	35.40	23.2%	99.28	0.07	99.35
150	20.20	13.2%	99.18	0.05	99.23
106	35.00	22.9%	99.02	0.11	99.13
75	23.80	15.6%	98.67	0.16	98.83
38	17.90	11.7%	99.08	0.33	99.41
-38	11.50	7.5%	93.64	1.27	94.91
Calc Head	152.70	100.0%	98.64	0.22	98.87

Elemental Distribution

Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	0.7	0.5%	0.5%	0.6%	0.5%
300	8.2	5.4%	5.4%	6.5%	5.4%
180	35.4	23.2%	23.3%	7.3%	23.3%
150	20.2	13.2%	13.3%	3.0%	13.3%
106	35.0	22.9%	23.0%	11.3%	23.0%
75	23.8	15.6%	15.6%	11.2%	15.6%
38	17.9	11.7%	11.8%	17.3%	11.8%
-38	11.5	7.5%	7.1%	42.9%	7.2%
Total	152.7	100.0%	100.0%	100.0%	100.0%

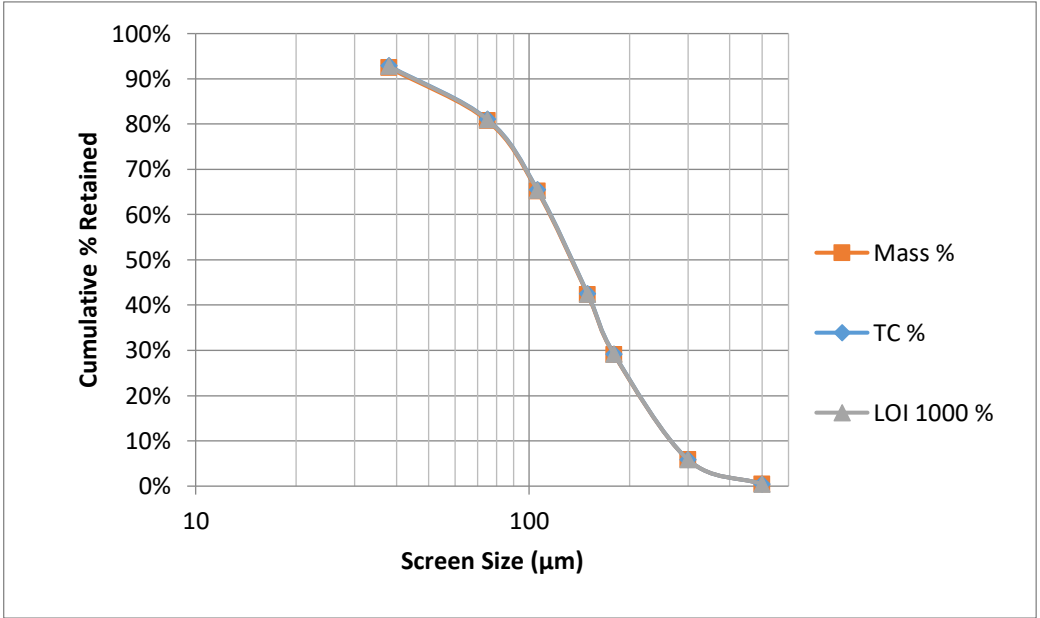
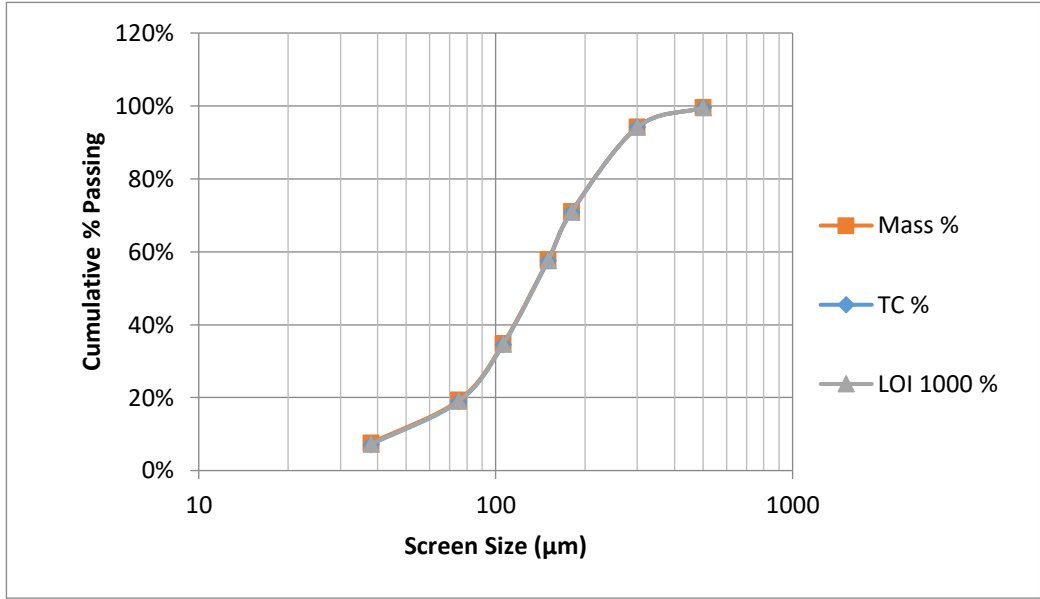
Combined for assay

Cumulative % Passing

Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	152.0	99.5%	99.5%	99.4%	99.5%
300	143.8	94.2%	94.2%	92.9%	94.2%
180	108.4	71.0%	70.8%	85.7%	70.9%
150	88.2	57.8%	57.5%	82.7%	57.6%
106	53.2	34.8%	34.5%	71.4%	34.6%
75	29.4	19.3%	18.9%	60.2%	19.0%
38	11.5	7.5%	7.1%	42.9%	7.2%
-38	0.0	0.0%			

Cumulative % Retained

Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	0.7	0.5%	0.5%	0.6%	0.5%
300	8.9	5.8%	5.8%	7.1%	5.8%
180	44.3	29.0%	29.2%	14.3%	29.1%
150	64.5	42.2%	42.5%	17.3%	42.4%
106	99.5	65.2%	65.5%	28.6%	65.4%
75	123.3	80.7%	81.1%	39.8%	81.0%
38	141.2	92.5%	92.9%	57.1%	92.8%
-38	152.7	100.0%	100.0%	100.0%	100.0%



	PROJECT	M815
	CLIENT	IMO - Sakariya Mines
	JOB REQUEST	JR005
	SAMPLE	M815 JR001 Comp 8 2kg (stage ground to 710µm)
	TEST No	FT04
	MASS	2000 g (DWE)
	GRIND SIZE	P100 710µm
	TEST DESCRIPTION	TGC 6-stage Cleaner Flotation
	DATE	28-Mar-25
	TECHNICIAN	Jermaine

REAGENT SCHEME

FT04	Time			Reagents					Monitors				
Operation	Grind Time (mins)	Cond Time (mins)	Float Time (mins)	Soda Ash	Ekofol			pH Target	pH - Start	pH - Finish	Eh (mV Ag /AgCl)	Cell Size (L)	Impellor Speed (rpm)
Concentration				g/t	g/t								
Rougher 1	0	1, 1	1	4905	123			8.50	8.48	8.32	112	8	900
Rougher 2	0	1, 1	2	270	67			8.50	8.45	8.39	140	8	900
Rougher 3	0	1, 1	5	180	67			8.50	8.47	8.20	21	8	900
Rougher 4	0	1, 1	7	290	34			8.50	8.46	8.47	109	8	900
Stirred Mill Regrind Stage 1	4	Obtain wet mass											
Cleaner 1	0	1, 1	8	185	20			8.50	8.45	8.32	172	2	900
Stirred Mill Regrind Stage 2	4	Obtain wet mass											
Cleaner 2	0	1, 1	14	55	19			8.50	8.46	8.26	184	2	900
Stirred Mill Regrind Stage 3	2	Obtain wet mass											
Cleaner 3	0	1, 1	18	25	20			8.50	8.48	8.22	192	2	900
Cleaner 4	0	1, 1	10	10	16			8.50	8.51	8.23	187	2	900
Cleaner 5	0	1, 1	9	10	16			8.50	8.55	8.24	187	2	900
Cleaner 6	0	1, 1	7	10	4			8.50	8.54	8.24	176	2	900

Product	Weight	
	(g)	(%)
Cleaner Con	139.40	7.23%
Cleaner Tail 6	0.10	0.01%
Cleaner Tail 5	0.44	0.02%
Cleaner Tail 4	2.40	0.12%
Cleaner Tail 3	6.70	0.35%
Cleaner Tail 2	32.00	1.66%
Cleaner Tail 1	128.80	6.68%
Rougher Tail	1618.60	83.93%
Total	1928.4	100%

DISTRIBUTION

Product	Weight		C		LOI 1000°C		LOI 425°C					
	(g)	(%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)
Cleaner Con	139.4	7.2%	97.37	89.1%	97.64	40.0%	0.26	1.3%				
Cleaner Tail 6	0.1	0.0%	8.58	0.0%	11.12	0.0%	1.32	0.0%				
Cleaner Tail 5	0.4	0.0%	8.58	0.0%	11.12	0.0%	1.32	0.0%				
Cleaner Tail 4	2.4	0.1%	8.58	0.1%	11.12	0.1%	1.32	0.1%				
Cleaner Tail 3	6.7	0.3%	2.38	0.1%	15.74	0.3%	3.42	0.8%				
Cleaner Tail 2	32.0	1.7%	1.07	0.2%	14.46	1.4%	2.21	2.6%				
Cleaner Tail 1	128.8	6.7%	0.33	0.3%	13.32	5.0%	2.03	9.5%				
Rougher Tail	1618.6	83.9%	0.95	10.1%	11.18	53.2%	1.45	85.6%				
Total	1928.4	16.1%		100%		100%		100%				
Calc. Head Grade			7.90		17.64		1.42					
Head Grade			7.70		17.87		1.36					
CUMULATIVE DATA												
Cleaner Con	139.4	7.2%	97.4	89.1%	97.6	40.0%	0.3	1.3%				
Cleaner Tail 6	139.5	7.2%	97.3	89.1%	97.6	40.0%	0.3	1.3%				
Cleaner Tail 5	139.9	7.3%	97.0	89.2%	97.3	40.0%	0.3	1.4%				
Cleaner Tail 4	142.3	7.4%	95.5	89.3%	95.8	40.1%	0.3	1.5%				
Cleaner Tail 3	149.0	7.7%	91.4	89.4%	92.2	40.4%	0.4	2.3%				
Cleaner Tail 2	181.0	9.4%	75.4	89.6%	78.5	41.8%	0.7	4.9%				
Cleaner Tail 1	309.8	16.1%	44.2	89.9%	51.4	46.8%	1.3	14.4%				
Total				89.9%		46.8%		14.4%				

Combined for assay



PROJECT	M815
CLIENT	IMO - Sakariya Mines
JOB REQUEST	JR005
SAMPLE	M815 JR001 Comp 8 2kg (stage ground to 710µm)
TEST No	FT04
MASS	2000 g (DWE)
GRIND SIZE	P100 710µm
TEST DESCRIPTION	TGC 6-stage Cleaner Flotation
Sizing Test	SxA

			Grade		
Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	19.60	14.1%	98.76	0.17	98.93
300	36.60	26.3%	98.37	0.18	98.55
180	35.00	25.1%	98.10	0.16	98.26
150	10.10	7.2%	97.29	0.21	97.50
106	13.70	9.8%	97.03	0.19	97.22
75	8.40	6.0%	96.87	0.25	97.12
38	7.20	5.2%	96.39	0.48	96.87
-38	8.80	6.3%	89.16	1.23	90.39
Calc Head	139.40	100.0%	97.37	0.26	97.64

Elemental Distribution

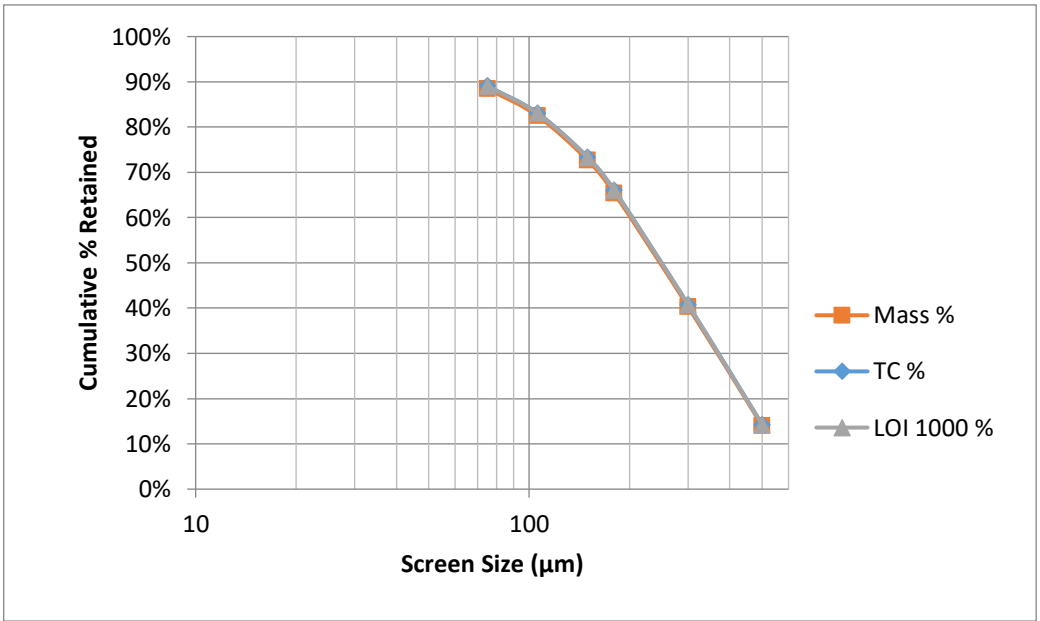
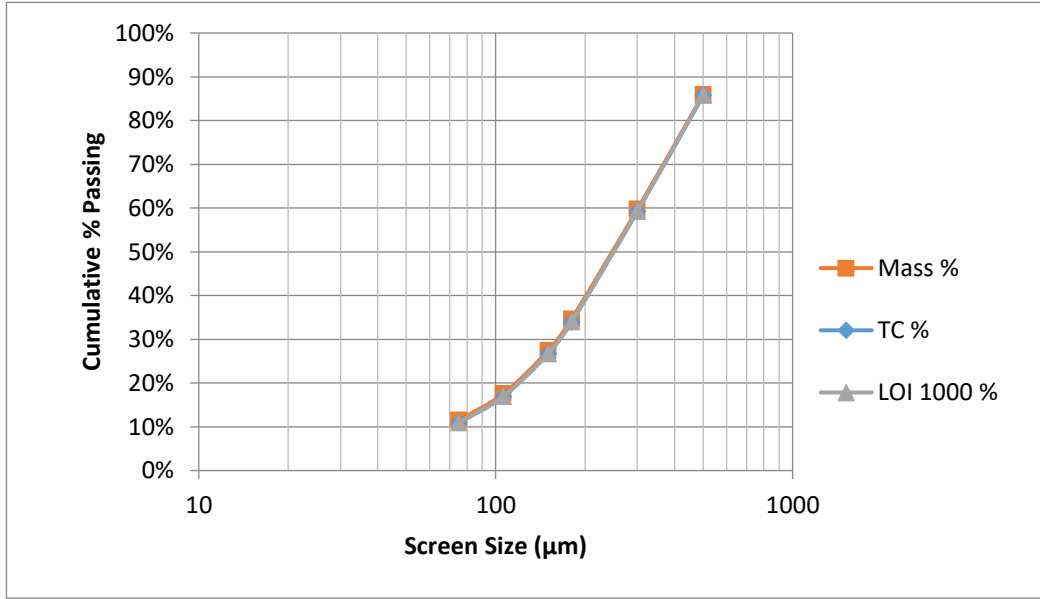
Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	19.6	14.1%	14.3%	9.1%	14.2%
300	36.6	26.3%	26.5%	18.0%	26.5%
180	35.0	25.1%	25.3%	15.3%	25.3%
150	10.1	7.2%	7.2%	5.8%	7.2%
106	13.7	9.8%	9.8%	7.1%	9.8%
75	8.4	6.0%	6.0%	5.7%	6.0%
38	7.2	5.2%	5.1%	9.4%	5.1%
-38	8.8	6.3%	5.8%	29.6%	5.8%
Total	139.4	100.0%	100.0%	100.0%	100.0%

Cumulative % Passing

Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	119.8	85.9%	85.7%	90.9%	85.8%
300	83.2	59.7%	59.2%	72.9%	59.3%
180	48.2	34.6%	33.9%	57.6%	34.0%
150	38.1	27.3%	26.7%	51.8%	26.7%
106	24.4	17.5%	16.9%	44.7%	17.0%
75	16.0	11.5%	10.9%	39.0%	11.0%
38	8.8	6.3%	5.8%	29.6%	5.8%
-38	0.0	0.0%			

Cumulative % Retained

Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	19.6	14.1%	14.3%	9.1%	14.2%
300	56.2	40.3%	40.8%	27.1%	40.7%
180	91.2	65.4%	66.1%	42.4%	66.0%
150	101.3	72.7%	73.3%	48.2%	73.3%
106	115.0	82.5%	83.1%	55.3%	83.0%
75	123.4	88.5%	89.1%	61.0%	89.0%
38	130.6	93.7%	94.2%	70.4%	94.2%
-38	139.4	100.0%	100.0%	100.0%	100.0%



	PROJECT	M815
	CLIENT	IMO - Sakariya Mines
	JOB REQUEST	JR006
	SAMPLE	M815 JR001 Comp 3 2kg (stage ground to 710µm)
	TEST No	FT05
	MASS	2000 g (DWE)
	GRIND SIZE	P100 710µm
	TEST DESCRIPTION	TGC 6-stage Cleaner Flotation
	DATE	0-Jan-00
	TECHNICIAN	Jermaine

REAGENT SCHEME

FT05	Time			Reagents					Monitors				
Operation	Grind Time (mins)	Cond Time (mins)	Float Time (mins)	Soda Ash	Ekofol			pH Target	pH - Start	pH - Finish	Eh (mV Ag /AgCl)	Cell Size (L)	Impellor Speed (rpm)
Concentration				g/t	g/t				7.61				
Rougher 1	0	1, 1	1	140	100			8.50	8.52	8.29	104	4	900
Rougher 2	0	1, 1	2	35	67			8.50	8.46	8.30	124	4	900
Rougher 3	0	1, 1	5	25	67			8.50	8.45	8.22	121	4	900
Rougher 4	0	1, 1	7	30	34			8.50	8.47	8.24	133	4	900
Stirred Mill Regrind Stage 1	4	Obtain wet mass											
Cleaner 1	0	1, 1	8	45	20			8.50	8.54	8.16	129	2	900
Stirred Mill Regrind Stage 2	4	Obtain wet mass											
Cleaner 2	0	1, 1	14	20	20			8.50	8.49	8.03	210	2	900
Stirred Mill Regrind Stage 3	2	Obtain wet mass											
Cleaner 3	0	1, 1	18	15	20			8.50	8.55	7.96	286	2	900
Cleaner 4	0	1, 1	10	15	16			8.50	8.66	8.02	269	2	900
Cleaner 5	0	1, 1	9	10	16			8.50	8.52	7.98	220	2	900
Cleaner 6	0	1, 1	7	10	4			8.50	8.57	8.00	218	2	900

Product	Weight	
	(g)	(%)
Cleaner Con	116.60	5.91%
Cleaner Tail 6	0.10	0.01%
Cleaner Tail 5	0.57	0.03%
Cleaner Tail 4	2.80	0.14%
Cleaner Tail 3	8.40	0.43%
Cleaner Tail 2	30.80	1.56%
Cleaner Tail 1	198.00	10.03%
Rougher Tail	1616.70	81.90%
Total	1974.0	100%

DISTRIBUTION

Product	Weight		C		LOI @ 1000°C		LOI @ 425°C					
	(g)	(%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)
Cleaner Con	116.6	5.9%	97.18	95.2%	97.61	71.4%	0.4	27.1%				
Cleaner Tail 6	0.1	0.0%	4.77	0.0%	12.97	0.0%	0.5	0.0%				
Cleaner Tail 5	0.6	0.0%	4.77	0.0%	12.97	0.0%	0.5	0.2%				
Cleaner Tail 4	2.8	0.1%	4.77	0.1%	12.97	0.2%	0.5	0.8%				
Cleaner Tail 3	8.4	0.4%	3.75	0.3%	10.93	0.6%	0.1	0.3%				
Cleaner Tail 2	30.8	1.6%	0.63	0.2%	6.88	1.3%	-0.6	-10.5%				
Cleaner Tail 1	198.0	10.0%	0.27	0.4%	5.15	6.4%	-0.6	-66.3%				
Rougher Tail	1616.7	81.9%	0.28	3.8%	1.97	20.0%	0.2	148.4%				
Total	1974.0	18.1%		100%		100%		100%				
Calc. Head Grade			6.03		8.07		0.09					
Head Grade			5.80		8.37		0.12					
CUMULATIVE DATA												
Cleaner Con	116.6	5.9%	97.2	95.2%	97.6	71.4%	0.4	27.1%				
Cleaner Tail 6	116.7	5.9%	97.1	95.2%	97.5	71.4%	0.4	27.1%				
Cleaner Tail 5	117.3	5.9%	96.7	95.2%	97.1	71.5%	0.4	27.3%				
Cleaner Tail 4	120.1	6.1%	94.5	95.3%	95.2	71.7%	0.4	28.1%				
Cleaner Tail 3	128.5	6.5%	88.6	95.6%	89.7	72.3%	0.4	28.4%				
Cleaner Tail 2	159.3	8.1%	71.6	95.7%	73.6	73.6%	0.2	17.9%				
Cleaner Tail 1	357.3	18.1%	32.1	96.2%	35.7	80.0%	-0.3	-48.4%				
Total				96.2%		80.0%		-48.4%				

Combined for Assay



PROJECT	M815
CLIENT	IMO - Sakariya Mines
JOB REQUEST	JR006
SAMPLE	M815 JR001 Comp 3 2kg (stage ground to 710µm)
TEST No	FT05
MASS	2000 g (DWE)
GRIND SIZE	P100 710µm
TEST DESCRIPTION	TGC 6-stage Cleaner Flotation
Sizing Test	SxA

			Grade		
Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	2.20	1.9%	97.86	0.72	98.58
300	21.70	18.6%	98.76	0.22	98.98
180	35.10	30.1%	98.98	0.15	99.13
150	11.60	9.9%	97.83	0.33	98.16
106	16.90	14.5%	97.33	0.27	97.60
75	10.00	8.6%	97.02	0.48	97.50
38	8.80	7.5%	97.17	0.72	97.89
-38	10.30	8.8%	86.76	1.85	88.61
Calc Head	116.60	100.0%	97.18	0.43	97.61

Elemental Distribution

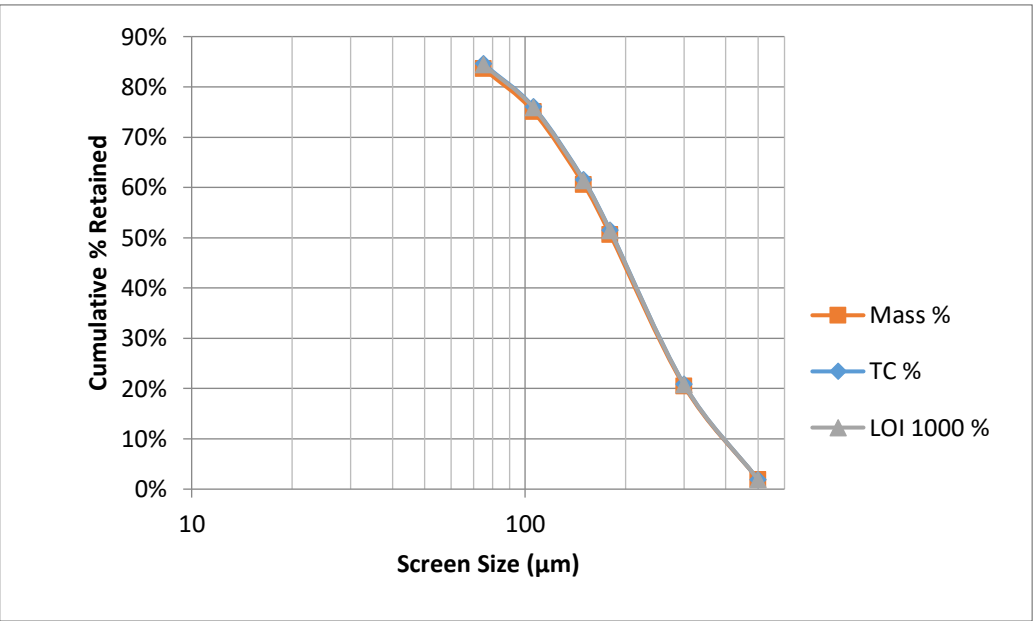
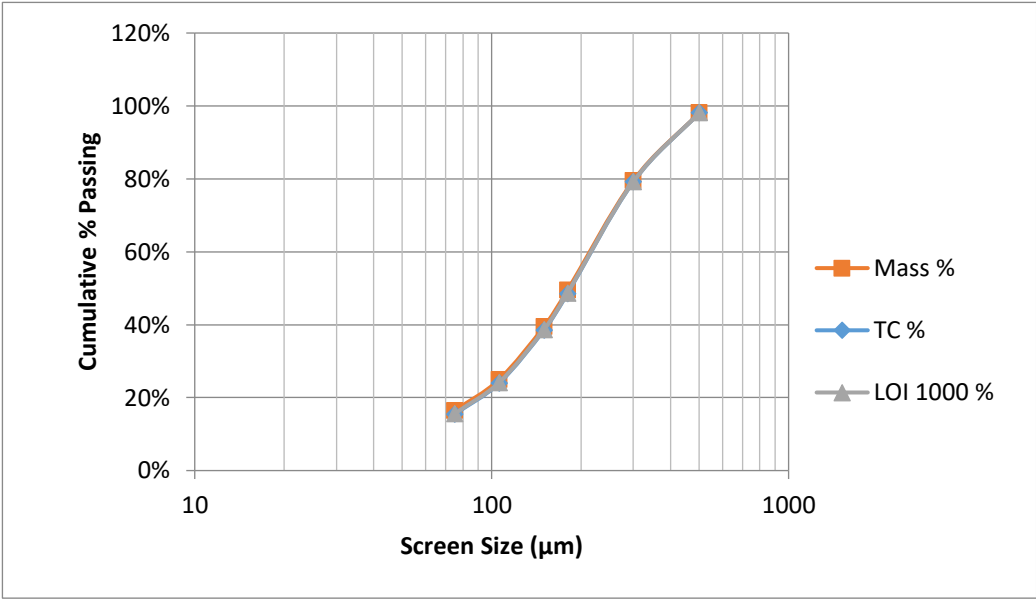
Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	2.2	1.9%	1.9%	3.2%	1.9%
300	21.7	18.6%	18.9%	9.5%	18.9%
180	35.1	30.1%	30.7%	10.5%	30.6%
150	11.6	9.9%	10.0%	7.6%	10.0%
106	16.9	14.5%	14.5%	9.1%	14.5%
75	10.0	8.6%	8.6%	9.6%	8.6%
38	8.8	7.5%	7.5%	12.6%	7.6%
-38	10.3	8.8%	7.9%	38.0%	8.0%
Total	116.6	100.0%	100.0%	100.0%	100.0%

Cumulative % Passing

Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	114.4	98.1%	98.1%	96.8%	98.1%
300	92.7	79.5%	79.2%	87.3%	79.2%
180	57.6	49.4%	48.5%	76.8%	48.7%
150	46.0	39.5%	38.5%	69.2%	38.6%
106	29.1	25.0%	24.0%	60.1%	24.2%
75	19.1	16.4%	15.4%	50.6%	15.6%
38	10.3	8.8%	7.9%	38.0%	8.0%
-38	0.0	0.0%			

Cumulative % Retained

Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	2.2	1.9%	1.9%	3.2%	1.9%
300	23.9	20.5%	20.8%	12.7%	20.8%
180	59.0	50.6%	51.5%	23.2%	51.3%
150	70.6	60.5%	61.5%	30.8%	61.4%
106	87.5	75.0%	76.0%	39.9%	75.8%
75	97.5	83.6%	84.6%	49.4%	84.4%
38	106.3	91.2%	92.1%	62.0%	92.0%
-38	116.6	100.0%	100.0%	100.0%	100.0%



	PROJECT	M815
	CLIENT	IMO - Sakariya Mines
	JOB REQUEST	JR007
	SAMPLE	M815 JR001 Comp 4 2kg (stage ground to 710µm)
	TEST No	FT06
	MASS	2000 g (DWE)
	GRIND SIZE	P100 710µm
	TEST DESCRIPTION	TGC 6-stage Cleaner Flotation
	DATE	28-Apr-25
	TECHNICIAN	Jermaine

REAGENT SCHEME

FT06	Time			Reagents					Monitors				
Operation	Grind Time (mins)	Cond Time (mins)	Float Time (mins)	Soda Ash	Ekofol			pH Target	pH - Start	pH - Finish	Eh (mV Ag /AgCl)	Cell Size (L)	Impellor Speed (rpm)
Concentration				g/t	g/t				8.50				
Rougher 1	0	1, 1	1	45	100			8.50	8.53	8.46	51	4	900
Rougher 2	0	1, 1	2	10	67			8.50	8.51	8.44	76	4	900
Rougher 3	0	1, 1	5	10	67			8.50	8.52	8.39	87	4	900
Rougher 4	0	1, 1	7	10	34			8.50	8.54	8.40	98	4	900
Stirred Mill Regrind Stage 1	4	Obtain wet mass											
Cleaner 1	0	1, 1	8	15	20			8.50	8.51	8.22	92	2	800
Stirred Mill Regrind Stage 2	4	Obtain wet mass											
Cleaner 2	0	1, 1	14	10	20			8.50	8.50	8.14	173	2	800
Stirred Mill Regrind Stage 3	2	Obtain wet mass											
Cleaner 3	0	1, 1	18	10	20			8.50	8.50	8.05	182	2	800
Cleaner 4	0	1, 1	10	10	16			8.50	8.50	8.05	186	2	800
Cleaner 5	0	1, 1	9	5	16			8.50	8.50	8.04	175	2	800
Cleaner 6	0	1, 1	7	10	4			8.50	8.56	8.05	175	2	800

Product	Weight	
	(g)	(%)
Cleaner Con	119.6	6.02%
Cleaner Tail 6	0.07	0.00%
Cleaner Tail 5	0.9	0.05%
Cleaner Tail 4	4.4	0.22%
Cleaner Tail 3	10.1	0.51%
Cleaner Tail 2	33.7	1.70%
Cleaner Tail 1	190.3	9.57%
Rougher Tail	1,629.1	81.94%
Total	1,988.2	100%

DISTRIBUTION

Product	Weight		C		LOI @ 1000°C		LOI @ 425°C					
	(g)	(%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)
Cleaner Con	119.6	6.0%	97.8	75.8%	98.1	47.6%	0.292	-31.7%				
Cleaner Tail 6	0.07	0.0%	9.45	0.0%								
Cleaner Tail 5	0.9	0.0%	9.45	0.1%								
Cleaner Tail 4	4.4	0.2%	3.79	0.1%	10.2	0.2%	0.0100	0.0%				
Cleaner Tail 3	10.1	0.5%	2.75	0.2%	7.78	0.3%	-0.110	1.0%				
Cleaner Tail 2	33.7	1.7%	1.60	0.3%	6.90	0.9%	-0.230	7.0%				
Cleaner Tail 1	190.3	9.6%	1.10	1.4%	6.94	5.4%	-0.630	108.9%				
Rougher Tail	1,629.1	81.9%	2.10	22.2%	6.90	45.6%	-0.0100	14.8%				
Total	1,988.2	18.1%		100%		100%		100%				
Calc. Head Grade			7.77		12.40		-0.06					
Head Grade (LOI Method)												
Head Grade			6.20		10.54		0.00					
CUMULATIVE DATA												
Cleaner Con	119.6	6.0%	97.8	75.8%	98.1	47.6%	0.292	-31.7%				
Cleaner Tail 5	119.7	6.0%	97.8	75.8%	98.1	47.6%	0.292	-31.7%				
Cleaner Tail 4	120.6	6.1%	97.1	75.9%	97.4	47.6%	0.289	-31.7%				
Cleaner Tail 3	125.0	6.3%	93.9	76.0%	94.3	47.8%	0.280	-31.7%				
Cleaner Tail 2	135.1	6.8%	87.0	76.1%	87.8	48.1%	0.250	-30.7%				
Cleaner Tail 1	168.8	8.5%	70.0	76.5%	71.7	49.1%	0.155	-23.7%				
Rougher 1-4 Con	359.1	18.1%	33.5	77.8%	37.4	54.4%	-0.261	85.2%				
Total				77.8%		54.4%		85.2%				

Combined for Assay. Insufficient sample for LOI.



PROJECT	M815
CLIENT	IMO - Sakariya Mines
JOB REQUEST	JR007
SAMPLE	M815 JR001 Comp 4 2kg (stage ground to 710µm)
TEST No	FT06
MASS	2000 g (DWE)
GRIND SIZE	P100 710µm
TEST DESCRIPTION	TGC 6-stage Cleaner Flotation
Sizing Test	SxA

			Grade		
Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	6.50	5.4%	99.04	0.25	99.29
300	24.20	20.2%	98.89	0.10	98.99
180	31.60	26.4%	98.64	0.17	98.81
150	10.10	8.4%	98.40	0.29	98.69
106	16.20	13.5%	98.25	0.24	98.49
75	10.20	8.5%	98.28	0.28	98.56
38	9.90	8.3%	97.36	0.43	97.79
-38	10.90	9.1%	91.46	1.06	92.52
Calc Head	119.60	100.0%	97.85	0.29	98.14

Elemental Distribution

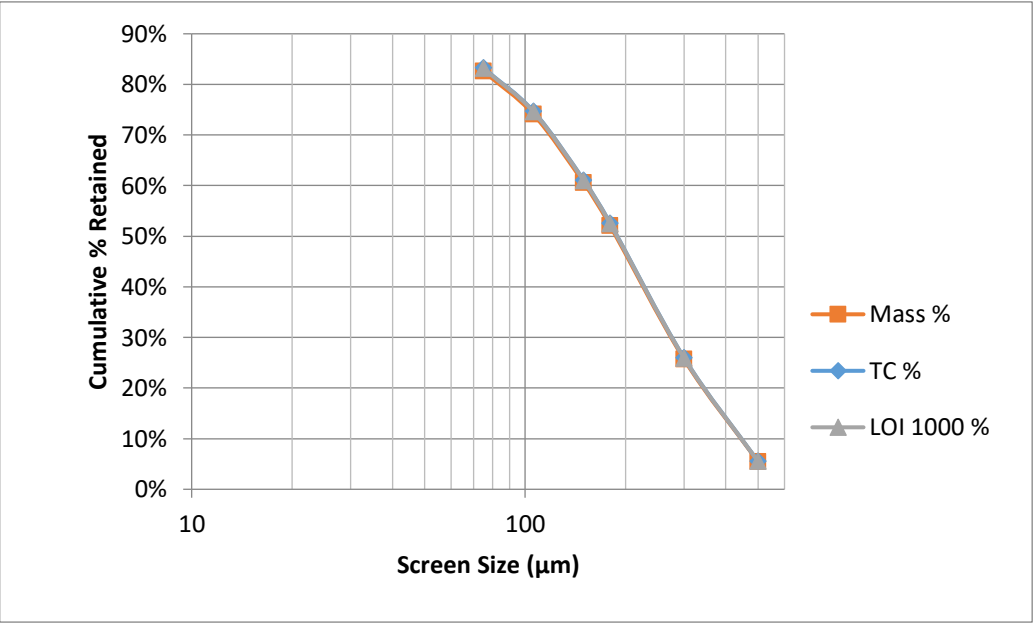
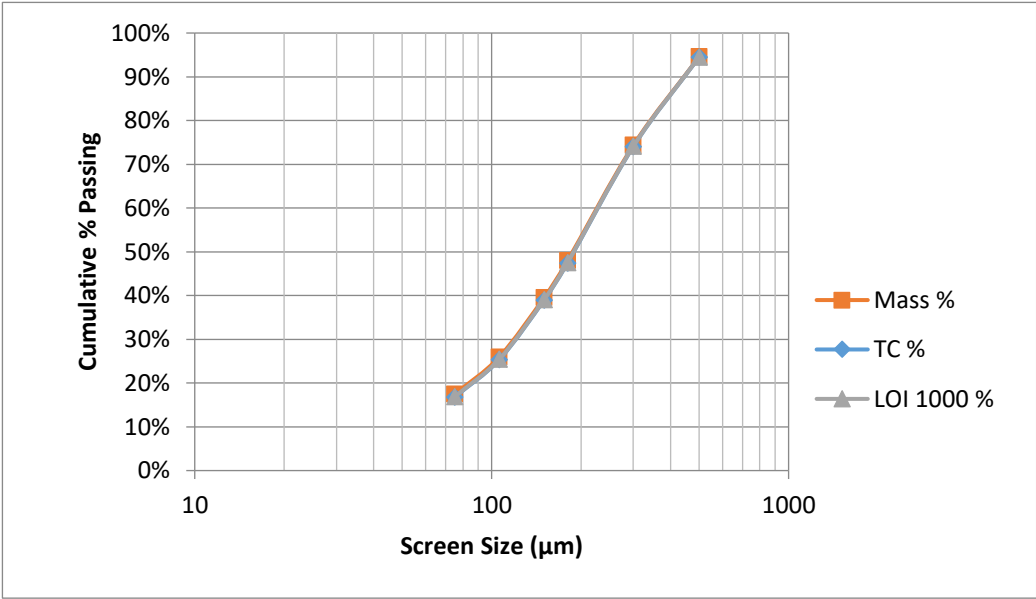
Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	6.5	5.4%	5.5%	4.7%	5.5%
300	24.2	20.2%	20.4%	6.9%	20.4%
180	31.6	26.4%	26.6%	15.4%	26.6%
150	10.1	8.4%	8.5%	8.4%	8.5%
106	16.2	13.5%	13.6%	11.1%	13.6%
75	10.2	8.5%	8.6%	8.2%	8.6%
38	9.9	8.3%	8.2%	12.2%	8.2%
-38	10.9	9.1%	8.5%	33.1%	8.6%
Total	119.6	100.0%	100.0%	100.0%	100.0%

Cumulative % Passing

Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	113.1	94.6%	94.5%	95.3%	94.5%
300	88.9	74.3%	74.0%	88.4%	74.1%
180	57.3	47.9%	47.4%	73.0%	47.5%
150	47.2	39.5%	38.9%	64.6%	39.0%
106	31.0	25.9%	25.3%	53.5%	25.4%
75	20.8	17.4%	16.8%	45.3%	16.8%
38	10.9	9.1%	8.5%	33.1%	8.6%
-38	0.0	0.0%			

Cumulative % Retained

Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	6.5	5.4%	5.5%	4.7%	5.5%
300	30.7	25.7%	26.0%	11.6%	25.9%
180	62.3	52.1%	52.6%	27.0%	52.5%
150	72.4	60.5%	61.1%	35.4%	61.0%
106	88.6	74.1%	74.7%	46.5%	74.6%
75	98.8	82.6%	83.2%	54.7%	83.2%
38	108.7	90.9%	91.5%	66.9%	91.4%
-38	119.6	100.0%	100.0%	100.0%	100.0%



	PROJECT	M815
	CLIENT	IMO - Sakariya Mines
	JOB REQUEST	JR008
	SAMPLE	M815 JR001 Comp 5 2kg (stage ground to 710µm)
	TEST No	FT07
	MASS	2000 g (DWE)
	GRIND SIZE	P100 710µm
	TEST DESCRIPTION	TGC 6-stage Cleaner Flotation
	DATE	24-Apr-25
	TECHNICIAN	Jermaine

REAGENT SCHEME

FT07	Time			Reagents					Monitors				
Operation	Grind Time (mins)	Cond Time (mins)	Float Time (mins)	Soda Ash	Ekofol			pH Target	pH - Start	pH - Finish	Eh (mV Ag /AgCl)	Cell Size (L)	Impellor Speed (rpm)
Concentration				g/t	g/t				8.50				
Rougher 1	0	1, 1	1	1035	100			8.50	8.49	8.36	103	4	900
Rougher 2	0	1, 1	2	130	67			8.50	8.49	8.37	93	4	900
Rougher 3	0	1, 1	5	75	67			8.50	8.48	8.33	93	4	900
Rougher 4	0	1, 1	7	65	34			8.50	8.48	8.31	90	4	900
Stirred Mill Regrind Stage 1	4	Obtain wet mass											
Cleaner 1	0	1, 1	8	190	20			8.50	8.47	8.29	143	2	800
Stirred Mill Regrind Stage 2	4	Obtain wet mass											
Cleaner 2	0	1, 1	14	115	20			8.50	8.48	8.34	222	2	800
Stirred Mill Regrind Stage 3	2	Obtain wet mass											
Cleaner 3	0	1, 1	18	60	20			8.50	8.48	8.27	223	2	800
Cleaner 4	0	1, 1	10	15	16			8.50	8.55	8.32	183	2	800
Cleaner 5	0	1, 1	9	15	16			8.50	8.50	8.29	253	2	800
Cleaner 6	0	1, 1	7	10	4			8.50	8.48	8.29	275	2	800

Product	Weight	
	(g)	(%)
Cleaner Con	148.5	7.54%
Cleaner Tail 6	3.2	0.16%
Cleaner Tail 5	9.2	0.47%
Cleaner Tail 4	34.2	1.74%
Cleaner Tail 3	92.0	4.67%
Cleaner Tail 2	164.2	8.34%
Cleaner Tail 1	186.8	9.49%
Rougher Tail	1,330.9	67.59%
Total	1,969.0	100%

DISTRIBUTION

Product	Weight		C		LOI @ 1000°C		LOI @ 425°C					
	(g)	(%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)
Cleaner Con	148.5	7.5%	98.17	90.9%	97.46	46.9%	0.236	1.4%				
Cleaner Tail 6	3.2	0.2%	7.88	0.2%								
Cleaner Tail 5	9.2	0.5%	3.81	0.2%	14.69	0.4%	1.95	0.7%				
Cleaner Tail 4	34.2	1.7%	0.64	0.1%	11.61	1.3%	1.51	2.0%				
Cleaner Tail 3	92.0	4.7%	0.35	0.2%	10.74	3.2%	1.50	5.4%				
Cleaner Tail 2	164.2	8.3%	0.22	0.2%	10.31	5.5%	1.34	8.5%				
Cleaner Tail 1	186.8	9.5%	0.17	0.2%	9.96	6.0%	1.28	9.3%				
Rougher Tail	1330.9	67.6%	0.96	8.0%	8.48	36.6%	1.41	72.8%				
Total	1969.0	32.4%		100%		100%		100%				
Calc. Head Grade			8.15		15.66		1.31					
Head Grade (LOI Method)												
Head Grade			7.40		15.26		1.11					
CUMULATIVE DATA												
Cleaner Con	148.5	7.5%	98.2	90.9%	97.5	46.9%	0.236	1.4%				
Cleaner Tail 5	151.7	7.7%	96.3	91.1%	95.4	46.9%	0.231	1.4%				
Cleaner Tail 4	160.9	8.2%	91.0	91.3%	90.8	47.4%	0.329	2.1%				
Cleaner Tail 3	195.1	9.9%	75.1	91.4%	76.9	48.7%	0.536	4.1%				
Cleaner Tail 2	287.1	14.6%	51.2	91.6%	55.7	51.9%	0.845	9.4%				
Cleaner Tail 1	451.3	22.9%	32.6	91.8%	39.2	57.4%	1.03	17.9%				
Rougher 1-4 Con	638.1	32.4%	23.1	92.0%	30.6	63.4%	1.10	27.2%				
Total				92.0%		63.4%		27.2%				

Insufficient sample for LOI.



PROJECT	M815
CLIENT	IMO - Sakariya Mines
JOB REQUEST	JR008
SAMPLE	M815 JR001 Comp 5 2kg (stage ground to 710µm)
TEST No	FT07
MASS	2000 g (DWE)
GRIND SIZE	P100 710µm
TEST DESCRIPTION	TGC 6-stage Cleaner Flotation
Sizing Test	SxA

			Grade		
Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	6.7	4.5%	98.17	0.32	98.49
300	37.6	25.3%	98.55	0.10	98.65
180	40.3	27.1%	98.33	0.13	98.46
150	12.2	8.2%	98.48	0.13	98.61
106	16.4	11.0%	98.06	0.17	98.23
75	10.0	6.7%	98.86	0.20	99.06
38	10.4	7.0%	97.24	0.29	97.53
-38	14.9	10.0%	87.39	0.97	88.36
Calc Head	148.5	100.0%	97.22	0.24	97.46

Elemental Distribution

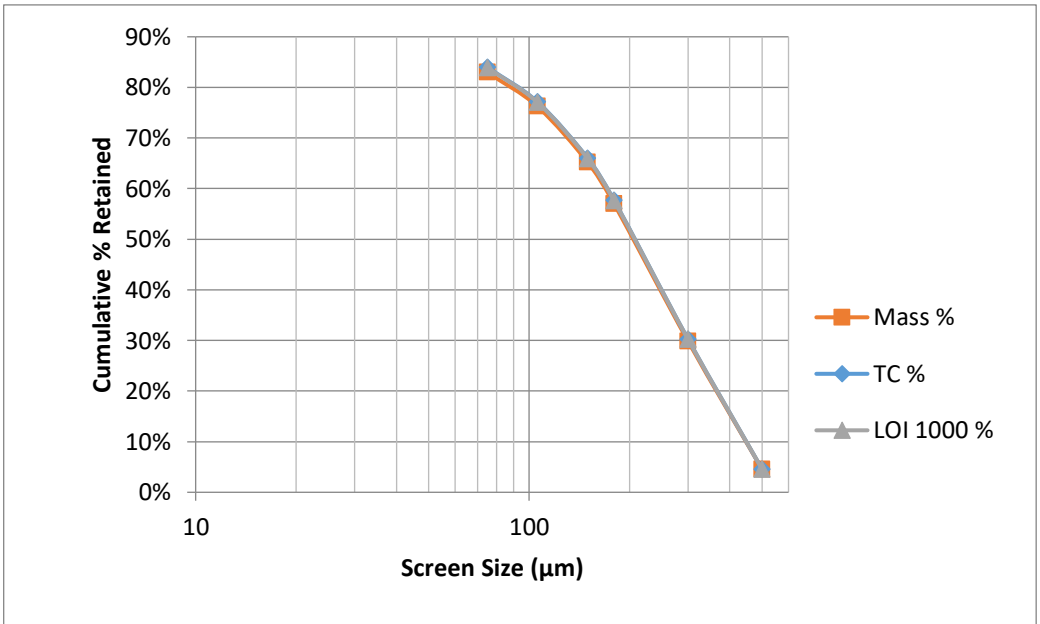
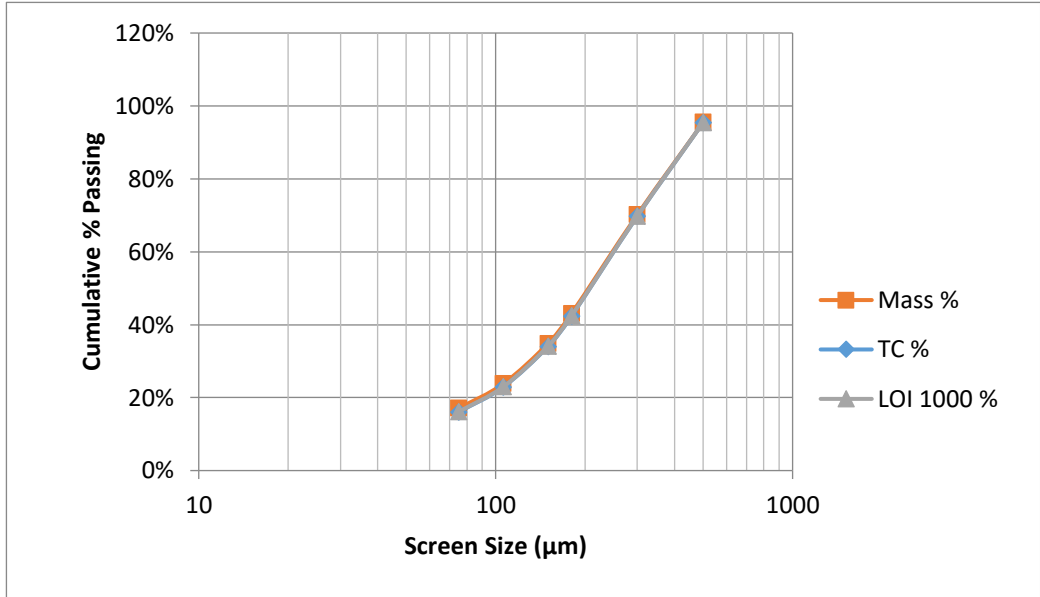
Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	6.7	4.5%	4.6%	6.1%	4.6%
300	37.6	25.3%	25.7%	10.7%	25.6%
180	40.3	27.1%	27.4%	15.0%	27.4%
150	12.2	8.2%	8.3%	4.5%	8.3%
106	16.4	11.0%	11.1%	8.0%	11.1%
75	10.0	6.7%	6.8%	5.7%	6.8%
38	10.4	7.0%	7.0%	8.6%	7.0%
-38	14.9	10.0%	9.0%	41.3%	9.1%
Total	148.5	100.0%	100.0%	100.0%	100.0%

Cumulative % Passing

Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	141.8	95.5%	95.4%	93.9%	95.4%
300	104.2	70.2%	69.8%	83.1%	69.8%
180	63.9	43.0%	42.3%	68.2%	42.4%
150	51.7	34.8%	34.0%	63.6%	34.1%
106	35.3	23.8%	22.9%	55.6%	23.0%
75	25.3	17.0%	16.0%	49.9%	16.1%
38	14.9	10.0%	9.0%	41.3%	9.1%
-38	0.0	0.0%			

Cumulative % Retained

Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	6.7	4.5%	4.6%	6.1%	4.6%
300	44.3	29.8%	30.2%	16.9%	30.2%
180	84.6	57.0%	57.7%	31.8%	57.6%
150	96.8	65.2%	66.0%	36.4%	65.9%
106	113.2	76.2%	77.1%	44.4%	77.0%
75	123.2	83.0%	84.0%	50.1%	83.9%
38	133.6	90.0%	91.0%	58.7%	90.9%
-38	148.5	100.0%	100.0%	100.0%	100.0%



 METALLURGY Quality Metallurgical Testwork	PROJECT	M815
	CLIENT	IMO - Sakariya Mines
	JOB REQUEST	JR009
	SAMPLE	M815 JR001 Comp 7 2kg (Stage ground to 710µm)
	TEST No	FT08
	MASS	2000 g (DWE)
	GRIND SIZE	P100 710µm
	TEST DESCRIPTION	TGC 6-stage Cleaner Flotation
	DATE	10-Apr-25
	TECHNICIAN	Jermaine

REAGENT SCHEME

FT08	Time			Reagents					Monitors				
Operation	Grind Time (mins)	Cond Time (mins)	Float Time (mins)	Soda Ash	Ekofol			pH Target	pH - Start	pH - Finish	Eh (mV Ag /AgCl)	Cell Size (L)	Impellor Speed (rpm)
Concentration				g/t	g/t				8.50				
Rougher 1	0	1, 1	2	2525	145			8.50	8.52	8.24	192	16	900
Rougher 2	0	1, 1	2	520	90			8.50	8.52	8.38	242	16	900
Rougher 3	0	1, 1	7	195	135			8.50	8.49	8.40	327	16	900
Rougher 4	0	1, 1	5	130	90			8.50	8.51	8.42	381	16	900
Stirred Mill Regrind Stage 1	4	Obtain wet mass											
Cleaner 1	0	1, 1	13	170	47			8.50	8.51	8.27	349	4	900
Stirred Mill Regrind Stage 2	4	Obtain wet mass											
Cleaner 2	0	1, 1	13	50	22			8.50	8.54	8.18	397	4	900
Stirred Mill Regrind Stage 3	4	Obtain wet mass											
Cleaner 3	0	1, 1	12	15	11			8.50	8.49	8.11	333	2	900
Cleaner 4	0	1, 1	10	10	4			8.50	8.52	8.12	277	2	900
Cleaner 5	0	1, 1	8	10	2			8.50	8.53	8.13	245	2	900
Cleaner 6	0	1, 1	7	5	0			8.50	8.49	8.13	223	2	900

Product	Weight	
	(g)	(%)
Cleaner Con	151.3	7.76%
Cleaner Tail 6	0.1	0.01%
Cleaner Tail 5	0.1	0.01%
Cleaner Tail 4	0.8	0.04%
Cleaner Tail 3	3.7	0.19%
Cleaner Tail 2	16.8	0.86%
Cleaner Tail 1	116.6	5.98%
Rougher Tail	1,660.7	85.16%
Total	1,950.1	100%

DISTRIBUTION

Product	Weight		C		LOI @ 1000°C		LOI @ 425°C					
	(g)	(%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)
Cleaner Con	151.3	7.8%	97.6	94.9%	97.9	48.8%	0.332	1.1%				
Cleaner Tail 6	0.1	0.01%	73.2	0.05%								
Cleaner Tail 5	0.1	0.01%	73.2	0.05%								
Cleaner Tail 4	0.8	0.04%	73.2	0.4%								
Cleaner Tail 3	3.7	0.2%	25.4	0.6%	33.7	0.4%	3.33	0.3%				
Cleaner Tail 2	16.8	0.9%	5.04	0.5%	17.0	0.9%	3.41	1.3%				
Cleaner Tail 1	116.6	6.0%	0.59	0.4%	12.6	4.8%	3.14	8.4%				
Rougher Tail	1,660.7	85.2%	0.28	3.0%	8.22	45.0%	2.34	88.9%				
Total	1,950.1	14.8%		100%		100%		100%				
Calc. Head Grade			7.97		15.6		2.24					
Head Grade (LOI Method)												
Head Grade			8.00		15.7		2.44					
CUMULATIVE DATA												
Cleaner Con	151.3	7.8%	97.6	94.9%	97.9	48.8%	0.332	1.1%				
Cleaner Tail 5	151.4	7.8%	97.6	95.0%	97.9	48.8%	0.332	1.1%				
Cleaner Tail 4	151.5	7.8%	97.6	95.0%	97.8	48.8%	0.332	1.1%				
Cleaner Tail 3	152.3	7.8%	97.4	95.4%	97.3	48.8%	0.330	1.1%				
Cleaner Tail 2	156.0	8.0%	95.7	96.0%	95.8	49.2%	0.401	1.4%				
Cleaner Tail 1	172.8	8.9%	86.9	96.6%	88.1	50.2%	0.694	2.7%				
Rougher 1-4 Con	289.4	14.8%	52.1	97.0%	57.7	55.0%	1.68	11.1%				
Total				97.0%		55.0%		11.1%				

Combined for Assay. Insufficient sample for LOI.



PROJECT	M815
CLIENT	IMO - Sakariya Mines
JOB REQUEST	JR009
SAMPLE	M815 JR001 Comp 7 2kg (Stage ground to 710µm)
TEST No	FT08
MASS	2000 g (DWE)
GRIND SIZE	P100 710µm
TEST DESCRIPTION	TGC 6-stage Cleaner Flotation
Sizing Test	SxA

			Grade		
Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	2.3	1.5%	97.99	0.65	98.64
300	18.9	12.7%	98.87	0.14	99.01
180	45.2	30.3%	97.08	0.62	97.70
150	18.9	12.7%	98.28	0.12	98.40
106	28.9	19.4%	97.92	0.14	98.06
75	17.4	11.7%	98.09	0.16	98.25
38	12.0	8.0%	97.43	0.20	97.63
-38	5.7	3.8%	91.95	1.04	92.99
Calc Head	149.3	100.0%	97.59	0.33	97.92

Elemental Distribution

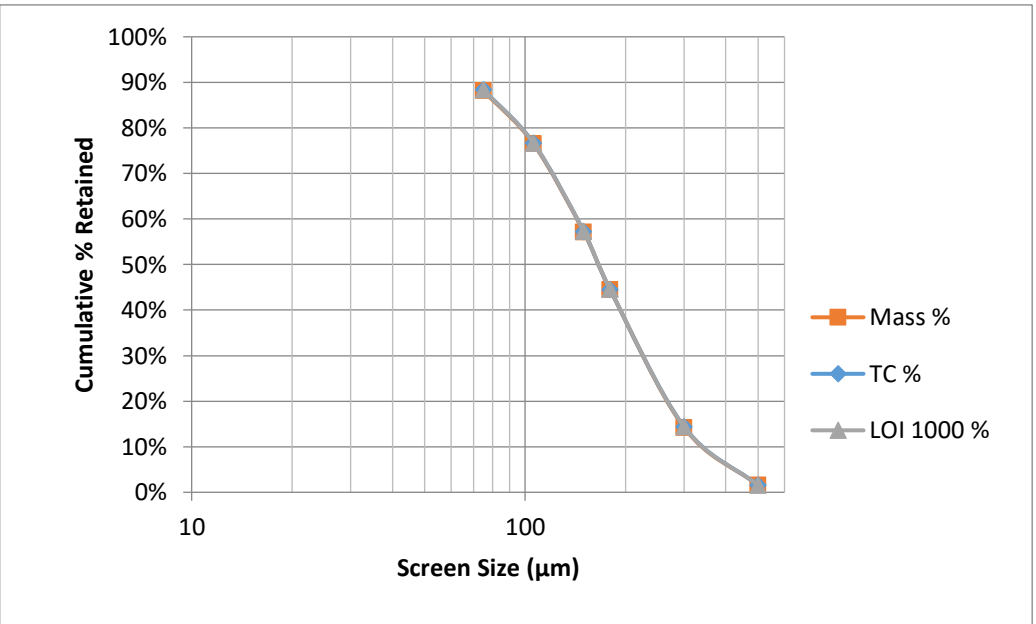
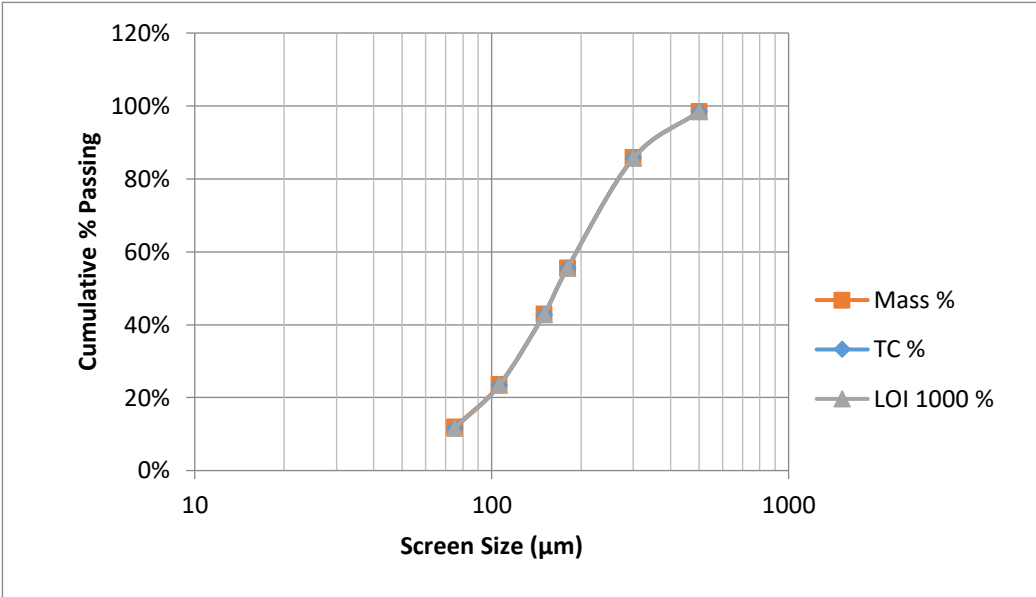
Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	2.3	1.5%	1.5%	3.0%	1.6%
300	18.9	12.7%	12.8%	5.3%	12.8%
180	45.2	30.3%	30.1%	56.5%	30.2%
150	18.9	12.7%	12.7%	4.6%	12.7%
106	28.9	19.4%	19.4%	8.2%	19.4%
75	17.4	11.7%	11.7%	5.6%	11.7%
38	12.0	8.0%	8.0%	4.8%	8.0%
-38	5.7	3.8%	3.6%	12.0%	3.6%
Total	149.3	100.0%	100.0%	100.0%	100.0%

Cumulative % Passing

Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	147.0	98.5%	98.5%	97.0%	98.4%
300	128.1	85.8%	85.6%	91.6%	85.6%
180	82.9	55.5%	55.5%	35.1%	55.4%
150	64.0	42.9%	42.8%	30.6%	42.7%
106	35.1	23.5%	23.3%	22.4%	23.3%
75	17.7	11.9%	11.6%	16.8%	11.6%
38	5.7	3.8%	3.6%	12.0%	3.6%
-38	0.0	0.0%			

Cumulative % Retained

Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	2.3	1.5%	1.5%	3.0%	1.6%
300	21.2	14.2%	14.4%	8.4%	14.4%
180	66.4	44.5%	44.5%	64.9%	44.6%
150	85.3	57.1%	57.2%	69.4%	57.3%
106	114.2	76.5%	76.7%	77.6%	76.7%
75	131.6	88.1%	88.4%	83.2%	88.4%
38	143.6	96.2%	96.4%	88.0%	96.4%
-38	149.3	100.0%	100.0%	100.0%	100.0%





PROJECT	M815
CLIENT	IMO - Sakariya Mines
JOB REQUEST	JR010
SAMPLE	Comp 5
TEST No	FT09
MASS	2000 g (DWE)
GRIND SIZE	P100 500µm
TEST DESCRIPTION	TGC 6-stage Cleaner Flotation
DATE	31-Jul-25
TECHNICIAN	BG

REAGENT SCHEME

FT09	Time			Reagents					Monitors				
Operation	Grind Time (mins)	Cond Time (mins)	Float Time (mins)	Soda Ash	Ekofol			pH Target	pH - Start	pH - Finish	Eh (mV Ag /AgCl)	Cell Size (L)	Impellor Speed (rpm)
Concentration				g/t	g/t								
Rougher 1	0	1, 1	1	1070	100			8.50	8.53	8.33	141	4	900
Rougher 2	0	1, 1	2	120	67			8.50	8.47	8.40	132	4	900
Rougher 3	0	1, 1	5	70	67			8.50	8.49	8.38	132	4	900
Rougher 4	0	1, 1	7	60	34			8.50	8.54	8.32	133	4	900
Stirred Mill Regrind Stage 1	Obtain wet mass												
Cleaner 1	0	1, 1	8	215	20			8.50	8.50	8.28	145	2	900
Stirred Mill Regrind Stage 2	Obtain wet mass												
Cleaner 2	0	1, 1	14	115	20			8.50	8.54	8.39	139	2	900
Stirred Mill Regrind Stage 3	Obtain wet mass												
Cleaner 3	0	1, 1	18	75	20			8.50	8.52	8.34	146	2	900
Cleaner 4	0	1, 1	10	25	16			8.50	8.63	8.41	141	2	900
Cleaner 5	0	1, 1	9	10	16			8.50	8.68	8.42	130	2	900
Cleaner 6	0	1, 1	7	5	4			8.50	8.66	8.36	120	2	900

Product	Weight	
	(g)	(%)
Cleaner Con	158.50	8.05%
Cleaner Tail 6	4.30	0.22%
Cleaner Tail 5	13.60	0.69%
Cleaner Tail 4	43.70	2.22%
Cleaner Tail 3	95.30	4.84%
Cleaner Tail 2	135.50	6.88%
Cleaner Tail 1	196.30	9.97%
Rougher Tail	1321.50	67.13%
Total	1968.7	100%

DISTRIBUTION

Product	Weight		C		LOI @ 1000°C		LOI @ 425°C					
	(g)	(%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)
Cleaner Con	158.5	8.1%	96.68	97.5%	97.06	49.9%	0.38	2.5%				
Cleaner Tail 6	4.3	0.2%	7.63	0.2%	18.87	0.3%	2.01	0.4%				
Cleaner Tail 5	13.6	0.7%	5.58	0.5%	16.74	0.7%	1.85	1.0%				
Cleaner Tail 4	43.7	2.2%	2.29	0.6%	13.52	1.9%	1.61	2.9%				
Cleaner Tail 3	95.3	4.8%	0.71	0.4%	11.55	3.6%	1.43	5.6%				
Cleaner Tail 2	135.5	6.9%	0.35	0.3%	10.81	4.8%	1.49	8.4%				
Cleaner Tail 1	196.3	10.0%	0.19	0.2%	10.42	6.6%	1.39	11.3%				
Rougher Tail	1321.5	67.1%	0.03	0.3%	7.51	32.2%	1.24	67.9%				
Total	1968.7	32.9%		100%		100%		100%				
Calc. Head Grade			7.99		15.65		1.23					
Head Grade (LOI Method)												
Head Grade			7.40		15.26		1.11					
CUMULATIVE DATA												
Cleaner Con	158.5	8.1%	96.7	97.5%	97.1	49.9%	0.4	2.5%				
Cleaner Tail 6	162.8	8.3%	94.3	97.7%	95.0	50.2%	0.4	2.9%				
Cleaner Tail 5	176.4	9.0%	87.5	98.1%	89.0	50.9%	0.5	3.9%				
Cleaner Tail 4	220.1	11.2%	70.6	98.8%	74.0	52.8%	0.7	6.8%				
Cleaner Tail 3	315.4	16.0%	49.5	99.2%	55.1	56.4%	1.0	12.5%				
Cleaner Tail 2	450.9	22.9%	34.7	99.5%	41.8	61.2%	1.1	20.8%				
Cleaner Tail 1	647.2	32.9%	24.2	99.7%	32.3	67.8%	1.2	32.1%				
Total				100.0%		100.0%		100.0%				



PROJECT	M815
CLIENT	IMO - Sakariya Mines
JOB REQUEST	JR010
SAMPLE	Comp 5
TEST No	FT09
MASS	2000 g (DWE)
GRIND SIZE	P100 500µm
TEST DESCRIPTION	TGC 6-stage Cleaner Flotation
Sizing Test	SxA

			Grade		
Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	9.80	6.2%	98.19	0.45	98.64
300	46.70	29.5%	98.04	0.21	98.25
180	40.50	25.6%	97.77	0.23	98.00
150	11.90	7.5%	97.46	0.32	97.78
106	15.70	9.9%	97.39	0.33	97.72
75	9.80	6.2%	97.13	0.47	97.60
38	9.00	5.7%	96.46	0.48	96.94
-38	15.10	9.5%	87.04	1.26	88.30
Calc Head	158.50	100.0%	96.68	0.38	97.06

Elemental Distribution

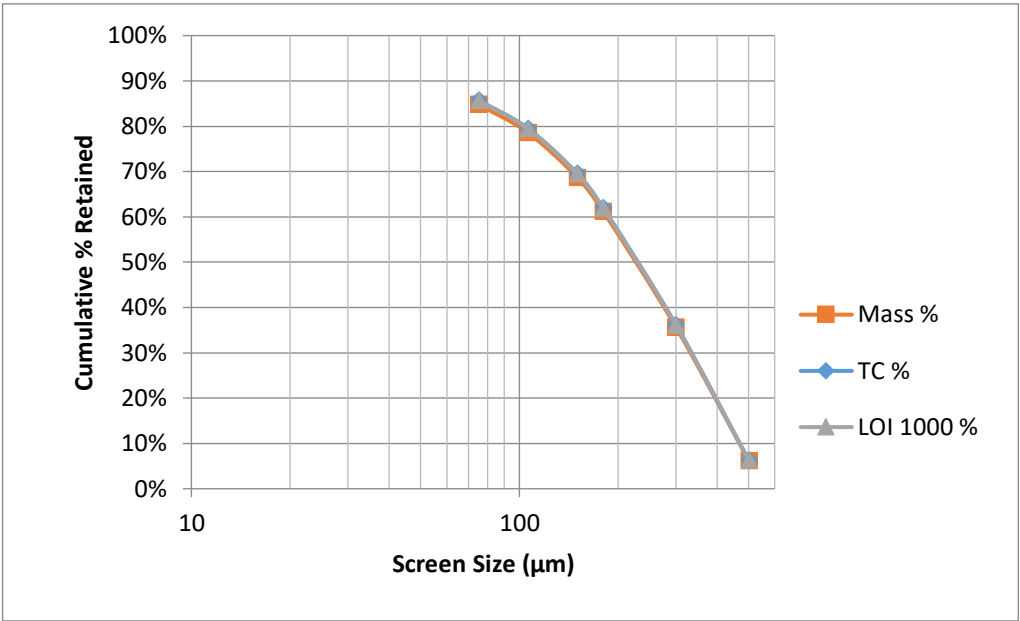
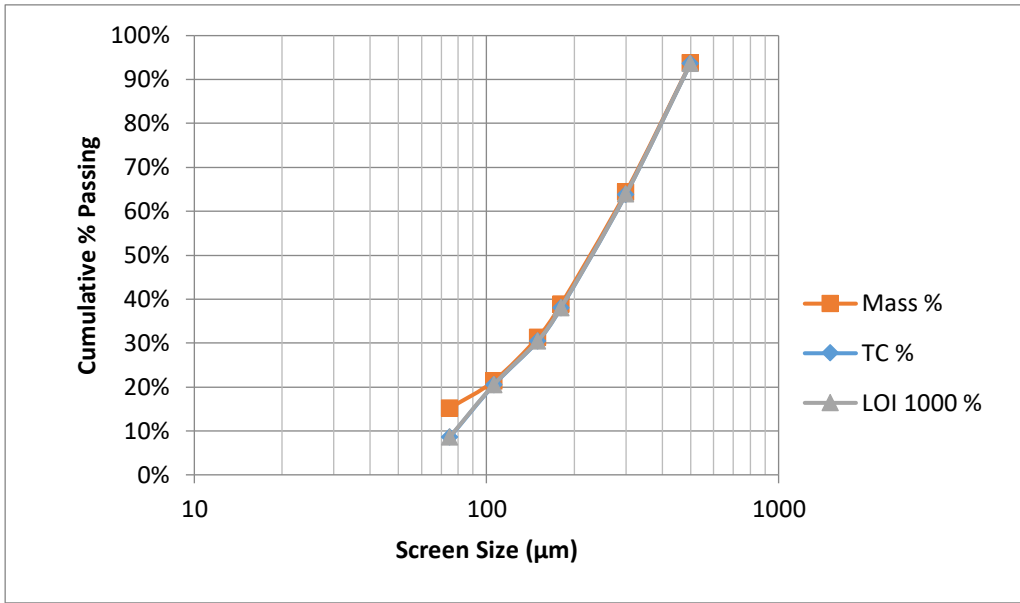
Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	9.8	6.2%	6.3%	7.3%	6.3%
300	46.7	29.5%	29.9%	16.2%	29.8%
180	40.5	25.6%	25.8%	15.4%	25.8%
150	11.9	7.5%	7.6%	6.3%	7.6%
106	15.7	9.9%	10.0%	8.6%	10.0%
75	9.8	6.2%	6.2%	7.6%	6.2%
38	9.0	5.7%	5.7%	7.1%	5.7%
-38	15.1	9.5%	8.6%	31.5%	8.7%
Total	158.5	100.0%	100.0%	100.0%	100.0%

Cumulative % Passing

Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	148.7	93.8%	93.7%	92.7%	93.7%
300	102.0	64.4%	63.8%	76.5%	63.9%
180	61.5	38.8%	38.0%	61.1%	38.1%
150	49.6	31.3%	30.4%	54.8%	30.5%
106	33.9	21.4%	20.5%	46.2%	20.6%
75	24.1	15.2%	8.6%	31.5%	8.7%
38	15.1	9.5%	108.6%	131.5%	108.7%
-38	0.0	0.0%			

Cumulative % Retained

Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	9.8	6.2%	6.3%	7.3%	6.3%
300	56.5	35.6%	36.2%	23.5%	36.1%
180	97.0	61.2%	62.0%	38.9%	61.9%
150	108.9	68.7%	69.6%	45.2%	69.5%
106	124.6	78.6%	79.5%	53.8%	79.4%
75	134.4	84.8%	85.8%	61.4%	85.7%
38	143.4	90.5%	91.4%	68.5%	91.3%
-38	158.5	100.0%	100.0%	100.0%	100.0%



 METALLURGY Quality Metallurgical Testwork	PROJECT	M815
	CLIENT	IMO - Sakariya Mines
	JOB REQUEST	JR011
	SAMPLE	M815 JR001 Comp 7 2kg (Stage ground to 710µm)
	TEST No	FT10
	MASS	2000 g (DWE)
	GRIND SIZE	P100 710µm
	TEST DESCRIPTION	TGC 6-stage Cleaner Flotation
	DATE	31-Jul-25
	TECHNICIAN	BG

REAGENT SCHEME

FT10	Time			Reagents					Monitors				
Operation	Grind Time (mins)	Cond Time (mins)	Float Time (mins)	Soda Ash	Ekofol			pH Target	pH - Start	pH - Finish	Eh (mV Ag /AgCl)	Cell Size (L)	Impellor Speed (rpm)
Concentration				g/t	g/t				6.48				
Rougher 1	0	1, 1	3	2260	145			8.50	8.52	8.21	334	8	900
Rougher 2	0	1, 1	4	450	90			8.50	8.50	8.39	394	8	900
Rougher 3	0	1, 1	14	285	135			8.50	8.48	8.02	522	8	900
Rougher 4	0	1, 1	10	260	90			8.50	8.60	8.51	596	8	900
Stirred Mill Regrind Stage 1		Obtain wet mass											
Cleaner 1	0	1, 1	13	260	47			8.50	8.50	8.34	206	4	900
Stirred Mill Regrind Stage 2		Obtain wet mass											
Cleaner 2	0	1, 1	13	90	22			8.50	8.61	8.39	121	4	900
Stirred Mill Regrind Stage 3		Obtain wet mass											
Cleaner 3	0	1, 1	12	30	11			8.50	8.51	8.39	203	2	900
Cleaner 4	0	1, 1	10	10	4			8.50	8.48	8.35	178	2	900
Cleaner 5	0	1, 1	8	10	2			8.50	8.53	8.30	348	2	900
Cleaner 6	0	1, 1	7	10	0			8.50	8.58	8.29	424	2	900

Product	Weight	
	(g)	(%)
Cleaner Con	152.6	7.98%
Cleaner Tail 6	0.6	0.03%
Cleaner Tail 5	1.9	0.10%
Cleaner Tail 4	5.3	0.28%
Cleaner Tail 3	21.7	1.14%
Cleaner Tail 2	64.0	3.35%
Cleaner Tail 1	194.8	10.19%
Rougher Tail	1,471.0	76.94%
Total	1,911.8	100%

DISTRIBUTION

Product	Weight		C		LOI @ 1000°C		LOI @ 425°C					
	(g)	(%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)	Assay (%)	Dist'n (%)
Cleaner Con	152.6	8.0%	96.59	96.7%	96.9	47.6%	0.303	1.0%				
Cleaner Tail 6	0.6	0.03%	23.54	0.09%	34.58	0.1%	4.46	0.1%				
Cleaner Tail 5	1.9	0.10%	23.54	0.29%	34.58	0.2%	4.46	0.2%				
Cleaner Tail 4	5.3	0.28%	6.16	0.2%	17.69	0.3%	3.73	0.4%				
Cleaner Tail 3	21.7	1.1%	3.28	0.5%	15.44	1.1%	3.46	1.6%				
Cleaner Tail 2	64.0	3.3%	0.87	0.4%	13.14	2.7%	3.39	4.6%				
Cleaner Tail 1	194.8	10.2%	0.30	0.4%	11.70	7.3%	3.32	13.7%				
Rougher Tail	1,471.0	76.9%	0.15	1.4%	8.59	40.7%	2.52	78.5%				
Total	1,911.8	23.1%		100%		100%		100%				
Calc. Head Grade			7.97		16.2		2.47					
Head Grade (LOI Method)												
Head Grade			8.00		15.7		2.44					
CUMULATIVE DATA												
Cleaner Con	152.6	8.0%	96.6	96.7%	96.9	47.6%	0.303	1.0%				
Cleaner Tail 5	153.2	8.0%	96.3	96.8%	96.7	47.7%	0.319	1.0%				
Cleaner Tail 4	155.0	8.1%	95.4	97.1%	95.9	47.9%	0.368	1.2%				
Cleaner Tail 3	160.3	8.4%	92.5	97.3%	93.3	48.2%	0.479	1.6%				
Cleaner Tail 2	182.0	9.5%	81.9	97.8%	84.1	49.3%	0.834	3.2%				
Cleaner Tail 1	246.0	12.9%	60.8	98.2%	65.6	52.0%	1.499	7.8%				
Rougher 1-4 Con	440.8	23.1%	34.1	98.6%	41.8	59.3%	2.30	21.5%				
Total				100.0%		100.0%		100.0%				

Combined for Assay. Insufficient sample for LOI.



PROJECT	M815
CLIENT	IMO - Sakariya Mines
JOB REQUEST	JR011
SAMPLE	M815 JR001 Comp 7 2kg (Stage ground to 710µm)
TEST No	FT10
MASS	2000 g (DWE)
GRIND SIZE	P100 710µm
TEST DESCRIPTION	TGC 6-stage Cleaner Flotation
Sizing Test	SxA

			Grade		
Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	2.9	1.9%	97.85	0.77	98.62
300	20.4	13.4%	97.45	0.18	97.63
180	46.1	30.2%	97.15	0.31	97.46
150	19.0	12.5%	96.73	0.21	96.94
106	27.0	17.7%	96.89	0.19	97.08
75	16.8	11.0%	96.60	0.23	96.83
38	12.8	8.4%	95.76	0.25	96.01
-38	7.6	5.0%	90.41	1.29	91.70
Calc Head	152.6	100.0%	96.59	0.30	96.89

Elemental Distribution

Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	2.9	1.9%	1.9%	4.8%	1.9%
300	20.4	13.4%	13.5%	8.0%	13.5%
180	46.1	30.2%	30.4%	30.9%	30.4%
150	19.0	12.5%	12.5%	8.6%	12.5%
106	27.0	17.7%	17.7%	11.1%	17.7%
75	16.8	11.0%	11.0%	8.4%	11.0%
38	12.8	8.4%	8.3%	6.9%	8.3%
-38	7.6	5.0%	4.7%	21.2%	4.7%
Total	152.6	100.0%	100.0%	100.0%	100.0%

Cumulative % Passing

Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	149.7	98.1%	98.1%	95.2%	98.1%
300	129.3	84.7%	84.6%	87.2%	84.6%
180	83.2	54.5%	54.2%	56.3%	54.2%
150	64.2	42.1%	41.7%	47.6%	41.8%
106	37.2	24.4%	24.0%	36.5%	24.0%
75	20.4	13.4%	13.0%	28.2%	13.0%
38	7.6	5.0%	4.7%	21.2%	4.7%
-38	0.0	0.0%			

Cumulative % Retained

Screen	Mass	Mass	TC	LOI 425	LOI 1000
µm	g	%	%	%	%
500	2.9	1.9%	1.9%	4.8%	1.9%
300	23.3	15.3%	15.4%	12.8%	15.4%
180	69.4	45.5%	45.8%	43.7%	45.8%
150	88.4	57.9%	58.3%	52.4%	58.2%
106	115.4	75.6%	76.0%	63.5%	76.0%
75	132.2	86.6%	87.0%	71.8%	87.0%
38	145.0	95.0%	95.3%	78.8%	95.3%
-38	152.6	100.0%	100.0%	100.0%	100.0%

